Dell EMC VMAX All Flash Product Guide

VMAX 250F, 450F, 850F, 950F with HYPERMAX OS

February 2021
Rev. 16

DELL EMC

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

© 2018 - 2021 Dell Inc. or its subsidiaries. All rights reserved. Dell, EMC, and other trademarks are trademarks of Dell Inc. or its subsidiaries. Other trademarks may be trademarks of their respective owners.

Contents

Figures **6**

Tables **7**

Preface 8
Revision history 14

Chapter 1: VMAX All Flash with HYPERMAX OS **16**
Introduction to VMAX All Flash with HYPERMAX OS 16
VMAX All Flash hardware specifications 18
Software packages 19
HYPERMAX OS 20
HYPERMAX OS emulations 21
Container applications 22
Data protection and integrity 25
Inline compression 30

Chapter 2: Management Interfaces **31**
Management interface versions 31
Unisphere for VMAX 31
Workload Planner 32
FAST Array Advisor 32
Unisphere 360 32
Solutions Enabler 32
Mainframe Enablers 33
Geographically Dispersed Disaster Restart (GDDR) 33
SMI-S Provider 34
VASA Provider 34
eNAS management interface 34
Storage Resource Management (SRM) 34
vStorage APIs for Array Integration 35
SRDF Adapter for VMware vCenter Site Recovery Manager 35
SRDF/Cluster Enabler 35
Product Suite for z/TPF 35
SRDF/TimeFinder Manager for IBM i 36
AppSync 36

Chapter 3: Open Systems Features **38**
HYPERMAX OS support for open systems 38
Backup and restore using PowerProtect Storage Direct and Data Domain 38
Backup 38
Restore 39
Storage Direct agents 40
Features used for Storage Direct backup and restore 40
Storage Direct and traditional backup 40
More information 40

VMware Virtual Volumes....41
vVol components....41
vVol scalability....41
vVol workflow....42

Chapter 4: Mainframe Features....43
HYPERMAX OS support for mainframe....43
IBM Z Systems functionality support....43
Global Mirror support....44
Transparent Cloud Tiering support....44
IBM 2107 support....46
Logical control unit capabilities....46
Disk drive emulations....46
Cascading configurations....46

Chapter 5: Provisioning....48
Thin provisioning....48
Pre-configuration for thin provisioning....49
Thin devices (TDEVs)....49
Thin device oversubscription....49
Open Systems-specific provisioning....50
Multi-array provisioning....51

Chapter 6: Native local replication with TimeFinder....53
About TimeFinder....53
Interoperability with legacy TimeFinder products....54
Targetless snapshots....54
Secure snaps....54
Provision multiple environments from a linked target....55
Cascading snapshots....55
Accessing point-in-time copies....56
Mainframe SnapVX and zDP....56
Snapshot policy....57

Chapter 7: Remote replication....59
Native remote replication with SRDF....59
SRDF 2-site solutions....60
SRDF multi-site solutions....62
Interfamily compatibility....63
SRDF device pairs....63
Dynamic device personalities....67
SRDF modes of operation....67
SRDF groups....68
Director boards, links, and ports....68
SRDF consistency....69
Data migration....69
More information....70
SRDF/Metro....71
Deployment options....71

SRDF/Metro Resilience ... 71
Disaster recovery facilities ... 73
More information ... 75
RecoverPoint ... 75
Remote replication using eNAS ... 75

Chapter 8: Blended local and remote replication ... 76
Integration of SRDF and TimeFinder ... 76
R1 and R2 devices in TimeFinder operations ... 76
SRDF/AR ... 76
SRDF/AR 2-site configurations ... 77
SRDF/AR 3-site configurations ... 78
TimeFinder and SRDF/A ... 78
TimeFinder and SRDF/S ... 79

Chapter 9: Data Migration ... 80
Overview ... 80
Data migration for open systems ... 81
Non-Disruptive Migration overview ... 81
Open Replicator ... 84
PowerPath Migration Enabler ... 85
Data migration using SRDF/Data Mobility ... 85
Space and zero-space reclamation ... 86
Data migration for mainframe ... 86
Volume migration using z/OS Migrator ... 87
Dataset migration using z/OS Migrator ... 87

Appendix A: Mainframe Error Reporting ... 89
Error reporting to the mainframe host ... 89
SIM severity reporting ... 89
Environmental errors ... 90
Operator messages ... 92

Appendix B: Licensing ... 94
eLicensing ... 94
Capacity measurements ... 95
Open systems licenses ... 96
License suites ... 96
Individual licenses ... 98
Ecosystem licenses ... 99

Figures

1 VMAX All Flash scale up and out 17
2 D@RE architecture 26
3 Inline compression and over-subscription 30
4 Data flow during a backup operation to Data Domain 39
5 Two-site Global Mirror 44
6 TCT environment with PowerMax and DLm 45
7 Auto-provisioning groups 51
8 SnapVX targetless snapshots 55
9 SnapVX cascaded snapshots 55
10 zDP operation 56
11 R1 and R2 devices 64
12 R11 device in concurrent SRDF 65
13 R21 device in cascaded SRDF 66
14 R22 devices in cascaded and concurrent SRDF/Star 66
15 Migrating data and removing a secondary (R2) array 70
16 SRDF/Metro 71
17 SRDF/Metro Smart DR 73
18 Disaster recovery for SRDF/Metro 74
19 SRDF/AR 2-site solution 77
20 SRDF/AR 3-site solution 78
21 Non-Disruptive Migration zoning 81
22 Open Replicator hot (or live) pull 85
23 Open Replicator cold (or point-in-time) pull 85
24 z/OS volume migration 87
25 z/OS Migrator dataset migration 87
26 z/OS IEA480E acute alert error message format (call home failure) 92
27 z/OS IEA480E service alert error message format (Disk Adapter failure) 93
28 z/OS IEA480E service alert error message format (SRDF Group lost/SIM presented against unrelated resource) 93
29 z/OS IEA480E service alert error message format (mirror-2 resynchronization) 93
30 z/OS IEA480E service alert error message format (mirror-1 resynchronization) 93
31 eLicensing process 94

Tables

1 Typographical conventions used in this content 13
2 Revision history 14
3 Symbol legend for VMAX All Flash software features/software package 19
4 VMAX All Flash software features by model 19
5 HYPERMAX OS emulations 21
6 eManagement resource requirements 22
7 eNAS configurations by array 24
8 Unisphere tasks 31
9 vVol architecture component management capability 41
10 vVol-specific scalability 41
11 Logical control unit maximum values 46
12 Maximum LPARs per port 46
13 RAID options 49
14 RAID options 49
15 SRDF 2-site solutions 60
16 SRDF multi-site solutions 62
17 SIM severity alerts 89
18 Environmental errors reported as SIM messages 90
19 VMAX All Flash product title capacity types 95
20 VMAX All Flash license suites 96
21 Individual licenses for open systems environment 98
22 Individual licenses for open systems environment 99

Preface

As part of an effort to improve its product lines, Dell EMC periodically releases revisions of its software and hardware. Functions that are described in this document may not be supported by all versions of the software or hardware. The product release notes provide the most up-to-date information about product features.

Contact your Dell EMC representative if a product does not function properly or does not function as described in this document.

NOTE: This document was accurate at publication time. New versions of this document might be released on Dell EMC Online Support (https://www.dell.com/support/home). Check to ensure that you are using the latest version of this document.

Purpose

This document introduces the features of the VMAX All Flash 250F, 450F, 850F, 950F arrays running HYPERMAX OS 5977.

Audience

This document is intended for use by customers and Dell EMC representatives.

Related documentation

The following documentation portfolios contain documents related to the hardware platform and manuals needed to manage your software and storage system configuration. Also listed are documents for external components that interact with the VMAX All Flash array.

Hardware platform documents:

Dell EMC VMAX All Flash Site Planning Guide for VMAX 250F, 450F, 850F, 950F with HYPERMAX OS	Provides planning information regarding the purchase and installation of a VMAX 250F, 450F, 850F, 950F with HYPERMAX OS.
Dell EMC VMAX Best Practices Guide for AC Power Connections	Describes the best practices to assure fault-tolerant power to a VMAX3 Family array or VMAX All Flash array.
Dell EMC VMAX Power-down/Power-up Procedure	Describes how to power-down and power-up a VMAX3 Family array or VMAX All Flash array.
Dell EMC VMAX Securing Kit Installation Guide	Describes how to install the securing kit on a VMAX3 Family array or VMAX All Flash array.
E-Lab™ Interoperability Navigator (ELN)	Provides a web-based interoperability and solution search portal. You can find the ELN at https://elabnavigator.EMC.com.

Unisphere documents:

EMC Unisphere for VMAX Release Notes	Describes new features and any known limitations for Unisphere for VMAX.
EMC Unisphere for VMAX Installation Guide	Provides installation instructions for Unisphere for VMAX.
EMC Unisphere for VMAX Online Help	Describes the Unisphere for VMAX concepts and functions.
EMC Unisphere for VMAX Performance Viewer Installation Guide	Provides installation instructions for Unisphere for VMAX Performance Viewer.
EMC Unisphere for VMAX Database Storage Analyzer Online Help	Describes the Unisphere for VMAX Database Storage Analyzer concepts and functions.
EMC Unisphere 360 for VMAX Release Notes	Describes new features and any known limitations for Unisphere 360 for VMAX.
EMC Unisphere 360 for VMAX Installation Guide	Provides installation instructions for Unisphere 360 for VMAX.
EMC Unisphere 360 for VMAX Online Help	Describes the Unisphere 360 for VMAX concepts and functions.

Solutions Enabler documents:

Dell EMC Solutions Enabler, VSS Provider, and SMI-S Provider Release Notes	Describes new features and any known limitations.
Dell EMC Solutions Enabler Installation and Configuration Guide	Provides host-specific installation instructions.
Dell EMC Solutions Enabler CLI Reference Guide	Documents the SYMCLI commands, daemons, error codes and option file parameters provided with the Solutions Enabler man pages.
Dell EMC Solutions Enabler Array Controls and Management CLI User Guide	Describes how to configure array control, management, and migration operations using SYMCLI commands for arrays running HYPERMAX OS and PowerMaxOS.
Dell EMC Solutions Enabler Array Controls and Management CLI User Guide	Describes how to configure array control, management, and migration operations using SYMCLI commands for arrays running Enginuity.
Dell EMC Solutions Enabler	Describes how to configure and manage SRDF environments using SYMCLI commands.

SRDF Family CLI User Guide	
Dell EMC Solutions Enabler SRDF Family State Tables Guide	Describes the applicable pair states for various SRDF operations.
SRDF Interfamily Connectivity Information	Defines the versions of PowerMaxOS, HYPERMAX OS and Enginuity that can make up valid SRDF replication and SRDF/Metro configurations, and can participate in Non-Disruptive Migration (NDM).
Dell EMC SRDF Introduction	Provides an overview of SRDF, its uses, configurations, and terminology.
Dell EMC Solutions Enabler TimeFinder SnapVX CLI User Guide	Describes how to configure and manage TimeFinder SnapVX environments using SYMCLI commands.
Dell EMC Solutions Enabler TimeFinder Family (Mirror, Clone, Snap, VP Snap) Version 8.2 and higher CLI User Guide	Describes how to configure and manage TimeFinder Mirror, Clone, Snap, VP Snap environments for Enginuity and HYPERMAX OS using SYMCLI commands.
Dell EMC Solutions Enabler SRM CLI User Guide	Provides Storage Resource Management (SRM) information that is related to various data objects and data handling facilities.
Dell EMC SRDF/ Metro vWitness Configuration Guide	Describes how to install, configure, and manage SRDF/Metro using vWitness.
Dell EMC Events and Alerts for PowerMax and VMAX User Guide	Documents the SYMAPI daemon messages, asynchronous errors and message events, SYMCLI return codes, and how to configure event logging.

Embedded NAS (eNAS) documents:

EMC VMAX Embedded NAS Release Notes	Describes the new features and identify any known functionality restrictions and performance issues that may exist in the current version.
EMC VMAX Embedded NAS Quick Start Guide	Describes how to configure eNAS on a VMAX3 or VMAX All Flash storage system.
EMC VMAX Embedded NAS File Auto Recovery with SRDF/S	Describes how to install and use EMC File Auto Recovery with SRDF/S.
Dell EMC PowerMax eNAS CLI Reference Guide	A reference for command-line users and script programmers that provides the syntax, error codes, and parameters of all eNAS commands.

PowerProtect Storage Direct documents:

Dell EMC PowerProtect Storage Direct Solutions Guide	Provides Storage Direct information that is related to various data objects and data handling facilities.
Dell EMC File System Agent Installation and Administration Guide	Shows how to install, configure, and manage the Storage Direct File System Agent.
Dell EMC Database Application Agent Installation and Administration Guide	Shows how to install, configure, and manage the Storage Direct Database Application Agent.
Dell EMC Microsoft Application Agent Installation and Administration Guide	Shows how to install, configure, and manage the Storage Direct Microsoft Application Agent.

NOTE: ProtectPoint has been renamed to Storage Direct and it is included in PowerProtect, Data Protection Suite for Apps, or Data Protection Suite Enterprise Software Edition.

Mainframe Enablers documents:

Dell EMC Mainframe Enablers Installation and Customization Guide	Describes how to install and configure Mainframe Enablers software.
Dell EMC Mainframe Enablers Release Notes	Describes new features and any known limitations.
Dell EMC Mainframe Enablers Message Guide	Describes the status, warning, and error messages generated by Mainframe Enablers software.
Dell EMC Mainframe Enablers ResourcePak Base for z/OS Product Guide	Describes how to configure VMAX system control and management using the EMC Symmetrix Control Facility (EMCSCF).
Dell EMC Mainframe Enablers AutoSwap for z/OS Product Guide	Describes how to use AutoSwap to perform automatic workload swaps between VMAX systems when the software detects a planned or unplanned outage.
Dell EMC Mainframe Enablers Consistency	Describes how to use Consistency Groups for z/OS (ConGroup) to ensure the consistency of data remotely copied by SRDF in the event of a rolling disaster.

Groups for z/OS Product Guide	
Dell EMC Mainframe Enablers SRDF Host Component for z/OS Product Guide	Describes how to use SRDF Host Component to control and monitor remote data replication processes.
Dell EMC Mainframe Enablers TimeFinder SnapVX and zDP Product Guide	Describes how to use TimeFinder SnapVX and zDP to create and manage space-efficient targetless snaps.
Dell EMC Mainframe Enablers TimeFinder/ Clone Mainframe Snap Facility Product Guide	Describes how to use TimeFinder/Clone, TimeFinder/Snap, and TimeFinder/CG to control and monitor local data replication processes.
Dell EMC Mainframe Enablers TimeFinder/ Mirror for z/OS Product Guide	Describes how to use TimeFinder/Mirror to create Business Continuance Volumes (BCVs) which can then be established, split, reestablished and restored from the source logical volumes for backup, restore, decision support, or application testing.
Dell EMC Mainframe Enablers TimeFinder Utility for z/OS Product Guide	Describes how to use the TimeFinder Utility to condition volumes and devices.

Geographically Dispersed Disaster Recovery (GDDR) documents:

Dell EMC GDDR for SRDF/S with ConGroup Product Guide	Describes how to use Geographically Dispersed Disaster Restart (GDDR) to automate business recovery following both planned outages and disaster situations.
Dell EMC GDDR for SRDF/S with AutoSwap Product Guide	Describes how to use Geographically Dispersed Disaster Restart (GDDR) to automate business recovery following both planned outages and disaster situations.
Dell EMC GDDR for SRDF/Star Product Guide	Describes how to use Geographically Dispersed Disaster Restart (GDDR) to automate business recovery following both planned outages and disaster situations.
Dell EMC GDDR for SRDF/Star with AutoSwap Product Guide	Describes how to use Geographically Dispersed Disaster Restart (GDDR) to automate business recovery following both planned outages and disaster situations.
Dell EMC GDDR for SRDF/SQAR with AutoSwap Product Guide	Describes how to use Geographically Dispersed Disaster Restart (GDDR) to automate business recovery following both planned outages and disaster situations.
Dell EMC GDDR for SRDF/A Product Guide	Describes how to use Geographically Dispersed Disaster Restart (GDDR) to automate business recovery following both planned outages and disaster situations.

Dell EMC GDDR Message Guide	Describes the status, warning, and error messages generated by GDDR.
Dell EMC GDDR Release Notes	Describes new features and any known limitations.
Dell EMC GDDR for Star-A Product Guide	Describes the basic concepts of Dell EMC Geographically Dispersed Disaster Restart (GDDR), how to install it, and how to implement its major features and facilities.

z/OS Migrator documents:

Dell EMC z/OS Migrator Product Guide	Describes how to use z/OS Migrator to perform volume mirror and migrator functions as well as logical migration functions.
Dell EMC z/OS Migrator Message Guide	Describes the status, warning, and error messages generated by z/OS Migrator.
Dell EMC z/OS Migrator Release Notes	Describes new features and any known limitations.

z/TPF documents:

Dell EMC ResourcePak for z/TPF Product Guide	Describes how to configure VMAX system control and management in the z/TPF operating environment.
Dell EMC SRDF Controls for z/TPF Product Guide	Describes how to perform remote replication operations in the z/TPF operating environment.
Dell EMC TimeFinder Controls for z/TPF Product Guide	Describes how to perform local replication operations in the z/TPF operating environment.
Dell EMC z/TPF Suite Release Notes	Describes new features and any known limitations.

Typographical conventions

Dell EMC uses the following type style conventions in this document:

Table 1. Typographical conventions used in this content

Bold	Used for names of interface elements Examples: Names of windows, dialog boxes, buttons, fields, tab names, key names, and menu paths (what the user selects or clicks)
Italic	Used for full titles of publications referenced in text
`Monospace`	Used for: • System code • System output, such as an error message or script • Pathnames, filenames, prompts, and syntax • Commands and options
Monospace italic	Used for variables
`Monospace bold`	Used for user input

Table 1. Typographical conventions used in this content (continued)

[ ]	Square brackets enclose optional values.
|	A vertical bar indicates alternate selections. The bar means "or".
{ }	Braces enclose content that the user must specify, such as x or y or z.
...	Ellipses indicate nonessential information that is omitted from the example.

Where to get help

Dell EMC support, product, and licensing information can be obtained as follows:

Product information

Dell EMC technical support, documentation, release notes, software updates, or information about Dell EMC products can be obtained at https://www.dell.com/support/home (registration required) or https://www.dellemc.com/en-us/documentation/vmax-all-flash-family.htm.

Technical support

To open a service request through the Dell EMC Online Support (https://www.dell.com/support/home) site, you must have a valid support agreement. Contact your Dell EMC sales representative for details about obtaining a valid support agreement or to answer any questions about your account.

Additional support options

- Support by Product — Dell EMC offers consolidated, product-specific information on the Web at: https://support.EMC.com/products

  The Support by Product web pages offer quick links to Documentation, White Papers, Advisories (such as frequently used Knowledgebase articles), and Downloads, as well as more dynamic content, such as presentations, discussion, relevant Customer Support Forum entries, and a link to Dell EMC Live Chat.

- Dell EMC Live Chat — Open a Chat or instant message session with a Dell EMC Support Engineer.

e-Licensing support

To activate your entitlements and obtain your license files, go to the Service Center on Dell EMC Online Support (https://www.dell.com/support/home). Follow the directions on your License Authorization Code (LAC) letter that is emailed to you.

- Expected functionality may be unavailable because it is not licensed. For help with missing or incorrect entitlements after activation, contact your Dell EMC Account Representative or Authorized Reseller.
- For help with any errors applying license files through Solutions Enabler, contact the Dell EMC Customer Support Center.
- Contact the Dell EMC worldwide LIcensing team if you are missing a LAC letter or require further instructions on activating your licenses through the Online Support site.
  - licensing@emc.com
  - North America, Latin America, APJK, Australia, New Zealand: SVC4EMC (800-782-4362) and follow the voice prompts.
  - EMEA: +353 (0) 21 4879862 and follow the voice prompts.

Your comments

Your suggestions help improve the accuracy, organization, and overall quality of the documentation. Send your comments and feedback to: VMAXContentFeedback@emc.com

Revision history

The following table lists the revision history of this document.

Table 2. Revision history

Revision	Description and/or change	Operating system
16	Updated with new and changed features related to the latest release of the PowerMax OS	PowerMax OS 5978.711.711

Table 2. Revision history (continued)

Revision	Description and/or change	Operating system
15	Updated with new and changed features related to the latest release of the PowerMax OS	PowerMax OS 5978.669.669
14	Updated with new and changed features related to the latest release of the PowerMax OS	PowerMax OS 5978.444.444
13	Revised content: • Updated links	PowerMax OS 5978.444.444
12	Updated with new and changed features related to the latest release of the PowerMax OS	PowerMax OS 5978.444.444
11	Revised content: • Clarify the recommended maximum distance between arrays using SRDF/S	HYPERMAX OS 5977.1125.1125
10	Revised content: • Update system capacities	HYPERMAX OS 5977.1125.1125
09	Revised content: • Update section on using ProtectPoint for backup and restore operations • Add hardware compression to table of SRDF features	HYPERMAX OS 5977.1125.1125
08	Revised content: • Update descriptions of the All Flash arrays • Add PowerMax and PowerMaxOS to the SRDF chapter	HYPERMAX OS 5977.1125.1125
07	New content: • RecoverPoint • VMAX 950F support • Secure snaps • Data at Rest Encryption	HYPERMAX OS 5977. 1125.1125
06	Revised content: • Power consumption and heat dissipation numbers for the VMAX 250F • SRDF/Metro array witness overview	HYPERMAX OS 5997.952. 892
05	New content: • VMAX 250F support • Inline Compression • Mainframe support • Non disruptive migration • Virtual Witness (vWitness)	HYPERMAX OS 5997.952. 892
04	Removed "RPQ" requirement from Third Party racking.	HYPERMAX 5977.810.784
03	Updated Licensing appendix.	HYPERMAX 5977.810.784
02	Updated values in the power and heat dissipation specification table.	HYPERMAX OS 5977.691.684 + Q1 2016 Service Pack
01	First release of the VMAX All Flash with EMC HYPERMAX OS 5977 for VMAX 450F, 450FX, 850F, and 850FX.	HYPERMAX OS 5977.691.684 + Q1 2016 Service Pack

1

VMAX All Flash with HYPERMAX OS

This chapter introduces VMAX All Flash systems and the HYPERMAX OS operating environment.

Topics:

- Introduction to VMAX All Flash with HYPERMAX OS
- Software packages
- HYPERMAX OS

Introduction to VMAX All Flash with HYPERMAX OS

VMAX All Flash is a range of storage arrays that use only high-density flash drives. The range contains four models that combine high scale, low latency and rich data services:

- VMAX 250F with a maximum capacity of 1.16 PBe (Petabytes effective)
- VMAX 450F with a maximum capacity of 2.3 PBe
- VMAX 850F with a maximum capacity of 4.4 PBe
- VMAX 950F with a maximum capacity of 4.42 PBe

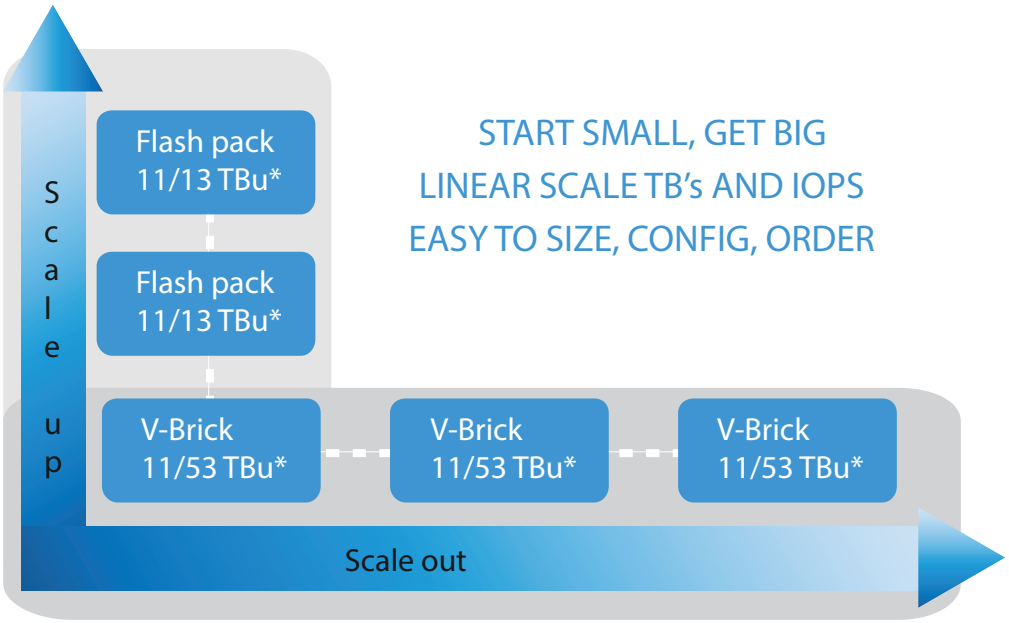
Each VMAX All Flash array is made up of one or more building blocks known as V-Bricks (in an open systems array) or zBricks (in a mainframe array). A V-Brick or zBrick consists of:

- An engine with two directors (the redundant data storage processing unit)
- Flash capacity in Drive Array Enclosures (DAEs):
  - VMAX 250F: Two 25-slot DAEs with a minimum base capacity of 13TBu
  - VMAX 450F, VMAX 850F: Two 120-slot DAEs with a minimum base capacity of 53TBu
  - VMAX 950F (open or mixed systems): Two 120-slot DAEs with a minimum base capacity of 53TBu
  - VMAX 950F (mainframe systems): Two 120-slot DAEs with a minimum base capacity of 13TBu
- Multiple software packages are available: F and FX packages for open system arrays and zF and zFX for mainframe arrays.

Customers can increase the initial configuration by adding 11 TBu (250F) or 13 TBu (450F, 850F, 950F) capacity packs that bundle all required flash capacity and software. In open system arrays, capacity packs are known as Flash capacity packs. In mainframe arrays, they are known as zCapacity packs. In addition, customers can also scale out the initial configuration by adding additional V-Bricks or zBricks to increase performance, connectivity, and throughput.

- VMAX 250F All Flash arrays scale from one to two V-Bricks
- VMAX 450F All Flash arrays scale from one to four V-Bricks/zBricks
- VMAX 850F/950F All Flash arrays scale from one to eight V-Bricks/zBricks

Independent and linear scaling of both capacity and performance enables VMAX All Flash to be extremely flexible at addressing varying workloads. The following illustrates scaling opportunities for VMAX All Flash open system arrays.



* Depending on the VMAX model

Figure 1. VMAX All Flash scale up and out

The All Flash arrays:

- Use the powerful Dynamic Virtual Matrix Architecture.
- Deliver high levels of performance and scale. For example, VMAX 950F arrays deliver 6.74M IOPS (RRH) with less than 0.5 ms latency at 150 GB/sec bandwidth. VMAX 250F, 450F, 850F, 950F arrays deliver consistently low response times (< 0.5ms).
- Provide mainframe (VMAX 450F, 850F, 950F) and open systems (including IBM i) host connectivity for mission critical storage needs
- Use the HYPERMAX OS hypervisor to provide file system storage with eNAS and embedded management services for Unisphere. Embedded Network Attached Storage (eNAS) on page 23 and Embedded Management on page 22, have more information on these features.
- Provide data services such as:
  - SRDF remote replication technology with the latest SRDF/Metro functionality
  - SnapVX local replication services based on SnapVX infrastructure
  - Data protection and encryption

  About TimeFinder on page 53 has more information on these features.
- Use the latest Flash drive technology in V-Bricks/zBricks and capacity packs to deliver a top-tier, diamond service level.

VMAX All Flash hardware specifications

Detailed specifications of the VMAX All Flash hardware, including capacity, cache memory, I/O protocols, and I/O connections are available at https://www.emc.com/collateral/specification-sheet/h16051-vmax-all-flash-250f-950f-ss.pdf

Software packages

VMAX All Flash arrays are available with multiple software packages (F/FX for open system arrays, and zF/zFX for mainframe arrays) containing standard and optional features.

Table 3. Symbol legend for VMAX All Flash software features/software package

✓	Standard feature with that model/software package.	✓+	Optional feature with that model/software package.

Table 4. VMAX All Flash software features by model

Software/Feature	VMAX model and software packages										
	250F		450F				850F, 950F				
	F	FX	F	FX	zF	zFX	F	FX	zF	zFX	See:
HYPERMAX OS	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	HYPERMAX OS on page 20
Embedded Management[a]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	Management Interfaces on page 31
Mainframe Essentials Plus					✓	✓			✓	✓	Mainframe Features on page 43
SnapVX	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	About TimeFinder on page 53
AppSync Starter Pack	✓	✓	✓	✓			✓	✓			AppSync on page 36
Compression	✓	✓	✓	✓			✓	✓			Inline compression on page 30
Non-Disruptive Migration	✓	✓	✓	✓			✓	✓			Non-Disruptive Migration overview on page 81
SRDF	✓+	✓	✓+	✓	✓+	✓	✓+	✓	✓+	✓	Remote replication on page 59
SRDF/Metro	✓+	✓	✓+	✓			✓+	✓			SRDF/Metro on page 71
Embedded Network Attached Storage (eNAS)	✓+	✓	✓+	✓			✓+	✓			Embedded Network Attached Storage (eNAS) on page 23
Unisphere 360	✓+	✓	✓+	✓	✓+	✓	✓+	✓	✓+	✓	Unisphere 360 on page 32
SRM	✓+	✓	✓+	✓			✓+	✓			Storage Resource Management (SRM) on page 34
Data at Rest Encryption (D@RE)	✓+	✓	✓+	✓	✓+	✓	✓+	✓	✓+	✓	Data at Rest Encryption on page 25
PowerPath	✓+	✓[b]	✓+	✓[b]			✓+	✓[b]			PowerPath Migration Enabler on page 85
AppSync Full Suite	✓+	✓+	✓+	✓+			✓+	✓+			AppSync on page 36
ProtectPoint	✓+	✓+	✓+	✓+			✓+	✓+			Storage Direct agents on page 40
AutoSwap and zDP					✓+	✓			✓+	✓	Mainframe SnapVX and zDP on page 56
GDDR					✓+	✓+			✓+	✓+	Geographically Dispersed Disaster Restart (GDDR) on page 33

a. eManagement includes: embedded Unisphere, Solutions Enabler, and SMI-S.
b. The FX package includes 75 PowerPath licenses.. Additional licensesare available separately..

HYPERMAX OS

This section highlights the features of the HYPERMAX OS.

HYPERMAX OS emulations

HYPERMAX OS provides emulations (executables) that perform specific data service and control functions in the HYPERMAX environment. The following table lists the available emulations.

Table 5. HYPERMAX OS emulations

Area	Emulation	Description	Protocol Speed[a]
Back-end	DS	Back-end connection in the array that communicates with the drives, DS is also known as an internal drive controller.	SAS 12 Gb/s (VMAX 250F) SAS 6 Gb/s (VMAX 450F, 850F, and 950F)
	DX	Back-end connections that are not used to connect to hosts. Used by ProtectPoint.	FC 16 or 8 Gb/s
		ProtectPoint links Data Domain to the array. DX ports must be configured for the FC protocol.	
Management	IM	Separates infrastructure tasks and emulations. By separating these tasks, emulations can focus on I/O-specific work only, while IM manages and executes common infrastructure tasks, such as environmental monitoring, Field Replacement Unit (FRU) monitoring, and vaulting.	N/A
	ED	Middle layer used to separate front-end and back-end I/O processing. It acts as a translation layer between the front-end, which is what the host knows about, and the back-end, which is the layer that reads, writes, and communicates with physical storage in the array.	N/A
Host connectivity	FA - Fibre Channel SE - iSCSI EF - FICON [b]	Front-end emulation that: • Receives data from the host (network) and commits it to the array • Sends data from the array to the host/network	FC - 16 or 8 Gb/s SE - 10 Gb/s EF - 16 Gb/s
Remote replication	RF - Fibre Channel RE - GbE	Interconnects arrays for SRDF.	RF - 8 Gb/s SRDF RE - 1 GbE SRDF RE - 10 GbE SRDF

a. The 8 Gb/s module auto-negotiates to 2/4/8 Gb/s and the 16 Gb/s module auto-negotiates to 16/8/4 Gb/s using optical SFP and OM2/OM3/OM4 cabling.
b. Only on VMAX 450F, 850F, and 950F arrays.

Container applications

HYPERMAX OS provides an open application platform for running data services. It includes a lightweight hypervisor that enables multiple operating environments to run as virtual machines on the storage array.

Application containers are virtual machines that provide embedded applications on the storage array. Each container virtualizes the hardware resources that are required by the embedded application, including:

- Hardware needed to run the software and embedded application (processor, memory, PCI devices, power management)
- VM ports, to which LUNs are provisioned
- Access to necessary drives (boot, root, swap, persist, shared)

Embedded Management

The eManagement container application embeds management software (Solutions Enabler, SMI-S, Unisphere for VMAX) on the storage array, enabling you to manage the array without requiring a dedicated management host.

With eManagement, you can manage a single storage array and any SRDF attached arrays. To manage multiple storage arrays with a single control pane, use the traditional host-based management interfaces: Unisphere and Solutions Enabler. To this end, eManagement allows you to link-and-launch a host-based instance of Unisphere.

eManagement is typically preconfigured and enabled at the factory. However, starting with HYPERMAX OS 5977.945.890, eManagement can be added to arrays in the field. Contact your support representative for more information.

Embedded applications require system memory. The following table lists the amount of memory unavailable to other data services.

Table 6. eManagement resource requirements

VMAX All Flash model	CPUs	Memory	Devices supported
VMAX 250F	4	16 GB	200K
VMAX 450F	4	16 GB	200K
VMAX 850F, 950F	4	20 GB	400K

Virtual machine ports

Virtual machine (VM) ports are associated with virtual machines to avoid contention with physical connectivity. VM ports are addressed as ports 32-63 on each director FA emulation.

LUNs are provisioned on VM ports using the same methods as provisioning physical ports.

A VM port can be mapped to one VM only. However, a VM can be mapped to multiple ports.

Embedded Network Attached Storage (eNAS)

eNAS is fully integrated into the VMAX All Flash array. eNAS provides flexible and secure multi-protocol file sharing (NFS 2.0, 3.0, 4.0/4.1, and CIFS/SMB 3.0) and multiple file server identities (CIFS and NFS servers). eNAS enables:

- File server consolidation/multi-tenancy
- Built-in asynchronous file level remote replication (File Replicator)
- Built-in Network Data Management Protocol (NDMP)
- VDM Synchronous replication with SRDF/S and optional automatic failover manager File Auto Recovery (FAR)
- Anti-virus

eNAS provides file data services for:

- Consolidating block and file storage in one infrastructure
- Eliminating the gateway hardware, reducing complexity and costs
- Simplifying management

Consolidated block and file storage reduces costs and complexity while increasing business agility. Customers can leverage data services across block and file storage including storage provisioning, dynamic Host I/O Limits, and Data at Rest Encryption.

eNAS solutions and implementation

eNAS runs on standard array hardware and is typically pre-configured at the factory. There is a one-off setup of the Control Station and Data Movers, containers, control devices, and required masking views as part of the factory pre-configuration. Additional front-end I/O modules are required to implement eNAS. However, starting with HYPERMAX OS 5977.945.890, eNAS can be added to arrays in the field. Contact your support representative for more information.

eNAS uses the HYPERMAX OS hypervisor to create virtual instances of NAS Data Movers and Control Stations on VMAX All Flash controllers. Control Stations and Data Movers are distributed within the VMAX All Flash array based upon the number of engines and their associated mirrored pair.

By default, VMAX All Flash arrays have:

- Two Control Station virtual machines
- Two or more Data Mover virtual machines. The number of Data Movers differs for each model of the array. All configurations include one standby Data Mover.

eNAS configurations

The storage capacity required for arrays supporting eNAS is at least 680 GB. This table lists eNAS configurations and front-end I/O modules.

Table 7. eNAS configurations by array

Component	Description	VMAX 250F	VMAX 450F	VMAX 850F, 950F
Data Movers[a] virtual machine	Maximum number	4	4	8[b]
	Max capacity/DM	512 TB	512 TB	512 TB
	Logical cores[c]	12/24	12/24	16/32/48/64[b]
	Memory (GB)[c]	48/96	48/96	48/96/144/192[b]
	I/O modules (Max)[c]	12	12[d]	24[d]
Control Station virtual machines (2)	Logical cores	2	2	2
	Memory (GB)	8	8	8
NAS Capacity/Array	Maximum	1.15 PB	1.5 PB	3.5 PB

a. Data Movers are added in pairs and must have the same configuration.
b. The 850F and 950F can be configured through Sizer with a maximum of four data movers. However, six and eight data movers can be ordered by RPQ. As the number of data movers increases, the maximum number of I/O cards , logical cores, memory, and maximum capacity also increases.
c. For 2, 4, 6, and 8 data movers, respectively.
d. A single 2-port 10GbE Optical I/O module is required by each Data Mover for initial All-Flash configurations. However, that I/O module can be replaced with a different I/O module (such as a 4-port 1GbE or 2-port 10GbE copper) using the normal replacement capability that exists with any eNAS Data Mover I/O module. In addition, additional I/O modules can be configured through a I/O module upgrade/add as long as standard rules are followed (no more than 3 I/O modules per Data Mover, all I/O modules must occupy the same slot on each director on which a Data Mover resides).

Replication using eNAS

The replication methods available for eNAS file systems are:

- Asynchronous file system level replication using VNX Replicator for File.
- Synchronous replication with SRDF/S using File Auto Recovery (FAR) with the optional File Auto Recover Manager (FARM).
- Checkpoint (point-in-time, logical images of a production file system) creation and management using VNX SnapSure.

NOTE: SRDF/A, SRDF/Metro, and TimeFinder are not available with eNAS.

Data protection and integrity

HYPERMAX OS provides facilities to ensure data integrity and to protect data in the event of a system failure or power outage:

- RAID levels
- Data at Rest Encryption
- Data erasure
- Block CRC error checks
- Data integrity checks
- Drive monitoring and correction
- Physical memory error correction and error verification
- Drive sparing and direct member sparing
- Vault to flash

RAID levels

VMAX All Flash arrays provide the following RAID levels:

- VMAX 250F: RAID5 (7+1) (Default), RAID5 (3+1), and RAID6 (6+2)
- VMAX 450F, 850F, 950F: RAID5 (7+1), and RAID6 (14+2)

Data at Rest Encryption

Securing sensitive data is an important IT issue, that has regulatory and legislative implications. Several of the most important data security threats relate to protection of the storage environment. Drive loss and theft are primary risk factors. Data at Rest Encryption (D@RE) protects data by adding back-end encryption to an entire array.

D@RE provides hardware-based encryption for VMAX All Flash arrays using I/O modules that incorporate AES-XTS inline data encryption. These modules encrypt and decrypt data as it is being written to or read from a drive. This protects your information from unauthorized access even when drives are removed from the array.

D@RE can use either an internal embedded key manager, or one of these external, enterprise-grade key managers:

- Gemalto SafeNet KeySecure
- IBM Security Key Lifecycle Manager

D@RE accesses an external key manager using the Key Management Interoperability Protocol (KMIP). The EMC E-Lab Interoperability Matrix (https://www.emc.com/products/interoperability/elab.htm) lists the external key managers for each version of HYPERMAX OS.

When D@RE is active, all configured drives are encrypted, including data drives, spares, and drives with no provisioned volumes. Vault data is encrypted on Flash I/O modules.

D@RE provides:

- Secure replacement for failed drives that cannot be erased—For some types of drive failures, data erasure is not possible. Without D@RE, if the failed drive is repaired, data on the drive may be at risk. With D@RE, deletion of the applicable keys makes the data on the failed drive unreadable.
- Protection against stolen drives—When a drive is removed from the array, the key stays behind, making data on the drive unreadable.
- Faster drive sparing—The drive replacement script destroys the keys associated with the removed drive, quickly making all data on that drive unreadable.
- Secure array retirement—Delete all copies of keys on the array, and all remaining data is unreadable.

In addition, D@RE:

- Is compatible with all array features and all supported drive types or volume emulations
- Provides encryption without degrading performance or disrupting existing applications and infrastructure

Enabling D@RE

D@RE is a licensed feature that is installed and configured at the factory. Upgrading an existing array to use D@RE is possible, but is disruptive. The upgrade requires re-installing the array, and may involve a full data back up and restore. Before upgrading, plan how to manage any data already on the array. Dell EMC Professional Services offers services to help you implement D@RE.

D@RE components

Embedded D@RE uses the following components, all of which reside on the primary Management Module Control Station (MMCS):

- EMC Key Trust Platform (KTP) (embedded)—This component adds embedded key management functionality to the KMIP Client.
- Lockbox—Hardware- and software-specific encrypted repository that securely stores passwords and other sensitive key manager configuration information. The lockbox binds to a specific MMCS.

External D@RE uses the same components as embedded D@RE, and adds the following:

- EMC Key Trust Platform (KTP)—Also known as the KMIP Client, this component resides on the MMCS and communicates with external key managers using the OASIS Key Management Interoperability Protocol (KMIP) to manage encryption keys.
- External Key Manager—Provides centralized encryption key management capabilities such as secure key generation, storage, distribution, audit, and enabling Federal Information Processing Standard (FIPS) 140-2.
- Cluster/Replication Group—Multiple external key managers sharing configuration settings and encryption keys. Configuration and key lifecycle changes made to one node are replicated to all members within the same cluster or replication group.

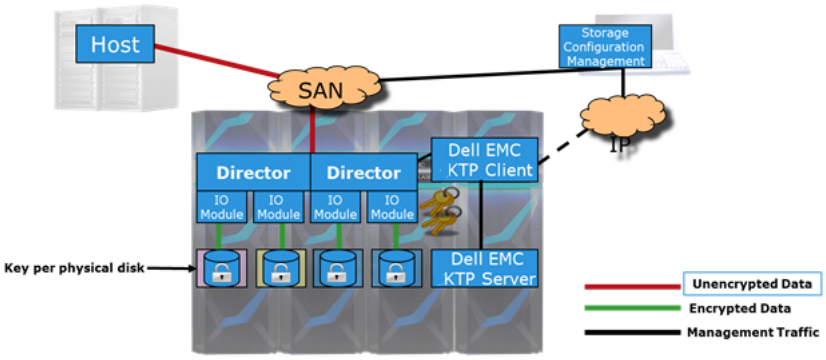


Figure 2. D@RE architecture

External Key Managers

D@RE's external key management is provided by Gemalto SafeNet KeySecure and IBM Security Key Lifecycle Manager. Keys are generated and distributed using industry standards (NIST 800-57 and ISO 11770). With D@RE, there is no need to replicate keys across volume snapshots or remote sites. D@RE external key managers can be used with FIPS 140-2 level 1 validated software.

Encryption keys must be highly available when they are needed, and tightly secured. Keys, and the information required to use keys (during decryption), must be preserved for the lifetime of the data. This is critical for encrypted data that is kept for many years.

Key accessibility is vital in high-availability environments. D@RE caches the keys locally. So connection to the Key Manager is necessary only for operations such as the initial installation of the array, replacement of a drive, or drive upgrades.

Lifecycle events involving keys (generation and destruction) are recorded in the array's Audit Log.

Key protection

The local keystore file is encrypted with a 256-bit AES key derived from a randomly generated password file. This password file is secured in the Lockbox. The Lockbox is protected using MMCS-specific stable system values (SSVs) of the primary MMCS. These are the same SSVs that protect Secure Service Credentials (SSC).

Compromising the MMCS drive or copying Lockbox and keystore files off the array causes the SSV tests to fail. Compromising the entire MMCS only gives an attacker access if they also successfully compromise SSC.

There are no backdoor keys or passwords to bypass D@RE security.

Key operations

D@RE provides a separate, unique Data Encryption Key (DEK) for each physical drive in the array, including spare drives. To ensure that D@RE uses the correct key for a given drive:

- DEKs stored in the array include a unique key tag and key metadata when they are wrapped (encrypted) for use by the array. This information is included with the key material when the DEK is wrapped (encrypted) for use in the array.

- During encryption I/O, the expected key tag associated with the drive is supplied separately from the wrapped key.
- During key unwrap, the encryption hardware checks that the key unwrapped correctly and that it matches the supplied key tag.
- Information in a reserved system LBA (Physical Information Block, or PHIB) verifies the key used to encrypt the drive and ensures the drive is in the correct location.
- During initialization, the hardware performs self-tests to ensure that the encryption/decryption logic is intact. The self-test prevents silent data corruption due to encryption hardware failures.

Audit logs

The audit log records major activities on an array, including:

- Host-initiated actions
- Physical component changes
- Actions on the MMCS
- D@RE key management events
- Attempts blocked by security controls (Access Controls)

The Audit Log is secure and tamper-proof so event contents cannot be altered. Users with the Auditor access can view, but not modify, the log.

Data erasure

Dell EMC Data Erasure uses specialized software to erase information on arrays. It mitigates the risk of information dissemination, and helps secure information at the end of the information lifecycle. Data erasure:

- Protects data from unauthorized access
- Ensures secure data migration by making data on the source array unreadable
- Supports compliance with internal policies and regulatory requirements

Data Erasure overwrites data at the lowest application-addressable level to drives. The number of overwrites is configurable from three (the default) to seven with a combination of random patterns on the selected arrays.

An optional certification service is available to provide a certificate of erasure. Drives that fail erasure are delivered to customers for final disposal.

For individual flash drives, Secure Erase operations erase all physical flash areas on the drive which may contain user data.

The available data erasure services are:

- Dell EMC Data Erasure for Full Arrays—Overwrites data on all drives in the system when replacing, retiring or re-purposing an array.
- Dell EMC Data Erasure/Single Drives—Overwrites data on individual drives.
- Dell EMC Disk Retention—Enables organizations that must retain all media to retain failed drives.
- Dell EMC Assessment Service for Storage Security—Assesses your information protection policies and suggests a comprehensive security strategy.

All erasure services are performed on-site in the security of the customer's data center and include a Data Erasure Certificate and report of erasure results.

Block CRC error checks

HYPERMAX OS provides:

- Industry-standard, T10 Data Integrity Field (DIF) block cyclic redundancy check (CRC) for track formats.

  For open systems, this enables host-generated DIF CRCs to be stored with user data by the arrays and used for end-to-end data integrity validation.

- Additional protections for address/control fault modes for increased levels of protection against faults. These protections are defined in user-definable blocks supported by the T10 standard.
- Address and write status information in the extra bytes in the application tag and reference tag portion of the block CRC.

Data integrity checks

HYPERMAX OS validates the integrity of data at every possible point during the lifetime of that data. From the time data enters an array, it is continuously protected by error detection metadata. This metadata is checked by hardware and software mechanisms any time data is moved within the array. This allows the array to provide true end-to-end integrity checking and protection against hardware or software faults.

The protection metadata is appended to the data stream, and contains information describing the expected data location as well as the CRC representation of the actual data contents. The expected values to be found in protection metadata are stored persistently in an area separate from the data stream. The protection metadata is used to validate the logical correctness of data being moved within the array any time the data transitions between protocol chips, internal buffers, internal data fabric endpoints, system cache, and system drives.

Drive monitoring and correction

HYPERMAX OS monitors medium defects by both examining the result of each disk data transfer and proactively scanning the entire disk during idle time. If a block on the disk is determined to be bad, the director:

1. Rebuilds the data in the physical storage, if necessary.
2. Rewrites the data in physical storage, if necessary.

The director keeps track of each bad block detected on a drive. If the number of bad blocks exceeds a predefined threshold, the array proactively invokes a sparing operation to replace the defective drive, and then alerts Customer Support to arrange for corrective action, if necessary. With the deferred service model, immediate action is not always required.

Physical memory error correction and error verification

HYPERMAX OS corrects single-bit errors and reports an error code once the single-bit errors reach a predefined threshold. In the unlikely event that physical memory replacement is required, the array notifies Customer Support, and a replacement is ordered.

Drive sparing and direct member sparing

When HYPERMAX OS 5977 detects a drive is about to fail or has failed, it starts a direct member sparing (DMS) process. Direct member sparing looks for available spares within the same engine that are of the same block size, capacity and speed, with the best available spare always used.

With direct member sparing, the invoked spare is added as another member of the RAID group. During a drive rebuild, the option to directly copy the data from the failing drive to the invoked spare drive is available. The failing drive is removed only when the copy process is complete. Direct member sparing is automatically initiated upon detection of drive-error conditions.

Direct member sparing provides the following benefits:

- The array can copy the data from the failing RAID member (if available), removing the need to read the data from all of the members and doing the rebuild. Copying to the new RAID member is less CPU intensive.
- If a failure occurs in another member, the array can still recover the data automatically from the failing member (if available).
- More than one spare for a RAID group is supported at the same time.

Vault to flash

VMAX All Flash arrays initiate a vault operation when the system is powered down, goes offline, or if environmental conditions occur, such as the loss of a data center due to an air conditioning failure.

Each array comes with Standby Power Supply (SPS) modules. On a power loss, the array uses the SPS power to write the system mirrored cache to flash storage. Vaulted images are fully redundant; the contents of the system mirrored cache are saved twice to independent flash storage.

The vault operation

When a vault operation starts:

- During the save part of the vault operation, the VMAX All Flash array stops all I/O. When the system mirrored cache reaches a consistent state, directors write the contents to the vault devices, saving two copies of the data. The array then completes the power down, or, if power down is not required, remains in the offline state.
- During the restore part of the operation, the array's startup program initializes the hardware and the environmental system, and restores the system mirrored cache contents from the saved data (while checking data integrity).

The system resumes normal operation when the SPS modules have sufficient charge to complete another vault operation, if required. If any condition is not safe, the system does not resume operation and notifies Customer Support for diagnosis and repair. This allows Customer Support to communicate with the array and restore normal system operations.

Vault configuration considerations

- To support vault to flash, the VMAX All Flash arrays require the following number of flash I/O modules:
  - VMAX 250F two to six per engine/V-Brick
  - VMAX 450F four to eight per engine/V-Brick/zBrick
  - VMAX 850F, 950F four to eight per engine/V-Brick/zBrick
- The size of the flash module is determined by the amount of system cache and metadata required for the configuration.
- The vault space is for internal use only and cannot be used for any other purpose when the system is online.
- The total capacity of all vault flash partitions is sufficient to keep two logical copies of the persistent portion of the system mirrored cache.

Inline compression

HYPERMAX OS 5977.945.890 introduced support for inline compression on VMAX All Flash arrays. Inline compression compresses data as it is written to flash drives.

Inline compression is a feature of storage groups. When enabled (this is the default setting), new I/O to a storage group is compressed when written to disk, while existing data on the storage group starts to compress in the background. After turning off compression, new I/O is no longer compressed, and existing data remains compressed until it is written again, at which time it decompresses.

Inline compression, deduplication, and over-subscription complement each other. Over-subscription allows presenting larger than needed devices to hosts without having the physical drives to fully allocate the space represented by the thin devices (Thin device oversubscription on page 49 has more information on over-subscription). Inline compression further reduces the data footprint by increasing the effective capacity of the array.

The example in Inline compression and over-subscription on page 30 shows this. Here, 1.3 PB of host attached devices (TDEVs) is over-provisioned to 1.0 PB of back-end (TDATs), that reside on 1.0 PB of Flash drives. Following data compression, the data blocks are compressed, by a ratio of 2:1, reducing the number of Flash drives by half. Basically, with compression enabled, the array requires half as many drives to support a given front-end capacity.

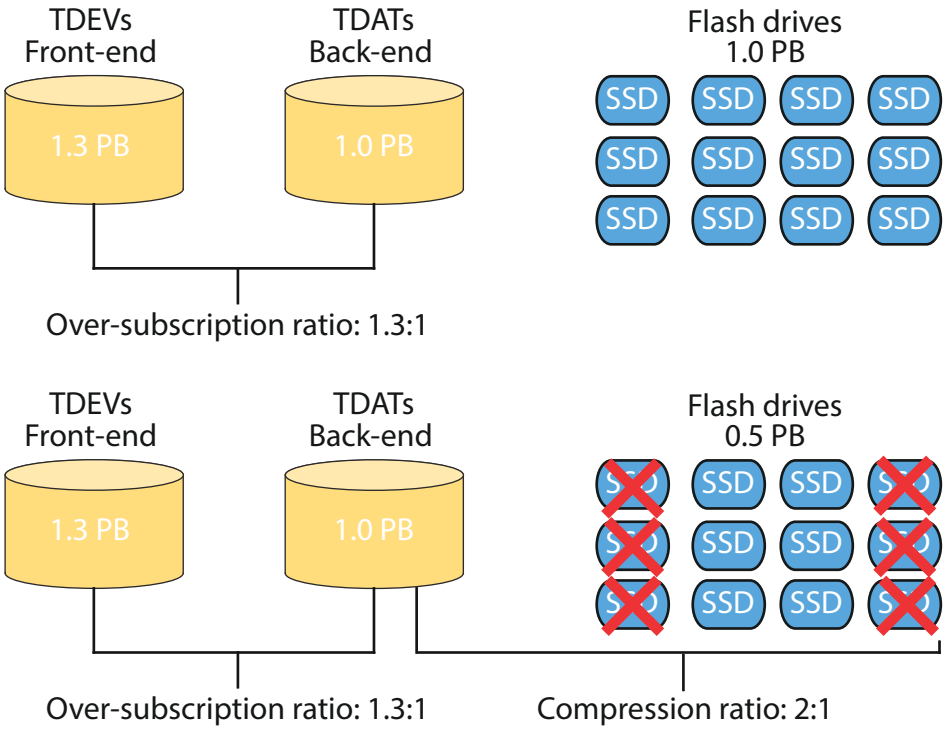


Figure 3. Inline compression and over-subscription

Compression is pre-configured on new VMAX All Flash arrays at the factory. Existing VMAX All Flash arrays in the field can have compression added to them. Contact your Support Representative for more information.

Further characteristics of compression are:

- All supported data services, such as SnapVX, SRDF, and encryption are supported with compression.
- Compression is available on open systems (FBA) only (including eNAS). It is not available for CKD arrays, including those with a mix of FBA and CKD devices. Any open systems array with compression enabled, cannot have CKD devices added to it.
- ProtectPoint operations are still supported to Data Domain arrays
- Compression is switched on and off through Solutions Enabler and Unisphere.
- Compression efficiency can be monitored for SRPs, storage groups, and volumes.
- Activity Based Compression: the most active tracks are held in cache and not compressed until they move from cache to disk. This feature helps improve the overall performance of the array while reducing wear on the flash drives.

2

Management Interfaces

This chapter introduces the tools for managing arrays.

Topics:

- Management interface versions
- Unisphere for VMAX
- Unisphere 360
- Solutions Enabler
- Mainframe Enablers
- Geographically Dispersed Disaster Restart (GDDR)
- SMI-S Provider
- VASA Provider
- eNAS management interface
- Storage Resource Management (SRM)
- vStorage APIs for Array Integration
- SRDF Adapter for VMware vCenter Site Recovery Manager
- SRDF/Cluster Enabler
- Product Suite for z/TPF
- SRDF/TimeFinder Manager for IBM i
- AppSync

Management interface versions

The following components provide management capabilities for HYPERMAX OS 5977.1125.1125:

- Unisphere for VMAX V8.4
- Solutions Enabler V8.4
- Mainframe Enablers V8.2
- GDDR V5.0
- SMI-S V8.4
- SRDF/CE V4.2.1
- SRA V6.3
- VASA Provider V8.4

Unisphere for VMAX

Unisphere for VMAX is a web-based application that provides provisioning, management, and monitoring of arrays.

With Unisphere you can perform the following tasks:

Table 8. Unisphere tasks

Section	Allows you to:
Home	View and manage functions such as array usage, alert settings, authentication options, system preferences, user authorizations, and link and launch client registrations.
Storage	View and manage storage groups and storage tiers.
Hosts	View and manage initiators, masking views, initiator groups, array host aliases, and port groups.

Table 8. Unisphere tasks (continued)

Section	Allows you to:
Data Protection	View and manage local replication, monitor and manage replication pools, create and view device groups, and monitor and manage migration sessions.
Performance	Monitor and manage array dashboards, perform trend analysis for future capacity planning, and analyze data. Set preferences, such as, general, dashboards, charts, reports, data imports, and alerts for performance management tasks.
Databases	Troubleshoot database and storage issues, and launch Database Storage Analyzer.
System	View and display dashboards, active jobs, alerts, array attributes, and licenses.
Support	View online help for Unisphere tasks.

Unisphere also has a Representational State Transfer (REST) API. With this API you can access performance and configuration information, and provision storage arrays. You can use the API in any programming environment that supports standard REST clients, such as web browsers and programming platforms that can issue HTTP requests.

Workload Planner

Workload Planner displays performance metrics for applications. Use Workload Planner to:

- Model the impact of migrating a workload from one storage system to another.
- Model proposed new workloads.
- Assess the impact of moving one or more workloads off of a given array running HYPERMAX OS.
- Determine current and future resource shortfalls that require action to maintain the requested workloads.

FAST Array Advisor

The FAST Array Advisor wizard guides you through the steps to determine the impact on performance of migrating a workload from one array to another.

If the wizard determines that the target array can absorb the added workload, it automatically creates all the auto-provisioning groups required to duplicate the source workload on the target array.

Unisphere 360

Unisphere 360 is an on-premises management solution that provides a single window across arrays running HYPERMAX OS at a single site. Use Unisphere 360 to:

- Add a Unisphere server to Unisphere 360 to allow for data collection and reporting of Unisphere management storage system data.
- View the system health, capacity, alerts and capacity trends for your Data Center.
- View all storage systems from all enrolled Unisphere instances in one place.
- View details on performance and capacity.
- Link and launch to Unisphere instances running V8.2 or higher.
- Manage Unisphere 360 users and configure authentication and authorization rules.
- View details of visible storage arrays, including current and target storage.

Solutions Enabler

Solutions Enabler provides a comprehensive command line interface (SYMCLI) to manage your storage environment.

SYMCLI commands are invoked from a management host, either interactively on the command line, or using scripts.

SYMCLI is built on functions that use system calls to generate low-level I/O SCSI commands. Configuration and status information is maintained in a host database file, reducing the number of enquiries from the host to the arrays.

Use SYMCLI to:

- Configure array software (For example, TimeFinder, SRDF, Open Replicator)
- Monitor device configuration and status
- Perform control operations on devices and data objects

Solutions Enabler also has a Representational State Transfer (REST) API. Use this API to access performance and configuration information, and provision storage arrays. It can be used in any programming environments that supports standard REST clients, such as web browsers and programming platforms that can issue HTTP requests.

Mainframe Enablers

The Dell EMC Mainframe Enablers are software components that allow you to monitor and manage arrays running HYPERMAX OS in a mainframe environment:

- ResourcePak Base for z/OS

  Enables communication between mainframe-based applications (provided by Dell EMC or independent software vendors) and PowerMax/VMAX arrays.

- SRDF Host Component for z/OS

  Monitors and controls SRDF processes through commands executed from a host. SRDF maintains a real-time copy of data at the logical volume level in multiple arrays located in physically separate sites.

- Dell EMC Consistency Groups for z/OS

  Ensures the consistency of data remotely copied by SRDF feature in the event of a rolling disaster.

- AutoSwap for z/OS

  Handles automatic workload swaps between arrays when an unplanned outage or problem is detected.

- TimeFinder SnapVX

  With Mainframe Enablers V8.0 and higher, SnapVX creates point-in-time copies directly in the Storage Resource Pool (SRP) of the source device, eliminating the concepts of target devices and source/target pairing. SnapVX point-in-time copies are accessible to the host through a link mechanism that presents the copy on another device. TimeFinder SnapVX and HYPERMAX OS support backward compatibility to traditional TimeFinder products, including TimeFinder/Clone, TimeFinder VP Snap, and TimeFinder/Mirror.

- Data Protector for z Systems (zDP™)

  With Mainframe Enablers V8.0 and higher, zDP is deployed on top of SnapVX. zDP provides a granular level of application recovery from unintended changes to data. zDP achieves this by providing automated, consistent point-in-time copies of data from which an application-level recovery can be conducted.

- TimeFinder/Clone Mainframe Snap Facility

  Produces point-in-time copies of full volumes or of individual datasets. TimeFinder/Clone operations involve full volumes or datasets where the amount of data at the source is the same as the amount of data at the target. TimeFinder VP Snap leverages clone technology to create space-efficient snaps for thin devices.

- TimeFinder/Mirror for z/OS

  Allows the creation of Business Continuance Volumes (BCVs) and provides the ability to ESTABLISH, SPLIT, RE-ESTABLISH and RESTORE from the source logical volumes.

- TimeFinder Utility

  Conditions SPLIT BCVs by relabeling volumes and (optionally) renaming and recataloging datasets. This allows BCVs to be mounted and used.

Geographically Dispersed Disaster Restart (GDDR)

GDDR automates business recovery following both planned outages and disaster situations, including the total loss of a data center. Using the VMAX All Flash architecture and the foundation of SRDF and TimeFinder replication families, GDDR eliminates any single point of failure for disaster restart plans in mainframe environments. GDDR intelligence automatically adjusts disaster restart plans based on triggered events.

GDDR does not provide replication and recovery services itself. Rather GDDR monitors and automates the services that other Dell EMC products and third-party products provide that are required for continuous operations or business restart. GDDR facilitates business continuity by generating scripts that can be run on demand. For example, scripts to restart business applications following a major data center incident, or resume replication following unplanned link outages.

Scripts are customized when invoked by an expert system that tailors the steps based on the configuration and the event that GDDR is managing. Through automatic event detection and end-to-end automation of managed technologies, GDDR removes human error from the recovery process and allows it to complete in the shortest time possible.

The GDDR expert system is also invoked to automatically generate planned procedures, such as moving compute operations from one data center to another. This is the gold standard for high availability compute operations, to be able to move from scheduled DR test weekend activities to regularly scheduled data center swaps without disrupting application workloads.

SMI-S Provider

Dell EMC SMI-S Provider supports the SNIA Storage Management Initiative (SMI), an ANSI standard for storage management. This initiative has developed a standard management interface that resulted in a comprehensive specification (SMI-Specification or SMI-S).

SMI-S defines the open storage management interface, to enable the interoperability of storage management technologies from multiple vendors. These technologies are used to monitor and control storage resources in multivendor or SAN topologies.

Solutions Enabler components required for SMI-S Provider operations are included as part of the SMI-S Provider installation.

VASA Provider

The VASA Provider enables VMAX All Flash management software to inform vCenter of how VMDK storage, including vVols, is configured and protected. These capabilities are defined by Dell EMC and include characteristics such as disk type, type of provisioning, storage tiering and remote replication status. This allows vSphere administrators to make quick and informed decisions about virtual machine placement. VASA offers the ability for vSphere administrators to complement their use of plugins and other tools to track how devices hosting vVols are configured to meet performance and availability needs. Details about VASA Provider replication groups can be viewed on the Unisphere vVols dashboard.

eNAS management interface

You manage eNAS block and file storage using the Unisphere File Dashboard. Link and launch enables you to run the block and file management GUI within the same session.

The configuration wizard helps you create storage groups (automatically provisioned to the Data Movers) quickly and easily. Creating a storage group creates a storage pool in Unisphere that can be used for file level provisioning tasks.

Storage Resource Management (SRM)

SRM provides comprehensive monitoring, reporting, and analysis for heterogeneous block, file, and virtualized storage environments.

Use SRM to:

- Visualize applications to storage dependencies
- Monitor and analyze configurations and capacity growth
- Optimize your environment to improve return on investment

Virtualization enables businesses to simplify management, control costs, and guarantee uptime. However, virtualized environments also add layers of complexity to the IT infrastructure that reduce visibility and can complicate the management of storage resources. SRM addresses these layers by providing visibility into the physical and virtual relationships to ensure consistent service levels.

As you build out a cloud infrastructure, SRM helps you ensure storage service levels while optimizing IT resources — both key attributes of a successful cloud deployment.

SRM is designed for use in heterogeneous environments containing multi-vendor networks, hosts, and storage devices. The information it collects and the functionality it manages can reside on technologically disparate devices in geographically diverse

locations. SRM moves a step beyond storage management and provides a platform for cross-domain correlation of device information and resource topology, and enables a broader view of your storage environment and enterprise data center.

SRM provides a dashboard view of the storage capacity at an enterprise level through Watch4net. The Watch4net dashboard view displays information to support decisions regarding storage capacity.

The Watch4net dashboard consolidates data from multiple ProSphere instances spread across multiple locations. It gives a quick overview of the overall capacity status in the environment, raw capacity usage, usable capacity, used capacity by purpose, usable capacity by pools, and service levels.

vStorage APIs for Array Integration

VMware vStorage APIs for Array Integration (VAAI) optimize server performance by offloading virtual machine operations to arrays running HYPERMAX OS.

The storage array performs the select storage tasks, freeing host resources for application processing and other tasks.

In VMware environments, storage arrays supports the following VAAI components:

- Full Copy — (Hardware Accelerated Copy) Faster virtual machine deployments, clones, snapshots, and VMware Storage vMotion® operations by offloading replication to the storage array.
- Block Zero — (Hardware Accelerated Zeroing) Initializes file system block and virtual drive space more rapidly.
- Hardware-Assisted Locking — (Atomic Test and Set) Enables more efficient meta data updates and assists virtual desktop deployments.
- UNMAP — Enables more efficient space usage for virtual machines by reclaiming space on datastores that is unused and returns it to the thin provisioning pool from which it was originally drawn.
- VMware vSphere Storage APIs for Storage Awareness (VASA).

VAAI is native in HYPERMAX OS and does not require additional software, unless eNAS is also implemented. If eNAS is implemented on the array, support for VAAI requires the VAAI plug-in for NAS. The plug-in is available from the Dell EMC support website.

SRDF Adapter for VMware vCenter Site Recovery Manager

Dell EMC SRDF Adapter is a Storage Replication Adapter (SRA) that extends the disaster restart management functionality of VMware vCenter Site Recovery Manager 5.x to arrays running HYPERMAX OS.

SRA allows Site Recovery Manager to automate storage-based disaster restart operations on storage arrays in an SRDF configuration.

SRDF/Cluster Enabler

Cluster Enabler (CE) for Microsoft Failover Clusters is a software extension of failover clusters functionality. Cluster Enabler enables Windows Server 2012 (including R2) Standard and Datacenter editions running Microsoft Failover Clusters to operate across multiple connected storage arrays in geographically distributed clusters.

SRDF/Cluster Enabler (SRDF/CE) is a software plug-in module to Dell EMC Cluster Enabler for Microsoft Failover Clusters software. The Cluster Enabler plug-in architecture consists of a CE base module component and separately available plug-in modules, which provide your chosen storage replication technology.

SRDF/CE supports:

- Synchronous and asynchronous mode (SRDF modes of operation on page 67 summarizes these modes)
- Concurrent and cascaded SRDF configurations (SRDF multi-site solutions on page 62 summarizes these configurations)

Product Suite for z/TPF

The Dell EMC Product Suite for z/TPF is a suite of components that monitor and manage arrays running HYPERMAX OS from a z/TPF host. z/TPF is an IBM mainframe operating system characterized by high-volume transaction rates with significant

communications content. The following software components are distributed separately and can be installed individually or in any combination:

- SRDF Controls for z/TPF

  Monitors and controls SRDF processes with functional entries entered at the z/TPF Prime CRAS (computer room agent set).

- TimeFinder Controls for z/TPF

  Provides a business continuance solution consisting of TimeFinder SnapVX, TimeFinder/Clone, and TimeFinder/Mirror.

- ResourcePak for z/TPF

  Provides PowerMax and VMAX configuration and statistical reporting and extended features for SRDF Controls for z/TPF and TimeFinder Controls for z/TPF.

SRDF/TimeFinder Manager for IBM i

Dell EMC SRDF/TimeFinder Manager for IBM i is a set of host-based utilities that provides an IBM i interface to SRDF and TimeFinder.

This feature allows you to configure and control SRDF or TimeFinder operations on arrays attached to IBM i hosts, including:

- SRDF: Configure, establish, and split SRDF devices, including:
  - SRDF/A
  - SRDF/S
  - Concurrent SRDF/A
  - Concurrent SRDF/S
- TimeFinder:
  - Create point-in-time copies of full volumes or individual data sets.
  - Create point-in-time snaphots of images.

Extended features

SRDF/TimeFinder Manager for IBM i extended features provide support for the IBM independent ASP (IASP) functionality.

IASPs are sets of switchable or private auxiliary disk pools (up to 223) that can be brought online/offline on an IBM i host without affecting the rest of the system.

When combined with SRDF/TimeFinder Manager for IBM i, IASPs let you control SRDF or TimeFinder operations on arrays attached to IBM i hosts, including:

- Display and assign TimeFinder SnapVX devices.
- Execute SRDF or TimeFinder commands to establish and split SRDF or TimeFinder devices.
- Present one or more target devices containing an IASP image to another host for business continuance (BC) processes.

Access to extended features control operations include:

- From the SRDF/TimeFinder Manager menu-driven interface.
- From the command line using SRDF/TimeFinder Manager commands and associated IBM i commands.

AppSync

Dell EMC AppSync offers a simple, SLA-driven, self-service approach for protecting, restoring, and cloning critical Microsoft and Oracle applications and VMware environments. After defining service plans, application owners can protect, restore, and clone production data quickly with item-level granularity by using the underlying Dell EMC replication technologies. AppSync also provides an application protection monitoring service that generates alerts when the SLAs are not met.

AppSync supports the following applications and storage arrays:

- Applications—Oracle, Microsoft SQL Server, Microsoft Exchange, and VMware vStorage VMFS and NFS datastores and File systems.
- Replication Technologies—SRDF, SnapVX, RecoverPoint, XtremIO Snapshot, VNX Advanced Snapshots, VNXe Unified Snapshot, and ViPR Snapshot.

NOTE: For VMAX All Flash arrays, AppSync is available in a starter bundle. The AppSync Starter Bundle provides the license for a scale-limited, yet fully functional version of AppSync. For more information, see the *AppSync Starter Bundle with VMAX All Flash Product Brief* available on the Dell EMC Online Support Website.

3

Open Systems Features

This chapter introduces the open systems features of VMAX All Flash arrays.

Topics:

- HYPERMAX OS support for open systems
- Backup and restore using PowerProtect Storage Direct and Data Domain
- VMware Virtual Volumes

HYPERMAX OS support for open systems

HYPERMAX OS provides FBA device emulations for open systems and D910 for IBM i.

Any logical device manager software installed on a host can be used with the storage devices.

HYPERMAX OS increases scalability limits from previous generations of arrays, including:

- Maximum device size is 64 TB
- Maximum host addressable devices is 64,000 for each array
- Maximum storage groups, port groups, initiator groups, and masking views is 16,000 for each object type for each array
- Maximum devices addressable through each port is 4,000

  HYPERMAX OS does not support meta devices, thus it is much more difficult to reach this limit.

Open Systems-specific provisioning on page 50 has more information on provisioning storage in an open systems environment.

The Dell EMC Support Matrix in the E-Lab Interoperability Navigator at http://elabnavigator.emc.com has the most recent information on HYPERMAX open systems capabilities.

Backup and restore using PowerProtect Storage Direct and Data Domain

Dell EMC Storage Direct provides data backup and restore facilities for a VMAX All Flash array. A remote Data Domain array stores the backup copies of the data.

Storage Direct uses existing features of the VMAX All Flash and Data Domain arrays to create backup copies and to restore backed up data if necessary. There is no need for any specialized or additional hardware and software.

This section is a high-level summary of Storage Direct backup and restore facilities. It also shows where to get detailed information about the product, including instructions on how to configure and manage it.

Backup

A LUN is the basic unit of backup in Storage Direct. For each LUN, Storage Direct creates a *backup image* on the Data Domain array. You can group backup images to create a *backup set*. One use of the backup set is to capture all the data for an application as a point-in-time image.

Backup process

To create a backup of a LUN, Storage Direct:

1. Uses SnapVX to create a local snapshot of the LUN on the VMAX All Flash array (the primary storage array).

After the snapshot is created, Storage Direct and the application can proceed independently each other and the backup process has no further impact on the application.

2. Copies the snapshot to a vdisk on the Data Domain array where it is deduplicated and cataloged.

   On the primary storage array, the vdisk appears as a FAST.X encapsulated LUN. The copy of the snapshot to the vdisk uses existing SnapVX link copy and VMAX All Flash destaging technologies.

When the vdisk contains all the data for the LUN, Data Domain converts the data into a *static image*. This image then has metadata added to it and Data Domain catalogs the resultant backup image.

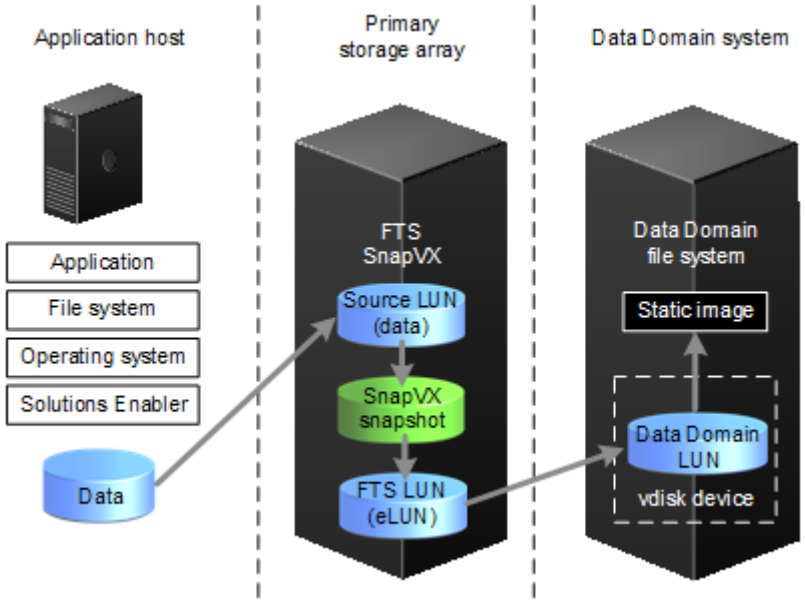


Figure 4. Data flow during a backup operation to Data Domain

Incremental data copy

The first time that Storage Direct backs up a LUN, it takes a complete copy of its contents using a SnapVX snapshot. While taking this snapshot, the application assigned to the LUN is paused for a short time. This ensures that Storage Direct has a copy of the LUN that is application consistent. To create the first backup image of the LUN, Storage Direct copies the entire snapshot to the Data Domain array.

For each subsequent backup of the LUN, Storage Direct copies only those parts of the LUN that have changed. This makes best use of the communication links and minimizes the time that is required to create the backup.

Restore

Storage Direct provides two forms of data restore:

- Object level restore from a selected backup image
- Full application rollback restore

Object level restore

For an object level restore, Data Domain puts the static image from the selected backup image on a vdisk. As with the backup process, this vdisk on the Data Domain array appears as a FAST.X encapsulated LUN on the VMAX All Flash array. The administrator can now mount the file system of the encapsulated LUN, and restore one or more objects to their final destination.

Full application rollback restore

In a full application rollback restore, all the static images in a selected backup set are made available as vdisks on the Data Domain array and available as FAST.X encapsulated LUNs on the VMAX All Flash array. From there, the administrator can restore data from the encapsulated LUNs to their original devices.

Storage Direct agents

Storage Direct has three agents, each responsible for backing up and restoring a specific type of data:

File system agent	Provides facilities to back up, manage, and restore application LUNs.
Database application agent	Provides facilities to back up, manage, and restore DB2 databases, Oracle databases, or SAP with Oracle database data.
Microsoft application agent	Provides facilities to back up, manage, and restore Microsoft Exchange and Microsoft SQL Server databases.

Features used for Storage Direct backup and restore

Storage Direct uses existing features of HYPERMAX OS and Data Domain to provide backup and restore services:

- HYPERMAX OS:
  - SnapVX
  - FAST.X encapsulated devices
- Data Domain:
  - Block services for Storage Direct
  - vdisk services
  - FastCopy

Storage Direct and traditional backup

The Storage Direct workflow can provide data protection in situations where more traditional approaches cannot successfully meet the business requirements. This is often due to small or nonexistent backup windows, demanding recovery time objective (RTO) or recovery point objective (RPO) requirements, or a combination of both.

Unlike traditional backup and recovery, Storage Direct does not rely on a separate process to identify the data that must be backed up and additional actions to move that data to backup storage. Instead of using dedicated hardware, host, and network resources, Storage Direct uses existing application and storage capabilities to create point-in-time copies of large datasets. The copies are transported across a storage area network (SAN) to Data Domain systems to protect the copies while providing deduplication to maximize storage efficiency.

Storage Direct minimizes the time that is required to protect large datasets, and allows backups to fit into the smallest of backup windows to meet demanding RTO or RPO requirements.

More information

More information about Storage Direct, its components, how to configure them, and how to use them is available in:

- *PowerProtect Storage Direct Solutions Guide*
- *File System Agent Installation and Administration Guide*
- *Database Application Agent Installation and Administration Guide*
- *Microsoft Application Agent Installation and Administration Guide*

VMware Virtual Volumes

VMware Virtual Volumes (vVols) are a storage object developed by VMware to simplify management and provisioning in virtualized environments.

With vVols, the management process moves from the LUN (data store) to the virtual machine (VM). This level of detail allows VMware and cloud administrators to assign specific storage attributes to each VM, according to its performance and storage requirements. Storage arrays running HYPERMAX OS implement vVols.

PowerMax 5978.669.669, and later, can remotely replicate vVols for disaster recovery purposes using SRDF/Asynchronous (SRDF/A).

For more information about vVols, see *Dell EMC SRDF Introduction* and *Dell EMC VASA Provider and Embedded VASA Provider for PowerMax Product Guide*.

vVol components

To support management capabilities of vVols, the storage/vCenter environment requires the following:

- EMC VMAX VASA Provider – The VASA Provider (VP) is a software plug-in that uses a set of out-of-band management APIs (VASA version 2.0). The VASA Provider exports storage array capabilities and presents them to vSphere through the VASA APIs. vVols are managed by way of vSphere through the VASA Provider APIs (create/delete) and not with the Unisphere for VMAX user interface or Solutions Enabler CLI. After vVols are setup on the array, Unisphere and Solutions Enabler only support vVol monitoring and reporting.
- Storage Containers (SC)—Storage containers are chunks of physical storage used to logically group vVols. SCs are based on the grouping of Virtual Machine Disks (VMDKs) into specific Service Levels. SC capacity is limited only by hardware capacity. At least one SC per storage system is required, but multiple SCs per array are allowed. SCs are created and managed on the array by the Storage Administrator. Unisphere and Solutions Enabler CLI support management of SCs.
- Protocol Endpoints (PE)—Protocol endpoints are the access points from the hosts to the array by the Storage Administrator. PEs are compliant with FC and replace the use of LUNs and mount points. vVols are "bound" to a PE, and the bind and unbind operations are managed through the VP APIs, not with the Solutions Enabler CLI. Existing multi-path policies and NFS topology requirements can be applied to the PE. PEs are created and managed on the array by the Storage Administrator. Unisphere and Solutions Enabler CLI support management of PEs.

Table 9. vVol architecture component management capability

Functionality	Component
vVol device management (create, delete)	VASA Provider APIs / Solutions Enabler APIs
vVol bind management (bind, unbind)	
Protocol Endpoint device management (create, delete)	Unisphere/Solutions Enabler CLI
Protocol Endpoint-vVol reporting (list, show)	
Storage Container management (create, delete, modify)	
Storage container reporting (list, show)	

vVol scalability

The vVol scalability limits are:

Table 10. vVol-specific scalability

Requirement	Value
Number of vVols/Array	64,000
Number of Snapshots/Virtual Machine[a]	12
Number of Storage Containers/Array	16
Number of Protocol Endpoints/Array	1/ESXi Host

Table 10. vVol-specific scalability (continued)

Requirement	Value
Maximum number of Protocol Endpoints/Array	1,024
Number of arrays supported /VP	1
Number of vCenters/VP	2
Maximum device size	16 TB

a. vVol Snapshots are managed through vSphere only. You cannot use Unisphere or Solutions Enabler to create them.

vVol workflow

Requirements

Install and configure these applications:

- Unisphere for VMAX V8.2 or later
- Solutions Enabler CLI V8.2 or later
- VASA Provider V8.2 or later

Instructions for installing Unisphere and Solutions Enabler are in their respective installation guides. Instructions on installing the VASA Provider are in the *Dell EMC PowerMax VASA Provider Release Notes* .

Procedure

The creation of a vVol-based virtual machine involves both the storage administrator and the VMware administrator:

Storage administrator

The storage administrator uses Unisphere or Solutions Enabler to create the storage and present it to the VMware environment:

1. Create one or more storage containers on the storage array.

   This step defines how much storage and from which service level the VMware user can provision.

2. Create Protocol Endpoints and provision them to the ESXi hosts.

VMware administrator

The VMware administrator uses the vSphere Web Client to deploy the VM on the storage array:

1. Add the VASA Provider to the vCenter.

   This allows vCenter to communicate with the storage array,

2. Create a vVol datastore from the storage container.
3. Create the VM storage policies.
4. Create the VM in the vVol datastore, selecting one of the VM storage policies.

4

Mainframe Features

This chapter introduces the mainframe-specific features of VMAX All Flash arrays.

Topics:

- HYPERMAX OS support for mainframe
- IBM Z Systems functionality support
- IBM 2107 support
- Logical control unit capabilities
- Disk drive emulations
- Cascading configurations

HYPERMAX OS support for mainframe

VMAX 450F, 850F, and 950F arrays can be ordered with the zF and zFX software packages to support mainframe.

VMAX All Flash arrays provide the following mainframe features:

- Mixed FBA and CKD drive configurations.
- Support for 64, 128, 256 FICON single and multi mode ports, respectively.
- Support for CKD 3380/3390 and FBA devices.
- Mainframe (FICON) and OS FC/iSCSI connectivity.
- High capacity flash drives.
- Up to 16 Gb/s FICON host connectivity.
- Support for Forward Error Correction, Query Host Access, and FICON Dynamic Routing.
- T10 DIF protection for CKD data along the data path (in cache and on disk) to improve performance for multi-record operations.
- D@RE external key managers. Data at Rest Encryption on page 25 provides more information on D@RE and external key managers.

IBM Z Systems functionality support

VMAX All Flash arrays support the latest IBM Z Systems enhancements, ensuring that the array can handle the most demanding mainframe environments:

- zHPF, including support for single track, multi track, List Prefetch, bi-directional transfers, QSAM/BSAM access, and Format Writes
- zHyperWrite
- Non-Disruptive State Save (NDSS)
- Compatible Native Flash (Flash Copy)
- Remote Pair Flash Copy (RPFC)
- Concurrent Copy
- Multi-subsystem Imaging
- Parallel Access Volumes (PAV)
- Dynamic Channel Management (DCM)
- Dynamic Parallel Access Volumes/Multiple Allegiance (PAV/MA)
- Peer-to-Peer Remote Copy (PPRC) SoftFence
- Extended Address Volumes (EAV)
- Persistent IU Pacing (Extended Distance FICON)
- HyperPAV
- PDS Search Assist
- Modified Indirect Data Address Word (MIDAW)

- Multiple Allegiance (MA)
- Sequential Data Striping
- Multi-Path Lock Facility
- Product Suite for z/TPF
- HyperSwap
- Global Mirror
- Transparent Cloud Tiering

Global Mirror support

Global Mirror is an IBM solution for long-distance replication that can be run through Geographically Dispersed Parallel Sysplex (GDPS) automation.

PowerMax (and VMAX) supports a two-site Global Mirror environment. A two-site Global Mirror configuration has Peer-to-Peer Remote Copy (PPRC) primary devices at Site A and PPRC secondary devices at Site B. There is also a relationship from the PPRC secondary devices to a third set of devices at Site B that is not usually active.

NOTE: Use of Global Mirror requires an RPQ.

Global Mirror requirements and restrictions are as follows:

- Global Mirror is supported on PowerMax 8000 and VMAX 950F arrays.
- Both PPRC primary and secondary sites run PowerMaxOS 5978.669.669, or later.
- One Global Mirror session is associated with one SRDF group.
- Up to 31 Global Mirror sessions can be run simultaneously.
- When multiple Global Mirror sessions are running on an array, each session maps to a unique SRDF group.
- All devices in the SRDF group are defined to the Global Mirror session.

NOTE: To add or remove devices from the Global Mirror session, stop or pause Global Mirror.

NOTE: A VMAX All Flash array can participate in a z/OS Global Mirror (XRC) configuration only as a secondary.

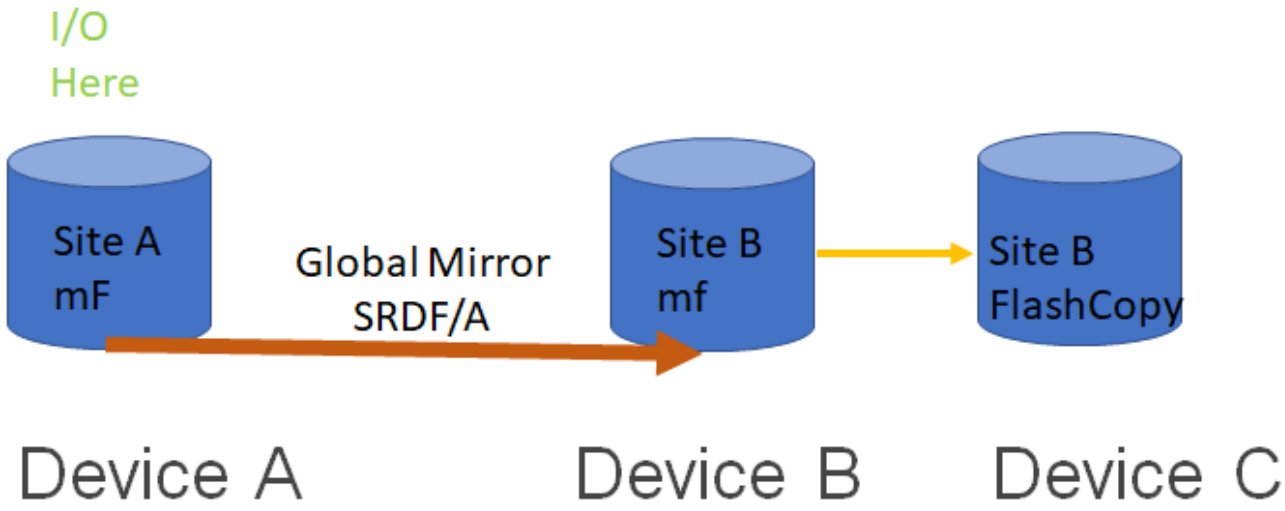


Figure 5. Two-site Global Mirror

Transparent Cloud Tiering support

PowerMax (or VMAX) can be used in an IBM Transparent Cloud Tiering (TCT) environment to store DASD files in, or retrieve them from, the cloud with minimal CPU requirements on the z/OS host.

PowerMax TCT support uses the Dell EMC Disk Library for mainframe (DLm) as its 'cloud'. Count Key Data (CKD) extents are stored as files on standard 3590 tape volumes. The DLm 3590 tape volumes and the DLm tape drives for TCT are separate from any z/OS-defined tape volumes and tape drives. TCT 3590 tapes are not accessible to, or managed, by z/OS. Tape volumes are

written to the DLm from the PowerMax over a FICON connection. DLm then stores the data on any backend storage that DLm supports. Optionally, the DLm Long-Term Retention feature can then be used, independent of TCT, to move the data to a Dell EMC Elastic Cloud Storage (ECS) solution.

A cloud object store is required for TCT support to operate. This cloud must support the OpenStack SWIFT protocol and is used to store cloud metadata. If ECS is used as the cloud, the ECS can be the same or a different ECS from any ECS deployed with the DLm.

A REST API proxy server is required in each z/OS image accessing a TCT-enabled PowerMax. This proxy server runs as a separate address space in z/OS.

The *ResourcePak Base for z/OS Product Guide* provides additional information about TCT support, including TCT support requirements and restrictions. It also discusses how to set up and run the Dell EMC REST API proxy and the Dell EMC REST API utility.

NOTE: Use of TCT in PowerMaxOS 5978 requires an RPQ.

PowerMax TCT environment

The PowerMax TCT environment includes the following components:

- z/OS host with DFSMS and Dell EMC REST API proxy:
  - The Data Facility Storage Management Subsystem (DFSMS) provides space management and data movement tools that are primary users of TCT services.
  - Dell EMC REST API proxy provides REST API services for DFSMS.

    The REST API proxy requires a connection to a cloud service. The cloud service can be the same ECS cloud that DLm uses or another supported service (SWIFT protocol is required).
- PowerMax with TCT support—A PowerMax mainframe host adapter runs the Cloud Services Stack. The Cloud Services Stack writes data to and reads data from DLm virtual tape volumes.
- Disk Library for mainframe (DLm)—The TCT-enabled PowerMax stores data in a DLm, which acts as a cloud for PowerMax. DLm can be connected to a cloud provider, such as ECS.

The following figure illustrates a TCT environment with PowerMax and DLm.

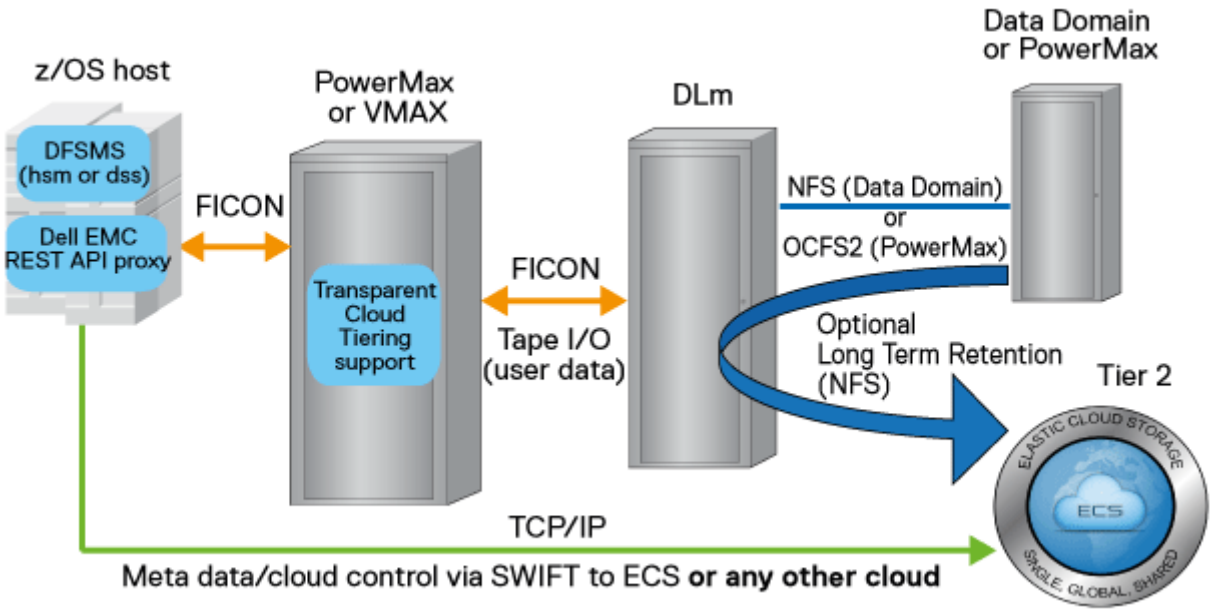


Figure 6. TCT environment with PowerMax and DLm

IBM 2107 support

When VMAX All Flash arrays emulate an IBM 2107, they externally represent the array serial number as an alphanumeric number in order to be compatible with IBM command output. Internally, the arrays retain a numeric serial number for IBM 2107 emulations. HYPERMAX OS handles correlation between the alphanumeric and numeric serial numbers.

Logical control unit capabilities

The following table lists logical control unit (LCU) maximum values:

Table 11. Logical control unit maximum values

Capability	Maximum value
LCUs per director slice (or port)	255 (within the range of 00 to FE)
LCUs per split[a]	255
Splits per array	16 (0 to 15)
Devices per split	65,280
LCUs per array	512
Devices per LCU	256
Logical paths per port	2,048
Logical paths per LCU per port (see Maximum LPARs per port on page 46)	128
Array system host address per array (base and alias)	64K
I/O host connections per array engine	32

a. A split is a logical partition of the storage array, identified by unique devices, SSIDs, and host serial number. The maximum storage array host address per array is inclusive of all splits.

The following table lists the maximum LPARs per port based on the number of LCUs with active paths:

Table 12. Maximum LPARs per port

LCUs with active paths per port	Maximum volumes supported per port	Array maximum LPARs per port
16	4K	128
32	8K	64
64	16K	32
128	32K	16
255	64K	8

Disk drive emulations

When VMAX All Flash arrays are configured to mainframe hosts, the data recording format is Extended CKD (ECKD). The supported CKD emulations are 3380 and 3390.

Cascading configurations

Cascading configurations greatly enhance FICON connectivity between local and remote sites by using switch-to-switch extensions of the CPU to the FICON network. These cascaded switches communicate over long distances using a small number

of high-speed lines called interswitch links (ISLs). A maximum of two switches may be connected together within a path between the CPU and the storage array.

Use of the same switch vendors is required for a cascaded configuration. To support cascading, each switch vendor requires specific models, hardware features, software features, configuration settings, and restrictions. Specific IBM CPU models, operating system release levels, host hardware, and HYPERMAX levels are also required.

The Dell EMC Support Matrix, available through E-Lab Interoperability Navigator (ELN) at http://elabnavigator.emc.com has the most up-to-date information on switch support.

5

Provisioning

This chapter introduces storage provisioning.

Topics:

- Thin provisioning
- Multi-array provisioning

Thin provisioning

VMAX All Flash arrays are configured in the factory with thin provisioning pools ready for use. Thin provisioning improves capacity utilization and simplifies storage management. It also enables storage to be allocated and accessed on demand from a pool of storage that services one or many applications. LUNs can be “grown” over time as space is added to the data pool with no impact to the host or application. Data is widely striped across physical storage (drives) to deliver better performance than standard provisioning.

NOTE: Data devices (TDATs) are provisioned/pre-configured/created while the host addressable storage devices TDEVs are created by either the customer or customer support, depending on the environment.

Thin provisioning increases capacity utilization and simplifies storage management by:

- Enabling more storage to be presented to a host than is physically consumed
- Allocating storage only as needed from a shared thin provisioning pool
- Making data layout easier through automated wide striping
- Reducing the steps required to accommodate growth

Thin provisioning allows you to:

- Create host-addressable thin devices (TDEVs) using Unisphere or Solutions Enabler
- Add the TDEVs to a storage group
- Run application workloads on the storage groups

When hosts write to TDEVs, the physical storage is automatically allocated from the default Storage Resource Pool.

Pre-configuration for thin provisioning

VMAX All Flash arrays are custom-built and pre-configured with array-based software applications, including a factory pre-configuration for thin provisioning that includes:

- *Data devices (TDAT)* — an internal device that provides physical storage used by thin devices.
- *Virtual provisioning pool* — a collection of data devices of identical emulation and protection type, all of which reside on drives of the same technology type and speed. The drives in a data pool are from the same disk group.
- *Disk group*— a collection of physical drives within the array that share the same drive technology and capacity. RAID protection options are configured at the disk group level. Dell Technologies strongly recommends that you use one or more of the RAID data protection schemes for all data devices.

Table 13. RAID options

RAID	Provides the following	Configuration considerations
RAID 5	Distributed parity and striped data across all drives in the RAID group. Options include: • RAID 5 (3 + 1) — Consists of four drives with parity and data striped across each device. • RAID-5 (7 + 1) — Consists of eight drives with parity and data striped across each device.	• RAID-5 (3 + 1) provides 75% data storage capacity. Only available with VMAX 250F arrays. • RAID-5 (7 + 1) provides 87.5% data storage capacity. • Withstands failure of a single drive within the RAID-5 group.
RAID 6	Striped drives with double distributed parity (horizontal and diagonal). The highest level of availability options include: • RAID-6 (6 + 2) — Consists of eight drives with dual parity and data striped across each device. • RAID-6 (14 + 2) — Consists of 16 drives with dual parity and data striped across each device.	• RAID-6 (6 + 2) provides 75% data storage capacity. Only available with VMAX 250F arrays. • RAID-6 (14 + 2) provides 87.5% data storage capacity. • Withstands failure of two drives within the RAID-6 group.

- *Storage Resource Pools* — one (default) Storage Resource Pool is pre-configured on the array. This process is automatic and requires no setup. You cannot modify Storage Resource Pools, but you can list and display their configuration. You can also generate reports detailing the demand storage groups are placing on the Storage Resource Pools.

Thin devices (TDEVs)

NOTE: On VMAX All Flash arrays the thin device is the only device type for front end devices.

Thin devices (TDEVs) have no storage allocated until the first write is issued to the device. Instead, the array allocates only a minimum allotment of physical storage from the pool, and maps that storage to a region of the thin device including the area targeted by the write.

These initial minimum allocations are performed in units called thin device extents. Each extent for a thin device is 1 track (128 KB).

When a read is performed on a device, the data being read is retrieved from the appropriate data device to which the thin device extent is allocated. Reading an area of a thin device that has not been mapped does not trigger allocation operations. Reading an unmapped block returns a block in which each byte is equal to zero.

When more storage is required to service existing or future thin devices, data devices can be added to existing thin storage groups.

Thin device oversubscription

A thin device can be presented for host use *before* mapping all of the reported capacity of the device.

The sum of the reported capacities of the thin devices using a given pool can exceed the available storage capacity of the pool. Thin devices whose capacity exceeds that of their associated pool are "oversubscribed".

Over-subscription allows presenting larger than needed devices to hosts and applications without having the physical drives to fully allocate the space represented by the thin devices.

Open Systems-specific provisioning

HYPERMAX host I/O limits for open systems

On open systems, you can define host I/O limits and associate a limit with a storage group. The I/O limit definitions contain the operating parameters of the I/O per second or bandwidth limitations.

When an I/O limit is associated with a storage group, the limit is divided equally among all the directors in the masking view that is associated with the storage group. All devices in that storage group share that limit.

When applications are configured, you can associate the limits with storage groups that contain a list of devices. A single storage group can only be associated with one limit, and a device can only be in one storage group that has limits associated.

There can be up to 4096 host I/O limits.

Consider the following when using host I/O limits:

- Cascaded host I/O limits controlling parent and child storage groups limits in a cascaded storage group configuration.
- Offline and failed director redistribution of quota that supports all available quota to be available instead of losing quota allocations from offline and failed directors.
- Dynamic host I/O limits support for dynamic redistribution of steady state unused director quota.

Initiator bandwidth limits

Slow SAN drain is a congestion issue that is caused by slow drain devices (host bus adapters (HBAs) that have a lower link speed (BW) compared to storage array SLIC speeds (BW)). This issue can lead to severe fabric-wide performance degradation, but can be mitigated by configuring per initiator bandwidth limits.

At the host group level you can configure initiator-based host I/O, or bandwidth, limits. To set the limits, there must be fibre channel connectivity to the host, and at least one initiator configured on the host.

Auto-provisioning groups on open systems

You can auto-provision groups on open systems to reduce complexity, execution time, labor cost, and the risk of error.

Auto-provisioning groups enables users to group initiators, front-end ports, and devices together, and to build masking views that associate the devices with the ports and initiators.

When a masking view is created, the necessary mapping and masking operations are performed automatically to provision storage.

After a masking view exists, any changes to its grouping of initiators, ports, or storage devices automatically propagate throughout the view, automatically updating the mapping and masking as required.

Components of an auto-provisioning group

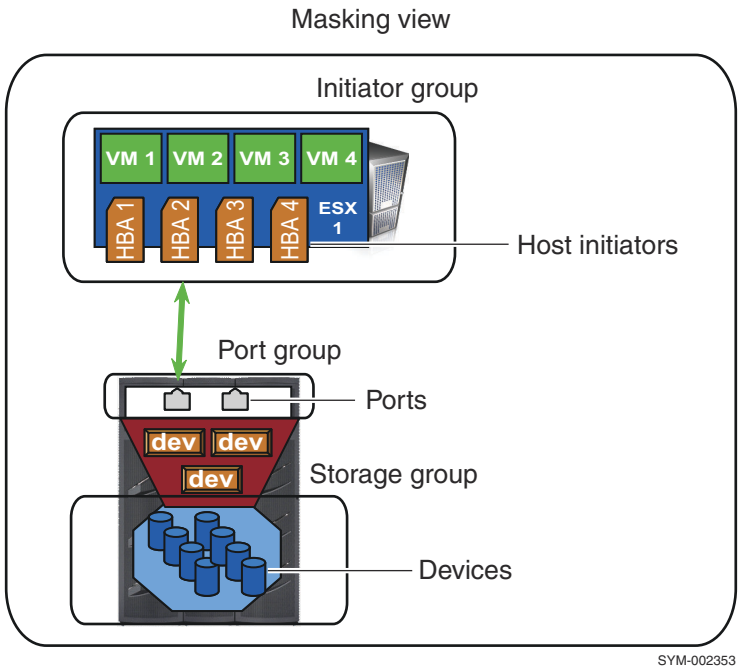


Figure 7. Auto-provisioning groups

Initiator group

A logical grouping of Fibre Channel initiators. An initiator group is limited to either a parent, which can contain other groups, or a child, which contains one initiator role. Mixing of initiators and child name in a group is not supported.

Port group

A logical grouping of Fibre Channel front-end director ports. A port group can contain up to 32 ports.

Storage group

A logical grouping of thin devices. LUN addresses are assigned to the devices within the storage group when the view is created if the group is either cascaded or stand alone. Often there is a correlation between a storage group and a host application. One or more storage groups may be assigned to an application to simplify management of the system. Storage groups can also be shared among applications.

Cascaded storage group

A parent storage group comprised of multiple storage groups (parent storage group members) that contain child storage groups comprised of devices. By assigning child storage groups to the parent storage group members and applying the masking view to the parent storage group, the masking view inherits all devices in the corresponding child storage groups.

Masking view

An association between one initiator group, one port group, and one storage group. When a masking view is created, the group within the view is a parent, the contents of the children are used. For example, the initiators from the children initiator groups and the devices from the children storage groups. Depending on the server and application requirements, each server or group of servers may have one or more masking views that associate a set of thin devices to an application, server, or cluster of servers.

Multi-array provisioning

The multi-array Provisioning Storage wizard simplifies the task of identifying the optimal target array and provisioning storage on that array.

Unisphere for PowerMax 9.2 provides a system-level provisioning launch point that takes array-independent inputs (storage group name, device count and size, and (optionally) response time target or initiator filter), selects ports that are based on

current utilization and port group best practices, and returns component impact scores for all locally connected arrays running HYPERMAX OS 5977 or PowerMaxOS 5978.

You can also select a provisioning template and provision new storage using the wizard. Storage group capacity information and response time targets that are already part of the provisioning template are populated when the wizard opens. The most suitable ports (based on specified options) are selected and a list of all locally connected arrays (V3 and higher) are returned. The list is sorted by the impact of the new workload on the target arrays.

Host I/O limits (quotas) can be used to limit the amount of Front End (FE) Bandwidth and I/O operations per second (IOPS) that can be consumed by a set of storage volumes over a set of director ports. Host I/O limits are defined as storage group attributes – the maximum bandwidth (in MB per second) and the maximum IOPS. The Host I/O limit for a storage group can be either active or inactive.

6

Native local replication with TimeFinder

This chapter introduces the local replication features.

Topics:

- About TimeFinder
- Mainframe SnapVX and zDP
- Snapshot policy

About TimeFinder

Dell EMC TimeFinder delivers point-in-time copies of volumes that can be used for backups, decision support, data warehouse refreshes, or any other process that requires parallel access to production data.

Previous VMAX families offered multiple TimeFinder products, each with their own characteristics and use cases. These traditional products required a target volume to retain snapshot or clone data.

HYPERMAX OS introduces TimeFinder SnapVX which provides the best aspects of the traditional TimeFinder offerings combined with increased scalability and ease-of-use.

TimeFinder SnapVX emulates the following legacy replication products:

- FBA devices:
  - TimeFinder/Clone
  - TimeFinder/Mirror
  - TimeFinder VP Snap
- Mainframe (CKD) devices:
  - TimeFinder/Clone
  - TimeFinder/Mirror
  - TimeFinder/Snap
  - Dell EMC Dataset Snap
  - IBM FlashCopy (Full Volume and Extent Level)

TimeFinder SnapVX dramatically decreases the impact of snapshots and clones:

- For snapshots, this is done by using redirect on write technology (ROW).
- For clones, this is done by storing changed tracks (deltas) directly in the Storage Resource Pool of the source device - sharing tracks between snapshot versions and also with the source device, where possible.

There is no need to specify a target device and source/target pairs. SnapVX supports up to 256 snapshots per volume. Each snapshot can have a name and an automatic expiration date.

Access to snapshots

With SnapVX, a snapshot can be accessed by *linking* it to a host accessible volume (known as a target volume). Target volumes are standard VMAX All Flash TDEVs. Up to 1024 target volumes can be linked to the snapshots of the source volumes. The 1024 links can all be to the same snapshot of the source volume, or they can be multiple target volumes linked to multiple snapshots from the same source volume. However, a target volume may be linked only to one snapshot at a time.

Snapshots can be cascaded from linked targets, and targets can be linked to snapshots of linked targets. There is no limit to the number of levels of cascading, and the cascade can be broken.

SnapVX links to targets in the following modes:

- Nocopy Mode (Default): SnapVX does not copy data to the linked target volume but still makes the point-in-time image accessible through pointers to the snapshot. The target device is modifiable and retains the full image in a space-efficient manner even after unlinking from the point-in-time.
- Copy Mode: SnapVX copies all relevant tracks from the snapshot's point-in-time image to the linked target volume. This creates a complete copy of the point-in-time image that remains available after the target is unlinked.

If an application needs to find a particular point-in-time copy among a large set of snapshots, SnapVX enables you to link and relink until the correct snapshot is located.

Interoperability with legacy TimeFinder products

TimeFinder SnapVX and HYPERMAX OS emulate legacy TimeFinder and IBM FlashCopy replication products to provide backwards compatibility. You can run your legacy replication scripts and jobs on VMAX All Flash arrays running TimeFinder SnapVX and HYPERMAX OS without altering them.

Arrays that run PowerMaxOS 5978.444.444 and later enable coexistence and interoperability of SnapVX with legacy TimeFinder products. On such an array, a device can simultaneously be the source of a SnapVX operation and the source of one of these legacy TimeFinder products:

- TimeFinder/Clone
- TimeFinder/Mirror
- TimeFinder VP Snap

The target device of a legacy TimeFinder product cannot be the source device for SnapVX. Similarly, the target device of SnapVX cannot be the source device for a legacy TimeFinder product.

Uses for the coexistence of SnapVX with legacy TimeFinder products include:

- A site wants to keep its current, legacy configuration in place while trying out SnapVX.
- Moving to SnapVX may require the deletion of existing legacy sessions and that violates local business policies.

NOTE: Coexistence of SnapVX and legacy TimeFinder products is not available when the source of a SnapVX session is undergoing a restore operation.

Targetless snapshots

With the TimeFinder SnapVX management interfaces you can take a snapshot of an entire VMAX All Flash Storage Group using a single command. With this in mind, VMAX All Flash supports up to 64K storage groups. The number of groups is enough even in the most demanding environment to provide one for each application. The storage group construct already exists in most cases as they are created for masking views. TimeFinder SnapVX uses this existing structure, so reducing the administration required to maintain the application and its replication environment.

Creation of SnapVX snapshots does not require preconfiguration of extra volumes. In turn, this reduces the amount of cache that SnapVX snapshots use and simplifies implementation. Snapshot creation and automatic termination can easily be scripted.

The following Solutions Enabler example creates a snapshot with a 2-day retention period. The command can be scheduled to run as part of a script to create multiple versions of the snapshot. Each snapshot shares tracks where possible with the other snapshots and the source devices. Use a cron job or scheduler to run the snapshot script on a schedule to create up to 256 snapshots of the source volumes; enough for a snapshot every 15 minutes with 2 days of retention:

```
symsnapvx -sid 001 -sg StorageGroup1 -name sg1_snap establish -ttl -delta 2
```

If a restore operation is required, any of the snapshots created by this example can be specified.

When the storage group transitions to a restored state, the restore session can be terminated. The snapshot data is preserved during the restore process and can be used again should the snapshot data be required for a future restore.

Secure snaps

Secure snaps prevent administrators or other high-level users from deleting snapshot data, intentionally or not. Also, Secure snaps are also immune to automatic failure resulting from running out of Storage Resource Pool (SRP) or Replication Data Pointer (RDP) space on the array.

When the administrator creates a secure snapshot, they assign it an expiration date and time. The administrator can express the expiration either as a delta from the current date or as an absolute date. Once the expiration date passes, and if the snapshot has no links, HYPERMAX OS automatically deletes the snapshot. Before its expiration, administrators can only extend the expiration date; they cannot shorten the date or delete the snapshot. If a secure snapshot expires, and it has a volume linked to it, or an active restore session, the snapshot is not deleted. However, it is no longer considered secure.

NOTE: Secure snapshots may only be terminated after they expire or by customer-authorized Dell EMC support. Refer to Knowledgebase article 498316 for more information.

Provision multiple environments from a linked target

Use SnapVX to create multiple test and development environments using linked snapshots. To access a point-in-time copy, create a link from the snapshot data to a host mapped target device.

Each linked storage group can access the same snapshot, or each can access a different snapshot version in either no copy or copy mode. Changes to the linked volumes do not affect the snapshot data. To roll back a test or development environment to the original snapshot image, perform a relink operation.

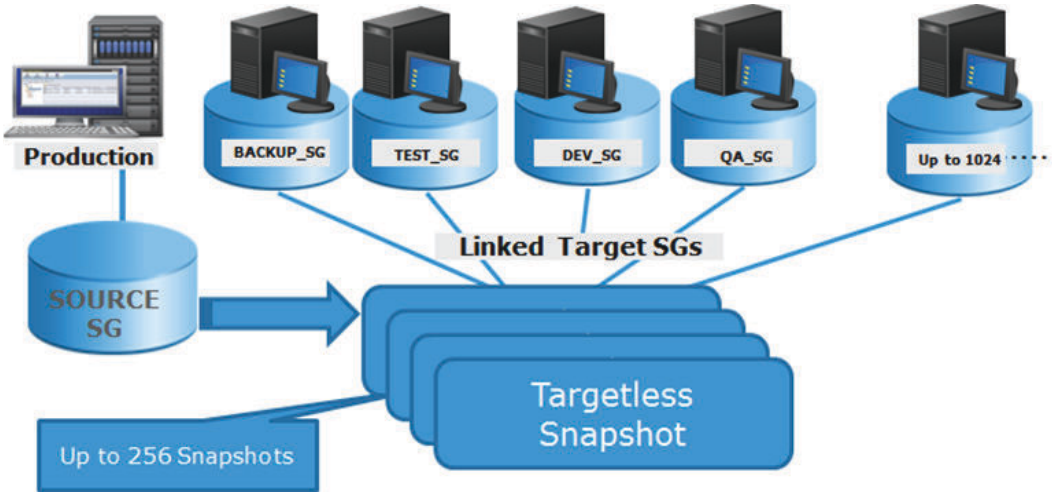


Figure 8. SnapVX targetless snapshots

NOTE: Unmount target volumes before issuing the relink command to ensure that the host operating system does not cache any filesystem data. If accessing through VPLEX, ensure that you follow the procedure outlined in the technical note *VPLEX: Leveraging Array Based and Native Copy Technologies*, available on the Dell EMC support website.

Once the relink is complete, volumes can be remounted.

Snapshot data is unchanged by the linked targets, so the snapshots can also be used to restore production data.

Cascading snapshots

Presenting sensitive data to test or development environments often requires that the source of the data be disguised beforehand. Cascaded snapshots provides this separation and disguise, as shown in the following image.

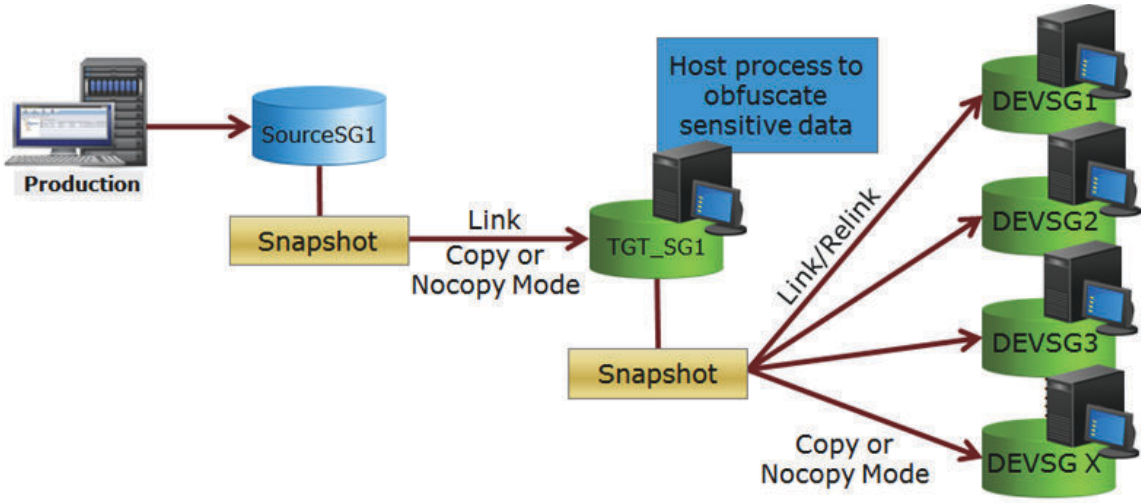


Figure 9. SnapVX cascaded snapshots

If no change to the data is required before presenting it to the test or development environments, there is no need to create a cascaded relationship.

Accessing point-in-time copies

To access a point-in time-copy, create a link from the snapshot data to a host mapped target device. The links may be created in Copy mode for a permanent copy on the target device, or in NoCopy mode for temporary use. Copy mode links create full-volume, full-copy clones of the data by copying it to the target device's Storage Resource Pool. NoCopy mode links are space-saving snapshots that only consume space for the changed data that is stored in the source device's Storage Resource Pool.

HYPERMAX OS supports up to 1,024 linked targets per source device.

NOTE: When a target is first linked, all of the tracks are undefined. This means that the target does not know where in the Storage Resource Pool the track is located, and host access to the target must be derived from the SnapVX metadata. A background process eventually defines the tracks and updates the thin device to point directly to the track location in the source device's Storage Resource Pool.

Mainframe SnapVX and zDP

Data Protector for z Systems (zDP) is a mainframe software solution that is layered on SnapVX on VMAX All Flash arrays. Using zDP you can recover from logical data corruption with minimal data loss. zDP achieves this by providing multiple, frequent, consistent point-in-time copies of data automatically. You can then use these copies to recover an application or the environment to a point prior to the logical corruption.

By providing easy access to multiple different point-in-time copies of data (with a granularity of minutes), precise recovery from logical data corruption can be performed using application-based recovery procedure. zDP results in minimal data loss compared to other methods such as restoring data from daily or weekly backups.

As shown in zDP operation on page 56, you can use zDP to create and manage multiple point-in-time snapshots of volumes. Each snapshot is a pointer-based, point-in-time image of a single volume. These images are created using the SnapVX feature of HYPERMAX OS. SnapVX is a space-efficient method for making snapshots of thin devices and consuming additional storage capacity only when changes are made to the source volume.

There is no need to copy each snapshot to a target volume as SnapVX separates the capturing of a point-in-time copy from its usage. Capturing a point-in-time copy does not require a target volume. Using a point-in-time copy from a host requires linking the snapshot to a target volume.

There can be up to 256 snapshots of each source volume.

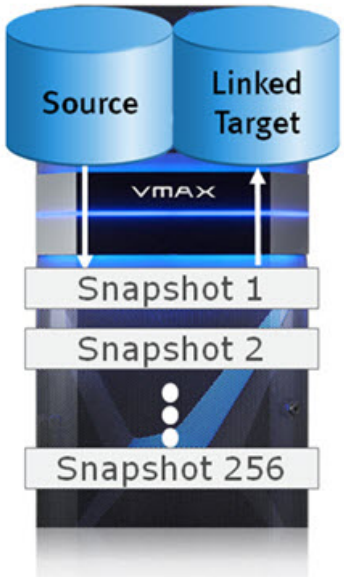


Figure 10. zDP operation

These snapshots share allocations to the same track image whenever possible while ensuring they each continue to represent a unique point-in-time image of the source volume. Despite the space efficiency achieved through shared allocation to unchanged data, additional capacity is required to preserve the pre-update images of changed tracks captured by each point-in-time snapshot.

The process of implementing zDP has two phases — the planning phase and the implementation phase.

- The planning phase is done in conjunction with your EMC representative who has access to tools that can help size the capacity needed for zDP if you are currently a VMAX All Flash user.
- The implementation phase uses the following methods for z/OS:
  - A batch interface that allows you to submit jobs to define and manage zDP.
  - A zDP run-time environment that executes under SCF to create snapsets.

For details on zDP usage, refer to the *TimeFinder SnapVX and zDP Product Guide*. For details on zDP usage in z/TPF, refer to the *TimeFinder Controls for z/TPF Product Guide*.

Snapshot policy

The Snapshot policy feature provides snapshot orchestration at scale (1024 snaps per storage group). The feature simplifies snapshot management for standard and cloud snapshots.

Snapshots can be used to recover from data corruption, accidental deletion, or other damage, offering continuous data protection. A large number of snapshots can be difficult to manage. The Snapshot policy feature provides an end to end solution to create, schedule and manage standard (local) and cloud snapshots.

The snapshot policy (Recovery Point Objective (RPO)) specifies how often the snapshot should be taken and how many of the snapshots should be retained. The snapshot may also be specified to be secure (these snapshots cannot be terminated by users before their time to live (TTL), derived from the snapshot policy's interval and maximum count, has expired.) Up to four policies can be associated with a storage group, and a snapshot policy can be associated with many storage groups.

The following rules apply to snapshot policies:

- The maximum number of snapshot policies that can be created on a storage system is 20. Multiple storage groups can be associated with a snapshot policy.
- A maximum of four snapshot policies can be associated with an individual storage group.
- A storage group or device can have a maximum of 256 manual snapshots.
- A storage group or device can have a maximum of 1024 snapshots.
- The oldest unused snapshots are removed or recycled in accordance with the specified policy max_count value.
- When devices are added to a snapshot policy storage group, snapshot policies that apply to the storage group are applied to the added devices.
- When devices are removed from a snapshot policy storage group, snapshot policies that apply to the storage group are no longer applied to the removed devices.
- If overlapping snapshot policies are applied to storage groups, they run and take snapshots independently.

Compliance information is provided for each snapshot policy that is directly associated with (not inherited to) a storage group.

Snapshot compliance for a storage group is taken as the lowest compliance value for any of the snapshot policies that are directly associated with the storage group.

Compliance for a snapshot policy that is associated with a storage group is based on the number of good snapshots within the retention count. The retention count is translated to a retention period for compliance calculation. The retention period is the snapshot interval multiplied by the snapshot maximum count. For example, a 1 hr interval with a 30 snapshot count means a 30-hour retention period.

The compliance threshold for green to yellow change is the maximum count, that is, all snapshots must be good and in place for the compliance to be green. If there is one snapshot short (missing or failed), then the compliance turns yellow.

The compliance threshold value for yellow to red is stored in the snapshot policy definition. Once the number of good snapshots falls below this value, compliance turns red.

Snapshot compliance is calculated by polling the storage system once an hour for SnapVX related information for storage groups which have snapshot policies that are associated with them. The returned snapshot information is summarized into the required information for the database compliance entries.

When the maximum count of snapshots for a snapshot policy is changed, this changes the compliance for the storage group or service level combination. Compliance values are updated accordingly simultaneously.

If compliance calculation is performed during the creation of a snapshot, then an establish-in-progress state may be detected. This is acceptable for the most recent snapshot but is considered failed for any older snapshot.

When a storage group and service level have only recently been associated and the full maximum count of snapshots has not yet been reached, the calculation is scaled to the number of snapshots that are available and represents compliance accordingly until the full maximum count of snapshots has been reached. If a snapshot failed to be taken for a reason, such as the storage group or service level was suspended, or a snapshot was manually terminated before the maximum snapshot count was reached, the compliance is reported as out of compliance appropriately.

When the service level interval is changed, the compliance window changes and the number of snapshots may not exist for correct compliance.

If a service level is suspended or a storage group or service level combination is suspended, snapshots are not created and older snapshots fall outside the compliance window and the maximum count of required snapshot is not found.

Manual termination of snapshots inside the compliance window results in the storage group or service level combination falling out of compliance.

7

Remote replication

This chapter introduces the remote replication facilities.

Topics:

- Native remote replication with SRDF
- SRDF/Metro
- RecoverPoint
- Remote replication using eNAS

Native remote replication with SRDF

The Dell EMC Symmetrix Remote Data Facility (SRDF) family of products offers a range of array-based disaster recovery, parallel processing, and data migration solutions for Dell EMC storage systems, including:

- PowerMaxOS for PowerMax 2000 and 8000 arrays and for VMAX All Flash 450F and 950F arrays
- HYPERMAX OS for VMAX All Flash 250F, 450F, 850F, and 950F arrays
- HYPERMAX OS for VMAX 100K, 200K, and 400K arrays
- Enginuity for VMAX 10K, 20K, and 40K arrays

SRDF disaster recovery solutions use “active, remote” mirroring and dependent-write logic to create consistent copies of data. Dependent-write consistency ensures transactional consistency when the applications are restarted at the remote location. You can tailor your SRDF solution to meet various Recovery Point Objectives and Recovery Time Objectives.

Using SRDF, you can create complete solutions to:

- Create real-time or dependent-write-consistent copies at 1, 2, or 3 remote arrays.
- Move data quickly over extended distances.
- Provide 3-site disaster recovery with zero data loss recovery, business continuity protection and disaster-restart.

You can integrate SRDF with other Dell EMC products to create complete solutions to:

- Restart operations after a disaster with zero data loss and business continuity protection.
- Restart operations in cluster environments. For example, Microsoft Cluster Server with Microsoft Failover Clusters.
- Monitor and automate restart operations on an alternate local or remote server.
- Automate restart operations in VMware environments.

SRDF 2-site solutions

The following table describes SRDF 2-site solutions.

Table 15. SRDF 2-site solutions

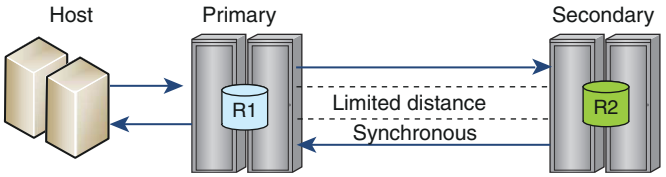
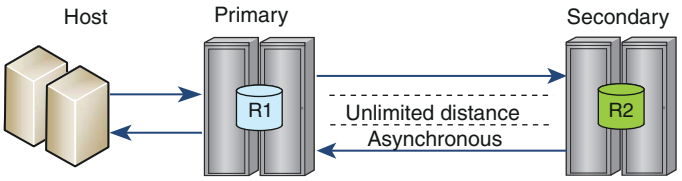
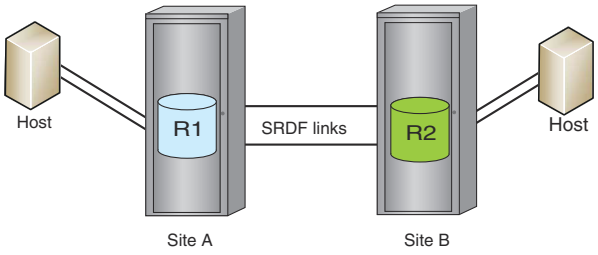
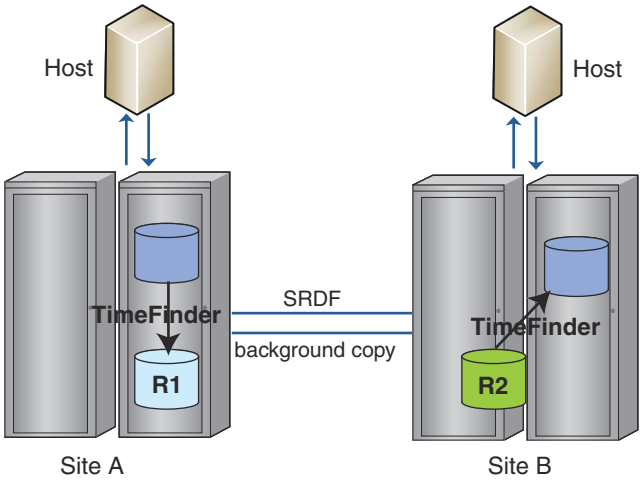
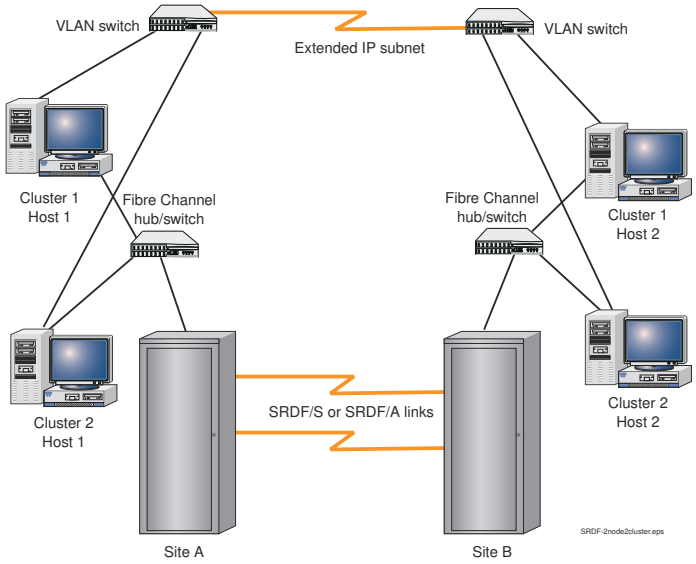
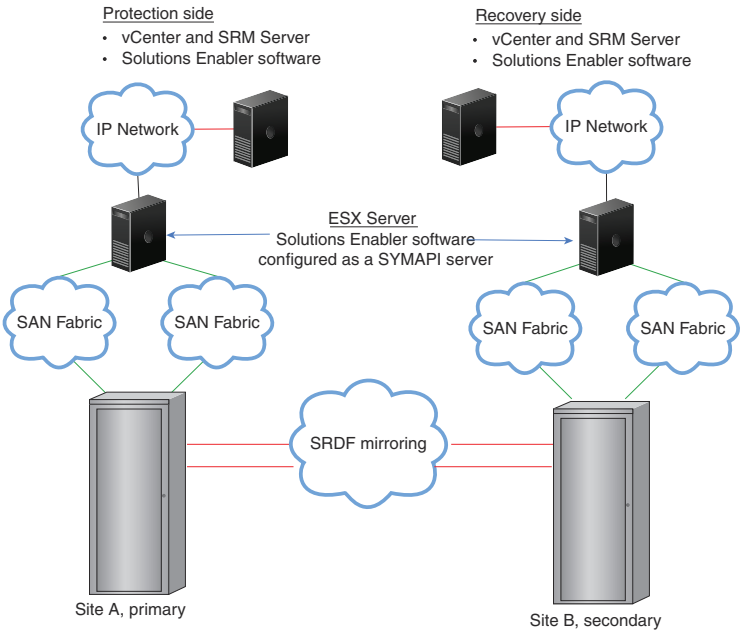
Solution highlights	Site topology
SRDF/Synchronous (SRDF/S) Maintains a real-time copy of production data at a physically separated array. • No data exposure. • Ensured consistency protection with SRDF/Consistency Group. • Recommended maximum distance of 200 km (125 miles) between arrays as application latency may rise to unacceptable levels at longer distances.[a]	
SRDF/Asynchronous (SRDF/A) Maintains a dependent-write consistent copy of the data on a remote secondary site. The sites can be an unlimited distance apart. The copy of the data at the secondary site is seconds behind the primary site.	
SRDF/Data Mobility (SRDF/DM) Enables the fast transfer of data from R1 to R2 devices over extended distances. • Uses adaptive copy mode to transfer data. • Designed for migration or data replication purposes, not for disaster restart solutions.	
SRDF/Automated Replication (SRDF/AR) • Combines SRDF and TimeFinder to optimize bandwidth requirements and provide a long-distance disaster restart solution. • Operates in 2-site solutions that use SRDF/DM in combination with TimeFinder.	

Table 15. SRDF 2-site solutions (continued)

Solution highlights	Site topology
SRDF/Cluster Enabler (CE) • Integrates SRDF/S or SRDF/A with Microsoft Failover Clusters (MSCS) to automate or semi-automate site failover. • Complete solution for restarting operations in cluster environments (MSCS with Microsoft Failover Clusters). • Expands the range of cluster storage and management capabilities while ensuring full protection of the SRDF remote replication.	
SRDF and VMware Site Recovery Manager Completely automates storage-based disaster restart operations for VMware environments in SRDF topologies. • The Dell EMC SRDF Adapter enables VMware Site Recovery Manager to automate storage-based disaster restart operations in SRDF solutions. • Can address configurations in which data are spread across multiple storage arrays or SRDF groups. • Requires that the adapter is installed on each array to facilitate the discovery of arrays and to initiate failover operations. • Implemented with: ○ SRDF/S ○ SRDF/A ○ SRDF/Star ○ TimeFinder	

a. In some circumstances, using SRDF/S over distances greater than 200 km may be feasible. Contact your Dell EMC representative for more information.

SRDF multi-site solutions

The following table describes SRDF multi-site solutions.

Table 16. SRDF multi-site solutions

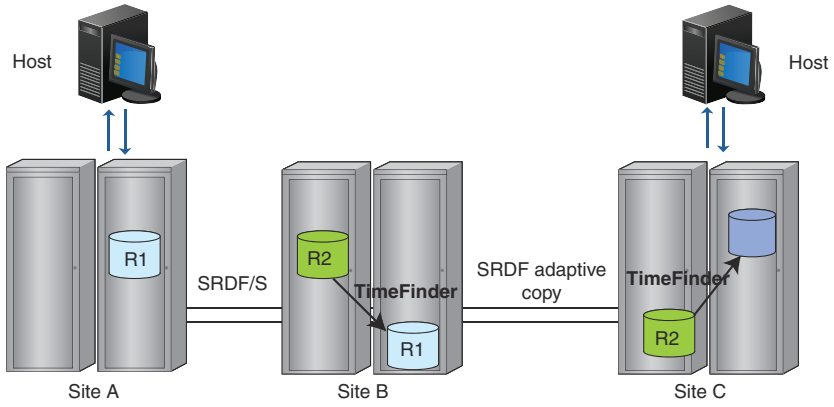
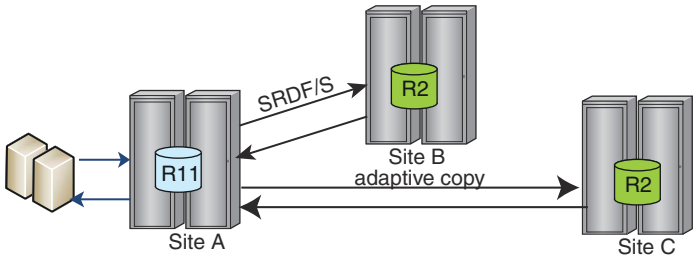
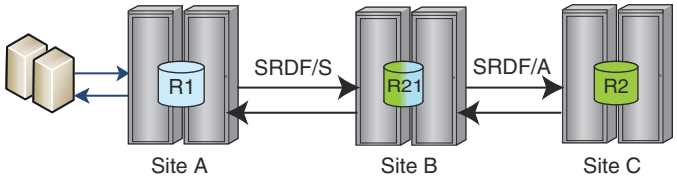
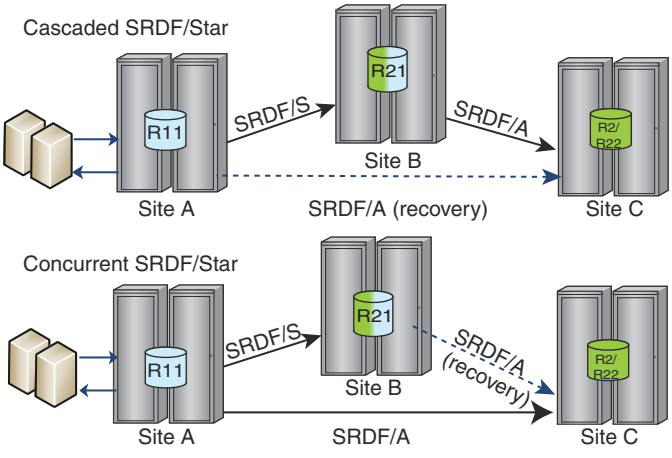
Solution highlights	Site topology
SRDF/Automated Replication (SRDF/AR) • Combines SRDF and TimeFinder to optimize bandwidth requirements and provide a long-distance disaster restart solution. • Operates in a 3-site environment that uses a combination of SRDF/S, SRDF/DM, and TimeFinder.	
Concurrent SRDF 3-site disaster recovery and advanced multi-site business continuity protection. • Data on the primary site is concurrently replicated to 2 secondary sites. • Replication to remote site can use SRDF/S, SRDF/A, or adaptive copy.	
Cascaded SRDF 3-site disaster recovery and advanced multi-site business continuity protection. Data on the primary site (Site A) is synchronously mirrored to a secondary site (Site B), and then asynchronously mirrored from the secondary site to a tertiary site (Site C).	
SRDF/Star 3-site data protection and disaster recovery configuration with zero data loss recovery, business continuity protection and disaster restart. • Available in 2 configurations: ○ Cascaded SRDF/Star ○ Concurrent SRDF/Star • Differential synchronization allows rapid reestablishment of mirroring among surviving sites in a multi-site disaster recovery implementation.	

Table 16. SRDF multi-site solutions (continued)

Solution highlights	Site topology
• Implemented using SRDF consistency groups (CG) with SRDF/S and SRDF/A.	

Interfamily compatibility

SRDF supports connectivity between different operating environments and arrays. Arrays running HYPERMAX OS can connect to legacy arrays running older operating environments. In mixed configurations where arrays are running different versions, SRDF features of the lowest version are supported.

VMAX All Flash arrays can connect to:

- PowerMax arrays running PowerMaxOS
- VMAX 250F, 450F, 850F, and 950F arrays running HYPERMAX OS
- VMAX 100K, 200K, and 400K arrays running HYPERMAX OS
- VMAX 10K, 20K, and 40K arrays running Enginuity 5876 with an Enginuity ePack

NOTE: When you connect between arrays running different operating environments, limitations may apply. Information about which SRDF features are supported, and applicable limitations for 2-site and 3-site solutions is in the *SRDF Interfamily Connectivity Information*.

This interfamily connectivity allows you to add the latest hardware platform/operating environment to an existing SRDF solution, enabling technology refreshes.

SRDF device pairs

An SRDF device pair is a logical device that is paired with another logical device that resides in a second array. The arrays are connected by SRDF links.

Encapsulated Data Domain devices that are used for Storage Direct cannot be part of an SRDF device pair.

R1 and R2 devices

An R1 device is the member of the device pair at the source (production) site. R1 devices are generally Read/Write accessible to the application host.

An R2 device is the member of the device pair at the target (remote) site. During normal operations, host I/O writes to the R1 device are mirrored over the SRDF links to the R2 device. In general, data on R2 devices is not available to the application host while the SRDF relationship is active. In SRDF synchronous mode, however, an R2 device can be in Read Only mode that allows a host to read from the R2.

In a typical environment:

- The application production host has Read/Write access to the R1 device.
- An application host connected to the R2 device has Read Only (Write Disabled) access to the R2 device.

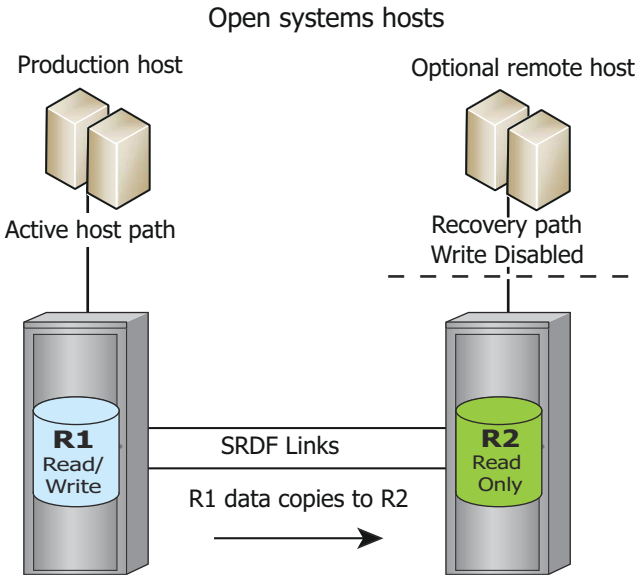


Figure 11. R1 and R2 devices

R11 devices

R11 devices operate as the R1 device for two R2 devices. Links to both R2 devices are active.

R11 devices are typically used in 3-site concurrent configurations where data on the R11 site is mirrored to two secondary (R2) arrays:

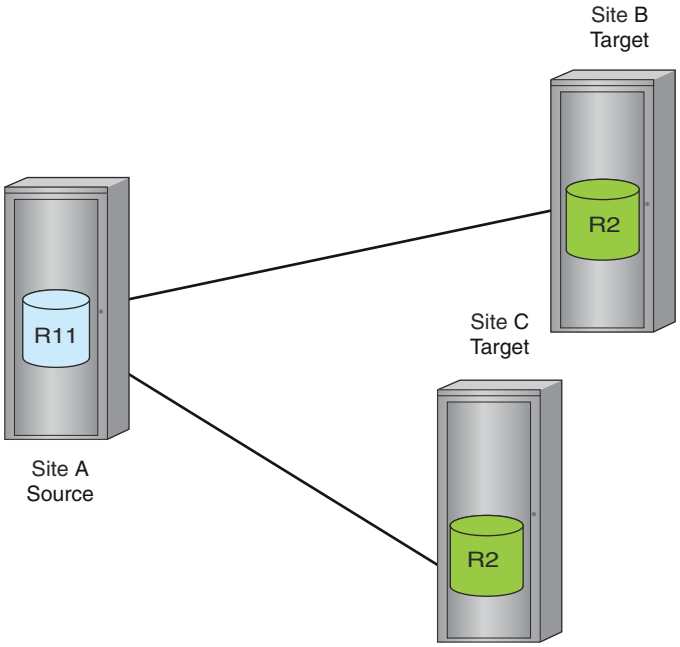


Figure 12. R11 device in concurrent SRDF

R21 devices

R21 devices have a dual role and are used in cascaded 3-site configurations where:

- Data on the R1 site is synchronously mirrored to a secondary (R21) site, and then
- Asynchronously mirrored from the secondary (R21) site to a tertiary (R2) site:

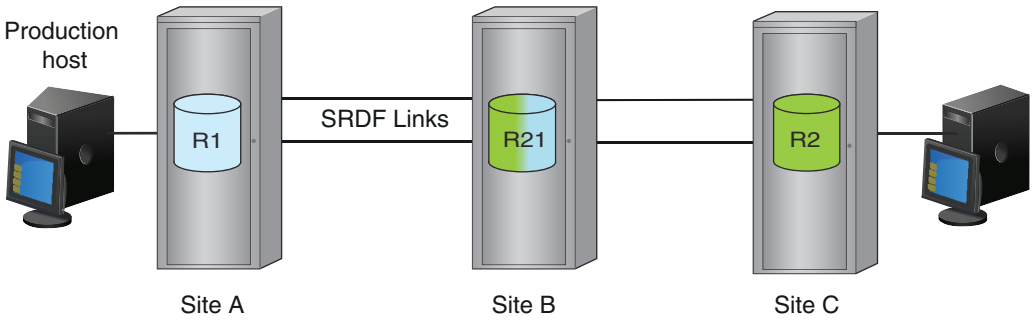


Figure 13. R21 device in cascaded SRDF

The R21 device acts as a R2 device that receives updates from the R1 device, and as a R1 device that sends updates to the R2 device.

When the R1->R21->R2 SRDF relationship is established, no host has write access to the R21 device.

In arrays that run Enginuity, the R21 device can be diskless. That is, it consists solely of cache memory and does not have any associated storage device. It acts purely to relay changes in the R1 device to the R2 device. This capability requires the use of thick devices. Systems that run PowerMaxOS or HYPERMAX OS contain thin devices only, so setting up a diskless R21 device is not possible on arrays running those environments.

R22 devices

R22 devices:

- Have two R1 devices, only one of which is active at a time.
- Are typically used in cascaded SRDF/Star and concurrent SRDF/Star configurations to decrease the complexity and time required to complete failover and failback operations.
- Let you recover without removing old SRDF pairs and creating new ones.

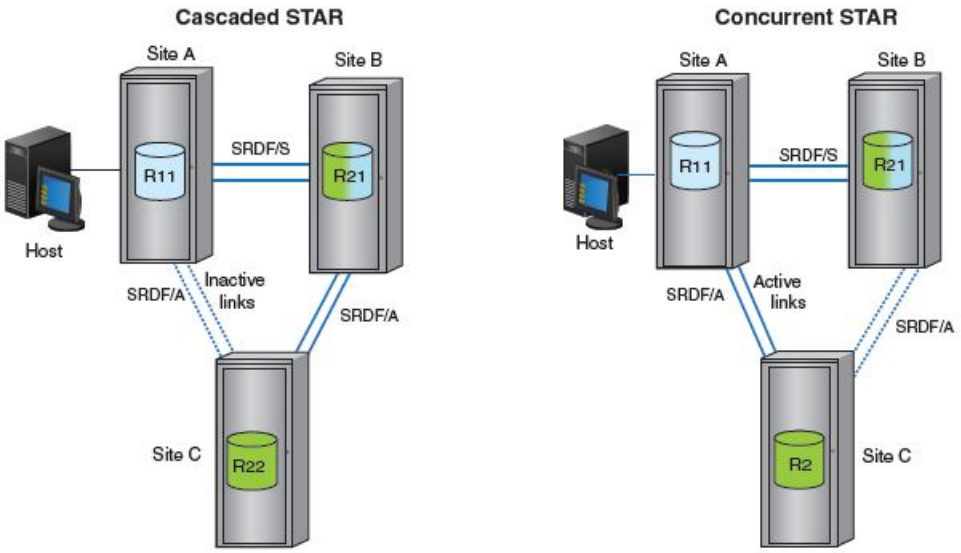


Figure 14. R22 devices in cascaded and concurrent SRDF/Star

Dynamic device personalities

SRDF devices can dynamically swap “personality” between R1 and R2. After a personality swap:

- The R1 in the device pair becomes the R2 device, and
- The R2 becomes the R1 device.

Swapping R1/R2 personalities allows the application to be restarted at the remote site without interrupting replication if an application fails at the production site. After a swap, the R2 side (now R1) can control operations while being remotely mirrored at the primary (now R2) site.

An R1/R2 personality swap is not supported:

- If the R2 device is larger than the R1 device.
- If the device to be swapped is participating in an active SRDF/A session.
- In SRDF/EDP topologies diskless R11 or R22 devices are not valid end states.
- If the device to be swapped is the target device of any TimeFinder or EMC Compatible flash operations.

SRDF modes of operation

The SRDF mode of operation determines:

- How R1 devices are remotely mirrored to R2 devices across the SRDF links
- How I/O operations are processed
- When the acknowledgment is returned to the application host that issued an I/O write command

In SRDF there are three principal modes:

- Synchronous
- Asynchronous
- Adaptive copy

Synchronous mode

Synchronous mode maintains a real-time mirror image of data between the R1 and R2 devices over distances up to 200 km (125 miles). Host data is written to both arrays in real time. The application host does not receive the acknowledgment until the data has been stored in the cache of both arrays.

Asynchronous mode

Asynchronous mode maintains a dependent-write consistent copy between the R1 and R2 device over unlimited distances. On receiving data from the application host, SRDF on the R1 side of the link writes that data to its cache. Also it batches the data received into *delta sets*. Delta sets are transferred to the R2 device in timed cycles. The application host receives the acknowledgment once data is successfully written to the cache on the R1 side.

Adaptive copy modes

Adaptive copy modes:

- Accumulate write requests that are destined for the R2 device on the R1 side, but not in cache memory.
- A background copy process sends the outstanding write requests to the R2 device.
- Allow the R1 and R2 devices to be out of synchronization by user-defined *maximum skew* value. Once the skew value is exceeded, SRDF transfers the batched data to the R2 device.
- Send the acknowledgment to the application host once the data is successfully written to cache on the R1 side.

Unlike asynchronous mode, the adaptive copy modes do not guarantee a dependent-write copy of data on the R2 devices.

SRDF groups

An SRDF group defines the logical relationship between SRDF devices and directors on both sides of an SRDF link.

Group properties

The properties of an SRDF group are:

- Label (name)
- Set of ports on the local array used to communicate over the SRDF links
- Set of ports on the remote array used to communicate over the SRDF links
- Local group number
- Remote group number
- One or more pairs of devices

The devices in the group share the ports and associated CPU resources of the port's directors.

Types of group

There are two types of SRDF group:

- Static: which are defined in the local array's configuration file.
- Dynamic: which are defined using SRDF management tools and their properties that are stored in the array's cache memory.

On arrays running PowerMaxOS or HYPERMAX OS all SRDF groups are dynamic.

Director boards, links, and ports

SRDF links are the logical connections between SRDF groups and their ports. The ports are physically connected by cables, routers, extenders, switches and other network devices.

NOTE: Two or more SRDF links per SRDF group are required for redundancy and fault tolerance.

The relationship between the resources on a director (CPU cores and ports) varies depending on the operating environment.

HYPERMAX OS

On arrays running HYPERMAX OS:

- The relationship between the SRDF emulation and resources on a director is configurable:
  - One director/multiple CPU cores/multiple ports
  - Connectivity (ports in the SRDF group) is independent of compute power (number of CPU cores). You can change the amount of connectivity without changing compute power.
- Each director has up to 16 front end ports, any or all of which can be used by SRDF. Both the SRDF Gigabit Ethernet and SRDF Fibre Channel emulations can use any port.
- The data path for devices in an SRDF group is not fixed to a single port. Instead, the path for data is shared across all ports in the group.

Mixed configurations: HYPERMAX OS and Enginuity 5876

For configurations where one array is running Enginuity 5876, and the other array is running HYPERMAX OS, the following rules apply:

- On the 5876 side, an SRDF group can have the full complement of directors, but no more than 16 ports on the HYPERMAX OS side.
- You can connect to 16 directors using one port each, 2 directors using 8 ports each or any other combination that does not exceed 16 per SRDF group.

SRDF consistency

Many applications, especially database systems, use dependent write logic to ensure data integrity. That is, each write operation must complete successfully before the next can begin. Without write dependency, write operations could get out of sequence resulting in irrecoverable data loss.

SRDF implements write dependency using the *consistency group* (also known as SRDF/CG). A consistency group consists of a set of SRDF devices that use write dependency. For each device in the group, SRDF ensures that write operations propagate to the corresponding R2 devices in the correct order.

However, if the propagation of any write operation to any R2 device in the group cannot complete, SRDF suspends propagation to all group's R2 devices. This suspension maintains the integrity of the data on the R2 devices. While the R2 devices are unavailable, SRDF continues to store write operations on the R1 devices. It also maintains a list of those write operations in their time order. When all R2 devices in the group become available, SRDF propagates the outstanding write operations, in the correct order, for each device in the group.

SRDF/CG is available for both SRDF/S and SRDF/A.

Data migration

Data migration is the one-time movement of data from one array to another. Once the movement is complete, the data is accessed from the secondary array. A common use of migration is to replace an older array with a new one.

Dell EMC support personnel can assist with the planning and implementation of migration projects.

SRDF multisite configurations enable migration to occur in any of these ways:

- Replace R2 devices.
- Replace R1 devices.
- Replace both R1 and R2 devices simultaneously.

For example, this diagram shows the use of concurrent SRDF to replace the secondary (R2) array in a 2-site configuration:

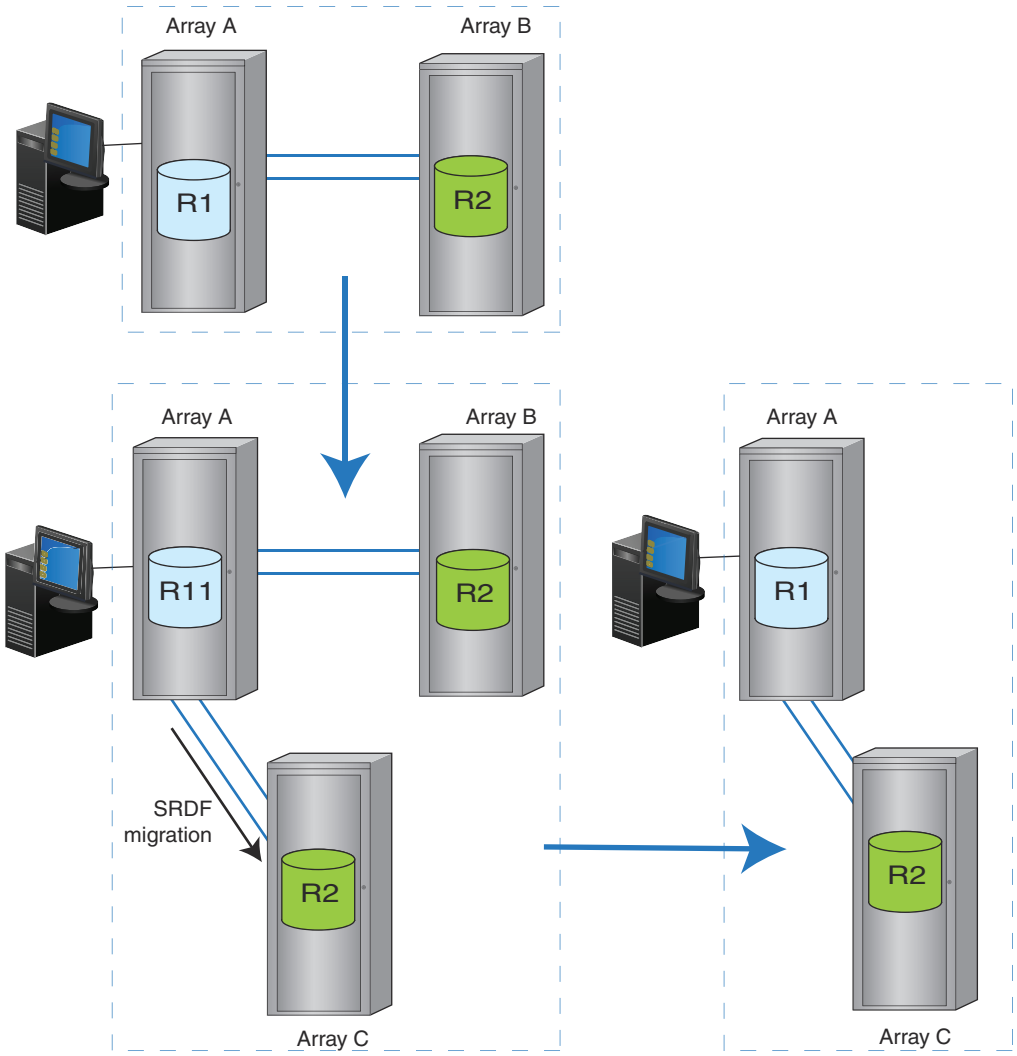


Figure 15. Migrating data and removing a secondary (R2) array

Here:

- The top section of the diagram shows the original, 2-site configuration.
- The lower left section of the diagram shows the interim, 3-site configuration with data being copied to two secondary arrays.
- The lower right section of the diagram shows the final, 2-site configuration where the new secondary array has replaced the original one.

The *Dell EMC SRDF Introduction* contains more information about using SRDF to migrate data.

More information

Here are other Dell EMC documents that contain more information about the use of SRDF in replication and migration:

SRDF Introduction

SRDF and NDM Interfamily Connectivity Information

SRDF/Cluster Enabler Plug-in Product Guide

Using the Dell EMC Adapter for VMWare Site Recovery Manager Technical Book

Dell EMC SRDF Adapter for VMware Site Recovery Manager Release Notes

SRDF/Metro

In traditional SRDF configurations, only the R1 devices are Read/Write accessible to the application hosts. The R2 devices are Read Only and Write Disabled.

In SRDF/Metro configurations, however:

- Both the R1 and R2 devices are Read/Write accessible to the application hosts.
- Application hosts can write to both the R1 and R2 side of the device pair.
- R2 devices assume the same external device identity as the R1 devices. The identity includes the device geometry and device WWN.

This shared identity means that R1 and R2 devices appear to application hosts as a single, virtual device across two arrays.

Deployment options

SRDF/Metro can be deployed in either a single, multipathed host environment or in a clustered host environment:

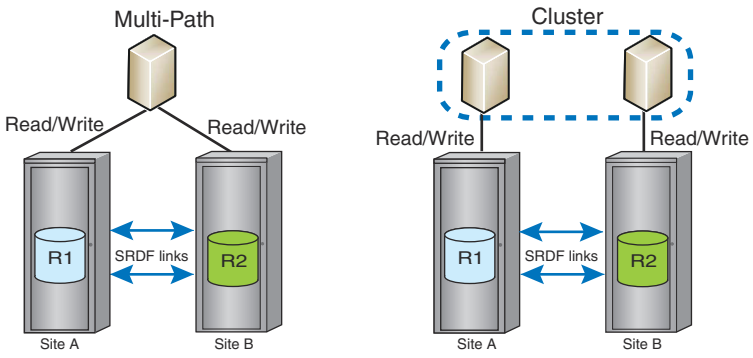


Figure 16. SRDF/Metro

Hosts can read and write to both the R1 and R2 devices:

- In a single host configuration, a single host issues I/O operations. Multipathing software directs parallel reads and writes to each array.
- In a clustered host configuration, multiple hosts issue I/O operations. Those hosts access both sides of the SRDF device pair. Each cluster node has dedicated access to one of the storage arrays.
- In both configurations, writes to the R1 and R2 devices are synchronously copied to the paired device in the other array. SRDF/Metro software resolves any write conflicts to maintain consistent images on the SRDF device pairs.

SRDF/Metro Resilience

If either of the devices in a SRDF/Metro configuration become Not Ready, or connectivity between the devices is lost, SRDF/Metro must decide which side remains available to the application host. There are two mechanisms that SRDF/Metro can use : Device Bias and Witness.

Device Bias

Device pairs for SRDF/Metro are created with a *bias* attribute. By default, the create pair operation sets the bias to the R1 side of the pair. That is, if a device pair becomes Not Ready (NR) on the SRDF link, the R1 (bias side) remains accessible to the hosts, and the R2 (nonbias side) becomes inaccessible. However, if there is a failure on the R1 side, the host loses all connectivity to the device pair. The Device Bias method cannot make the R2 device available to the host.

Witness

A witness is a third party that mediates between the two sides of a SRDF/Metro pair to help:

- Decide which side remains available to the host

- Avoid a "split brain" scenario when both sides attempt to remain accessible to the host despite the failure

The witness method allows for intelligently choosing on which side to continue operations when the bias-only method may not result in continued host availability to a surviving, nonbiased array.

There are two forms of the Witness mechanism:

- **Array Witness:** The operating environment of a third array is the mediator.
- **Virtual Witness (vWitness):** A daemon running on a separate, virtual machine is the mediator.

When both sides run PowerMaxOS 5978 SRDF/Metro takes these criteria into account when selecting the side to remain available to the hosts (in priority order):

1. The side that has connectivity to the application host (requires PowerMaxOS 5978.444.444or later)
2. The side that has a SRDF/A DR leg
3. Whether the SRDF/A DR leg is synchronized
4. The side that has more than 50% of the RA or FA directors that are available
5. The side that is currently the bias side

The first of these criteria that one array has, and the other does not, stops the selection process. The side with the matched criteria is the preferred winner.

Disaster recovery facilities

Devices in SRDF/Metro groups can simultaneously be in other groups that replicate data to a third, disaster recovery site. There are two replication solutions. The number available in any SRDF/Metro configuration depends on the version of the operating environment that the participating arrays run:

- Highly-available disaster recovery – in configurations that consist of arrays that run PowerMaxOS 5978.669.669 and later
- Independent disaster recovery – in configurations that run all supported versions of PowerMaxOS 5978 and HYPERMAX OS 5977

Highly available disaster recovery (SRDF/Metro Smart DR)

SRDF/Metro Smart DR maintains a single, disaster recovery (DR) copy of the data in a SRDF/Metro pair on a third, remote array. This diagram shows the SRDF/Metro Smart DR configuration:

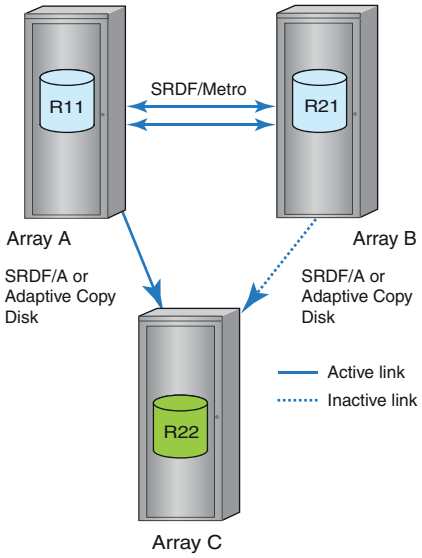


Figure 17. SRDF/Metro Smart DR

Notice that the device names differ from a standard SRDF/Metro configuration. This difference reflects the change in the device functions when SRDF/Metro Smart DR is in operation. For instance, as the diagram shows the R1 side of the SRDF/Metro on Array A now has the name R11, because it is the R1 device to both the:

- R21 device on Array B in the SRDF/Metro configuration
- R22 device on Array C in the SRDF/Metro Smart DR configuration

Arrays A and B both have SRDF/Asynchronous or Adaptive Copy Disk connections to the DR array (Array C). However, only one of those connections is active at a time (in this example the connection between Array A and Array C). The two SRDF/A connections are known as the active and standby connections.

If a problem prevents Array A replicating data to Array C, the standby link between Array B and Array C becomes active and replication continues. Array A and Array B keep track of the data replicated to Array C to enable replication and avoid data loss.

Independent disaster recovery

Devices in SRDF/Metro groups can simultaneously be part of device groups that replicate data to a third, disaster-recovery site.

Either or both sides of the Metro region can be replicated. You can choose which ever configuration that suits your business needs. The following diagram shows the possible configurations:

NOTE: When the SRDF/Metro session is using a witness, the R1 side of the Metro pair can change based on the witness determination of the preferred side.

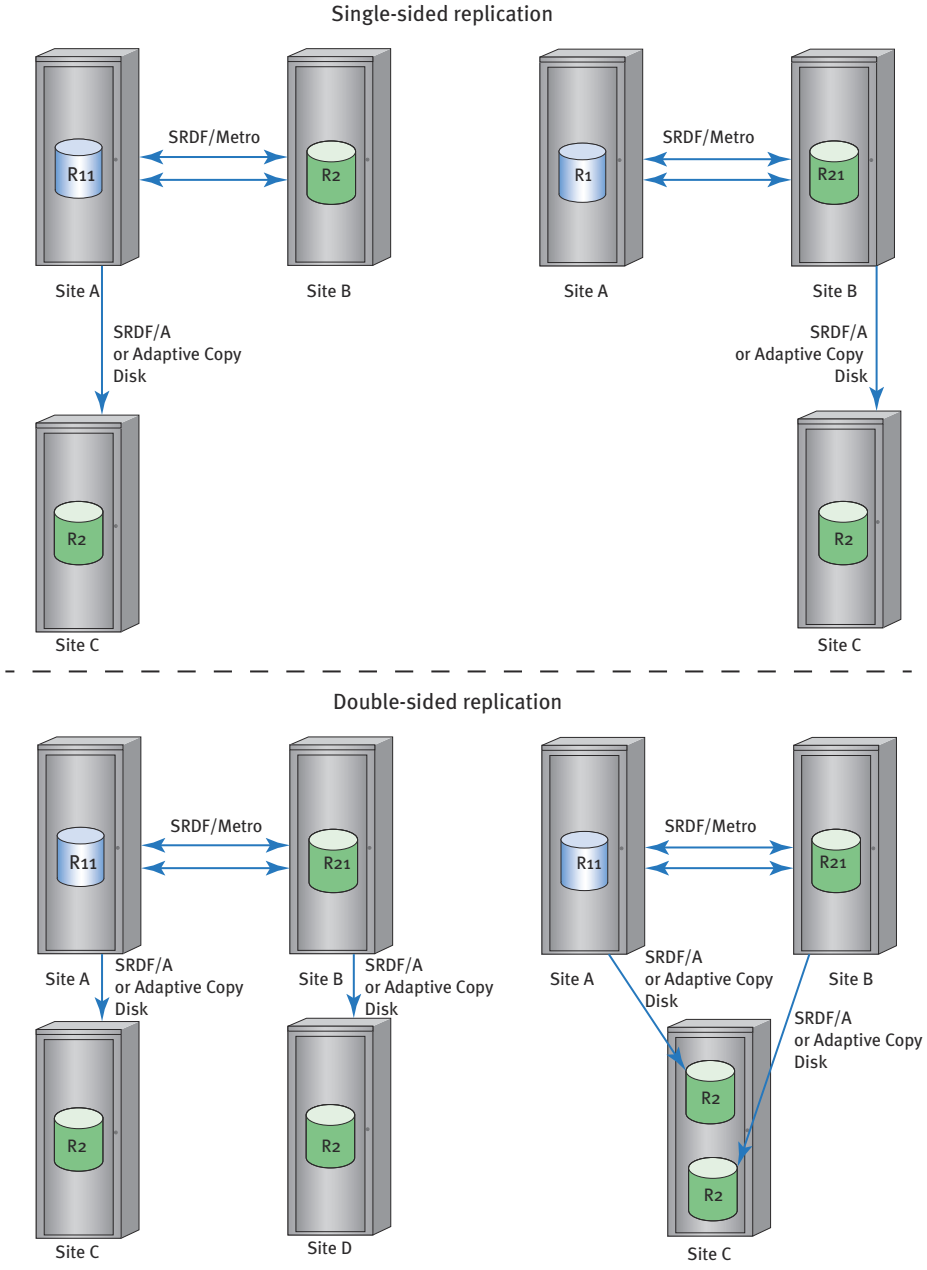


Figure 18. Disaster recovery for SRDF/Metro

The device types differ from a stand-alone SRDF/Metro configuration. This difference reflects the change in the devices' function when disaster recovery facilities are in place. For instance, when the R2 side is replicated to a disaster recovery site, its type changes to R21 because it is both the:

- R2 device in the SRDF/Metro configuration
- R1 device in the disaster-recovery configuration

When an SRDF/Metro uses a witness for resilience protection, the two sides periodically renegotiate the winning and losing sides. If the winning and losing sides switch as a result of renegotiation:

- An R11 device becomes an R21 device. That device was the R1 device for both the SRDF/Metro and disaster recovery configurations. Now the device is the R2 device of the SRDF/Metro configuration but it remains the R1 device of the disaster recovery configuration.
- An R21 device becomes and R11 device. That device was the R2 device in the SRDF/Metro configuration and the R1 device of the disaster recovery configuration. Now the device is the R1 device of both the SRDF/Metro and disaster recovery configurations.

More information

Here are other Dell EMC documents that contain more information on SRDF/Metro:

SRDF Introduction

SRDF/Metro vWitness Configuration Guide

SRDF Interfamily Connectivity Information

RecoverPoint

HYPERMAX OS 5977.1125.1125 introduced support for RecoverPoint on VMAX storage arrays. RecoverPoint is a comprehensive data protection solution designed to provide production data integrity at local and remote sites. RecoverPoint also provides the ability to recover data from a point in time using journaling technology.

The primary reasons for using RecoverPoint are:

- Remote replication to heterogeneous arrays
- Protection against Local and remote data corruption
- Disaster recovery
- Secondary device repurposing
- Data migrations

RecoverPoint systems support local and remote replication of data that applications are writing to SAN-attached storage. The systems use existing Fibre Channel infrastructure to integrate seamlessly with existing host applications and data storage subsystems. For remote replication, the systems use existing Fibre Channel connections to send the replicated data over a WAN, or use Fibre Channel infrastructure to replicate data aysnchronously. The systems provide failover of operations to a secondary site in the event of a disaster at the primary site.

Previous implementations of RecoverPoint relied on a splitter to track changes made to protected volumes. The current implementation relies on a cluster of RecoverPoint nodes, provisioned with one or more RecoverPoint storage groups, leveraging SnapVX technology, on the storage array. Volumes in the RecoverPoint storage groups are visible to all the nodes in the cluster, and available for replication to other storage arrays.

RecoverPoint allows data replication of up to 8,000 LUNs for each RecoverPoint cluster and up to eight different RecoverPoint clusters attached to one array. Supported array types include PowerMax, VMAX All Flash, VMAX3, VMAX, VNX, VPLEX, and XtremIO.

RecoverPoint is licensed and sold separately. For more information about RecoverPoint and its capabilities see the *Dell EMC RecoverPoint Product Guide*.

Remote replication using eNAS

File Auto Recovery (FAR) allows you to manually failover or move a virtual Data Mover (VDM) from a source eNAS system to a destination eNAS system. The failover or move leverages block-level SRDF synchronous replication, so it incurs zero data loss in the event of an unplanned operation. This feature consolidates VDMs, file systems, file system checkpoint schedules, CIFS servers, networking, and VDM configurations into their own separate pools. This feature works for a recovery where the source is unavailable. For recovery support in the event of an unplanned failover, there is an option to recover and clean up the source system and make it ready as a future destination.

The manually initiated failover and reverse operations can be performed using EMC File Auto Recovery Manager (FARM). FARM can automatically failover a selected sync-replicated VDM on a source eNAS system to a destination eNAS system. FARM can also monitor sync-replicated VDMs and trigger automatic failover based on Data Mover, File System, Control Station, or IP network unavailability that would cause the NAS client to lose access to data.

8

Blended local and remote replication

This chapter introduces TimeFinder integration with SRDF.

Topics:

- Integration of SRDF and TimeFinder
- R1 and R2 devices in TimeFinder operations
- SRDF/AR
- TimeFinder and SRDF/A
- TimeFinder and SRDF/S

Integration of SRDF and TimeFinder

You can use TimeFinder and SRDF products to complement each other when you require both local and remote replication. For example, you can use TimeFinder to create local gold copies of SRDF devices for recovery operations and for testing disaster recovery solutions.

The key benefits of TimeFinder integration with SRDF include:

- Remote controls simplify automation—Use Dell EMC host-based control software to transfer commands across the SRDF links. A single command from the host to the primary array can initiate TimeFinder operations on both the primary and secondary arrays.
- Consistent data images across multiple devices and arrays—SRDF/CG guarantees that a dependent-write consistent image of production data on the R1 devices is replicated across the SRDF links.

You can use TimeFinder/CG in an SRDF configuration to create dependent-write consistent local and remote images of production data across multiple devices and arrays.

NOTE: Using a SRDF/A single session guarantees dependent-write consistency across the SRDF links and does not require SRDF/CG. SRDF/A MSC mode requires host software to manage consistency among multiple sessions.

NOTE: Some TimeFinder operations are not supported on devices that SRDF protects. The *Dell EMC Solutions Enabler TimeFinder SnapVX CLI User Guide* has further information.

The rest of this chapter summarizes the ways of integrating SRDF and TimeFinder.

R1 and R2 devices in TimeFinder operations

You can use TimeFinder to create local replicas of R1 and R2 devices. The following rules apply:

- You can use R1 devices and R2 devices as TimeFinder source devices.
- R1 devices can be the target of TimeFinder operations as long as there is no host accessing the R1 during the operation.
- R2 devices can be used as TimeFinder target devices if SRDF replication is not active (writing to the R2 device). To use R2 devices as TimeFinder target devices, first suspend the SRDF replication session.

SRDF/AR

SRDF/AR combines SRDF and TimeFinder to provide a long-distance disaster restart solution. SRDF/AR can be deployed over 2 or 3 sites:

- In 2-site configurations, SRDF/DM is deployed with TimeFinder.
- In 3-site configurations, SRDF/DM is deployed with a combination of SRDF/S and TimeFinder.

The time to create the new replicated consistent image is determined by the time that it takes to replicate the deltas.

SRDF/AR 2-site configurations

The following image shows a 2-site configuration where the production device (R1) on the primary array (Site A) is also a TimeFinder target device:

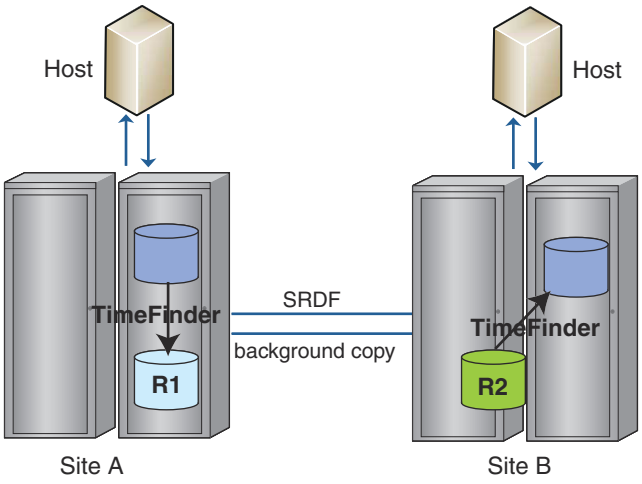


Figure 19. SRDF/AR 2-site solution

In this configuration, data on the SRDF R1/TimeFinder target device is replicated across the SRDF links to the SRDF R2 device.

The SRDF R2 device is also a TimeFinder source device. TimeFinder replicates this device to a TimeFinder target device. You can map the TimeFinder target device to the host connected to the secondary array at Site B.

In a 2-site configuration, SRDF operations are independent of production processing on both the primary and secondary arrays. You can utilize resources at the secondary site without interrupting SRDF operations.

Use SRDF/AR 2-site configurations to:

- Reduce required network bandwidth using incremental resynchronization between the SRDF target sites.
- Reduce network cost and improve resynchronization time for long-distance SRDF implementations.

SRDF/AR 3-site configurations

SRDF/AR 3-site configurations provide a zero data loss solution at long distances in the event that the primary site is lost.

The following image shows a 3-site configuration where:

- Site A and Site B are connected using SRDF in synchronous mode.
- Site B and Site C are connected using SRDF in adaptive copy mode.

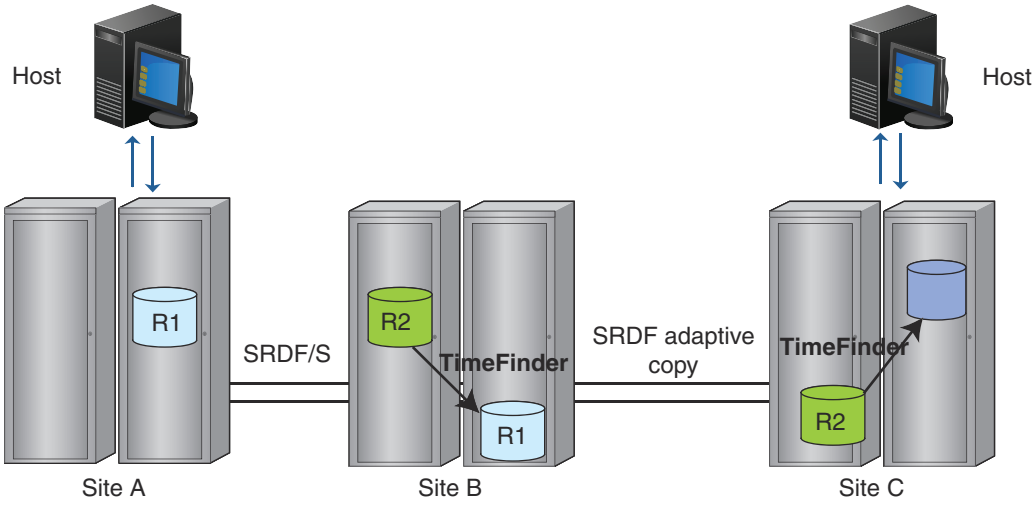


Figure 20. SRDF/AR 3-site solution

If Site A (primary site) fails, the R2 device at Site B provides a restartable copy with zero data loss. Site C provides an asynchronous restartable copy.

If both Site A and Site B fail, the device at Site C provides a restartable copy with controlled data loss. The amount of data loss is a function of the replication cycle time between Site B and Site C.

SRDF and TimeFinder control commands to R1 and R2 devices for all sites can be issued from Site A. No controlling host is required at Site B.

Use SRDF/AR 3-site configurations to:

- Reduce required network bandwidth using incremental resynchronization between the secondary SRDF target site and the tertiary SRDF target site.
- Reduce network cost and improve resynchronization time for long-distance SRDF implementations.
- Provide disaster recovery testing, point-in-time backups, decision support operations, third-party software testing, and application upgrade testing or the testing of new applications.

Requirements/restrictions

In a 3-site SRDF/AR multi-hop configuration, SRDF/S host I/O to Site A is not acknowledged until Site B has acknowledged it. This can cause a delay in host response time.

TimeFinder and SRDF/A

In SRDF/A solutions, device pacing:

- Prevents cache utilization bottlenecks when the SRDF/A R2 devices are also TimeFinder source devices.
- Allows R2 or R22 devices at the middle hop to be used as TimeFinder source devices.

NOTE: Device write pacing is not required in configurations that include HYPERMAX OS and Enginuity 5876.

TimeFinder and SRDF/S

SRDF/S solutions support any type of TimeFinder copy sessions running on R1 and R2 devices as long as the conditions described in R1 and R2 devices in TimeFinder operations on page 76 are met.

9

Data Migration

This chapter introduces data migration solutions.

Topics:

- Overview
- Data migration for open systems
- Data migration for mainframe

Overview

Data migration is a one-time movement of data from one array (the source) to another array (the target). Typical examples are data center refreshes where data is moved from an old array after which the array is retired or re-purposed. Data migration is *not* data movement due to replication (where the source data is accessible after the target is created) or data mobility (where the target is continually updated).

After a data migration operation, applications that access the data reference it at the new location.

To plan a data migration, consider the potential impact on your business, including the:

- Type of data to be migrated
- Site location(s)
- Number of systems and applications
- Amount of data to be moved
- Business needs and schedules

PowerMaxOS provides migration facilities for:

- Open systems
- IBM System i
- Mainframe

Data migration for open systems

The data migration features available for open system environments are:

- Non-disruptive migration
- Open Replicator
- PowerPath Migration Enabler
- Data migration using SRDF/Data Mobility
- Space and zero-space reclamation

Non-Disruptive Migration overview

Non-Disruptive Migration (NDM) provides a method for migrating data from a source array to a target array without application host downtime across a metro distance, typically within a data center. For NDM array operating system version support, please consult the NDM support matrix, or the *SRDF Interfamily Connectivity Guide*.

If regulatory or business requirements for DR (disaster recovery) dictate the use of SRDF/S during migration, contact Dell EMC for required ePacks for SRDF/S configuration.

The NDM operations involved in a typical migration are:

- Environment setup – Configures source and target array infrastructure for the migration process.
- Create – Duplicates the application storage environment from source array to target array.
- Cutover – Switches the application data access form the source array to the target array and duplicates the application data on the source array to the target array.
- Commit – Removes application resources from the source array and releases the resources used for migration. Application permanently runs on the target array.
- Enviroment remove –Removes the migration infrastructure created by the environmental setup.

Some key features of NDM are:

- Simple process for migration:
  1. Select storage group to migrate.
  2. Create the migration session.
  3. Discover paths to the host.
  4. Cutover or readytgt storage group to VMAX3 or VMAX All Flash array.
  5. Monitor for synchronization to complete.
  6. Commit the migration.
- Allows for inline compression on VMAX All Flash array during migration.
- Maintains snapshot and disaster recovery relationships on source array, but are not migrated.
- Allows for non-disruptive revert to source array.
- Allows up to 50 concurrent migration sessions.
- Requires no license since it is part of HYPERMAX OS.
- Requires no additional hardware in the data path.

The following graphic shows the connections required between the host (single or cluster) and the source and target array, and the SRDF connection between the two arrays.

Figure 21. Non-Disruptive Migration zoning



The App host connection to both arrays uses FC, and the SRDF connection between arrays uses FC or GigE .

The migration controls should be run from a control host and not from the application host. The control host should have visibility to both the source array and target array.

The following devices and components are not supported with NDM:

- CKD devices
- eNAS data
- Storage Direct, FAST.X relationships and associated data

Environmental requirements for Non-Disruptive Migration

The following configurations are required for a successful data migration:

Array configuration

- The target array must be running HYPERMAX OS 5977.811.784 or higher. This includes VMAX3 Family arrays and VMAX All Flash arrays.
- The source array must be a VMAX array running Enginuity 5876 with required ePack (contact Dell EMC for required ePack).
- SRDF is used for data migration, so zoning of SRDF ports between the source and target arrays is required. Note that an SRDF license is not required, as there is no charge for NDM.
- The NDM RDF group is configured with a minimum of two paths on different directors for redundancy and fault tolerance. If more paths are found up to eight paths will be configured.
- If SRDF is not normally used in the migration environment, it may be necessary to install and configure RDF directors and ports on both the source and target arrays and physically configure SAN connectivity.

Host configuration

- The migration controls should be run from a control host and not from the application host.
- Both the source and the target array have be visible to the controlling host that runs the migration commands.

Pre-migration rules and restrictions for NDM

In addition to general configuration requirements of the migration environment, the following rules and restrictions apply before starting a migration:

- A storage group is the data container that is migrated, and the requirements that apply to the group and its devices are:
  - Storage groups must have masking views. All devices in the group on the source array must be visible only through a masking view. Each device must be mapped only to a port that is part of the masking view.

- Multiple masking views on a storage group using the same initiator group are valid only when:
  - Port groups on the target array exist for each masking view, and
  - Ports in the port group are selected
- A storage group must be a parent or stand-alone group. A child storage group with a masking view on the child group is not supported.
- If the selected storage group is a parent, its child groups are also migrated.
- The names of storage groups and their children (if any) must not exist on the target array.
- Gatekeeper devices in a storage group are not migrated.

- Devices cannot:
  - Have a mobility ID
  - Have a nonbirth identity, when the source array runs Enginuity 5876
  - Have the BCV attribute
  - Be encapsulated
  - Be RP devices
  - Be Data Domain devices
  - Be vVOL devices
  - Be R2 or Concurrent SRDF devices
  - Be masked to FCoE (in the case of source arrays), iSCSI, non-ACLX, or NVMe over FC ports
  - Be part of another data migration operation
  - Be part of an ORS relationship
  - Be in other masked storage groups
  - Have a device status of Not Ready
- Devices can be part of TimeFinder sessions.
- Devices can act as R1 devices but cannot be part of a SRDF/Star or SRDF/SQAR configuration.
- The names of masking groups to migrate must not exist on the target array.
- The names of initiator groups to migrate may exist on the target array. However, the aggregate set of host initiators in the initiator groups that the masking groups use must be the same. Also, the effective ports flags on the host initiators must have the same setting on both arrays.
- The names of port groups to migrate may exist on the target array, as long as the groups on the target array are in the logging history table for at least one port.
- The status of the target array must be as follows:
  - If a target-side Storage Resource Pool (SRP) is specified for the migration, that SRP must exist on the target array.
  - The SRP to be used for target-side storage must have enough free capacity to support the migration.
  - The target side must be able to support the additional devices required to receive the source-side data.
  - All initiators provisioned to an application on the source array must also be logged into ports on the target array.

Migration infrastructure - RDF device pairing

RDF device pairing is done during the create operation, with the following actions occurring on the device pairs.

- NDM creates RDF device pairs, in a DM RDF group, between devices on the source array and the devices on the target array.
- Once device pairing is complete NDM controls the data flow between both sides of the migration process.
- Once the migration is complete, the RDF pairs are deleted when the migration is committed.
- Other RDF pairs may exist in the DM RDF group if another migration is still in progress.

Due to differences in device attributes between the source and target array, the following rules apply during migration:

- Any source array device that has an odd number of cylinders is migrated to a device on the target array that has Geometry Compatibility Mode (GCM).
- Any source array meta device is migrated to a non-meta device on the target array.

Once the copying of data to the target array has begun, the target devices can have SRDF mirrors (R2 devices) added to them for remote replication. However, the mirror devices cannot be:

- Enabled for MSC or Synchronous SRDF Consistency
- Part of a SRDF/Star, SRDF/SQAR, or SRDF/Metro configuration

Open Replicator

Open Replicator enables copying data (full or incremental copies) from qualified arrays within a storage area network (SAN) infrastructure to or from arrays running HYPERMAX OS. Open Replicator uses the Solutions Enabler SYMCLI `symrcopy` command.

Use Open Replicator to migrate and back up/archive existing data between arrays running HYPERMAX OS. Open Replicator uses the Solutions Enabler SYMCLI `symrcopy` and third-party storage arrays within the SAN infrastructure without interfering with host applications and ongoing business operations.

Use Open Replicator to:

- Pull from source volumes on qualified remote arrays to a volume on an array running HYPERMAX OS. Open Replicator uses the Solutions Enabler SYMCLI `symrcopy`.
- Perform online data migrations from qualified storage to an array running HYPERMAX OS. Open Replicator uses the Solutions Enabler SYMCLI `symrcopy` with minimal disruption to host applications.

NOTE: Open Replicator cannot copy a volume that is in use by TimeFinder.

Open Replicator operations

Open Replicator uses the following terminology:

Control

The recipent array and its devices are referred to as the control side of the copy operation.

Remote

The donor Dell EMC arrays or third-party arrays on the SAN are referred to as the remote array/devices.

Hot

The Control device is Read/Write online to the host while the copy operation is in progress.

NOTE: Hot push operations are not supported on arrays running HYPERMAX OS. Open Replicator uses the Solutions Enabler SYMCLI `symrcopy`.

Cold

The Control device is Not Ready (offline) to the host while the copy operation is in progress.

Pull

A pull operation copies data to the control device from the remote device(s).

Push

A push operation copies data from the control device to the remote device(s).

Pull operations

On arrays running HYPERMAX OS, Open Replicator uses the Solutions Enabler SYMCLI `symrcopy` support up to 4096 pull sessions.

For pull operations, the volume can be in a live state during the copy process. The local hosts and applications can begin to access the data as soon as the session begins, even before the data copy process has completed.

These features enable rapid and efficient restoration of remotely vaulted volumes and migration from other storage platforms.

Copy on First Access ensures the appropriate data is available to a host operation when it is needed. The following image shows an Open Replicator hot pull.

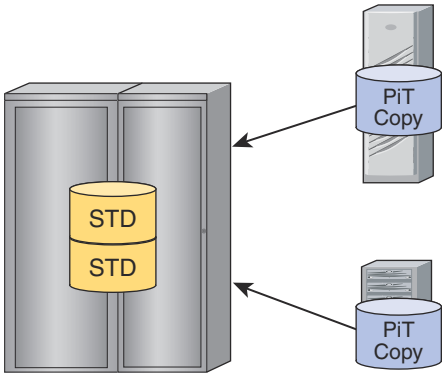


Figure 22. Open Replicator hot (or live) pull

The pull can also be performed in cold mode to a static volume. The following image shows an Open Replicator cold pull.

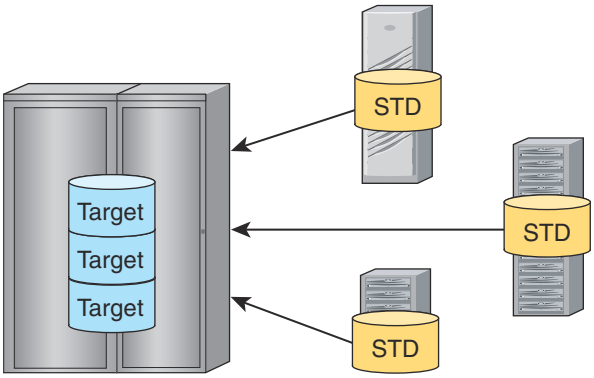


Figure 23. Open Replicator cold (or point-in-time) pull

PowerPath Migration Enabler

Dell EMC PowerPath is host-based software that provides automated data path management and load-balancing capabilities for heterogeneous server, network, and storage deployed in physical and virtual environments. PowerPath includes a migration tool called PowerPath Migration Enabler (PPME). PPME enables non-disruptive or minimally disruptive data migration between storage systems or within a single storage system.

PPME allows applications continued data access throughout the migration process. PPME integrates with other technologies to minimize or eliminate application downtime during data migration.

PPME works in conjunction with underlying technologies, such as Open Replicator, SnapVX, and Host Copy.

NOTE: PowerPath Multipathing must be installed on the host machine.

The following documentation provides additional information:

- *Dell EMC Support Matrix PowerPath Family Protocol Support*
- *Dell EMC PowerPath Migration Enabler User Guide*

Data migration using SRDF/Data Mobility

SRDF/Data Mobility (DM) uses SRDF's adaptive copy mode to transfer large amounts of data without impact to the host.

SRDF/DM supports data replication or migration between two or more arrays running HYPERMAX OS. Adaptive copy mode enables applications using the primary volume to avoid propagation delays while data is transferred to the remote site. SRDF/DM can be used for local or remote transfers.

Data migration on page 69 has a more information about using SRDF to migrate data.

Space and zero-space reclamation

Space reclamation reclaims unused space following a replication or migration activity from a regular device to a thin device in which software tools, such as Open Replicator and Open Migrator, copied-all-zero, unused space to a target thin volume.

Space reclamation deallocates data chunks that contain all zeros. Space reclamation is most effective for migrations from standard, fully provisioned devices to thin devices. Space reclamation is non-disruptive and can be executed while the targeted thin device is fully available to operating systems and applications.

Zero-space reclamations provides instant zero detection during Open Replicator and SRDF migration operations by reclaiming all-zero space, including both host-unwritten extents (or chunks) and chunks that contain all zeros due to file system or database formatting.

Solutions Enabler and Unisphere can be used to initiate and monitor the space reclamation process.

Data migration for mainframe

For mainframe environments, z/OS Migrator provides non-disruptive migration from any vendor storage to VMAX All Flash or VMAX arrays. z/OS Migrator can also migrate data from one VMAX or VMAX All Flash array to another VMAX or VMAX All Flash array. With z/OS Migrator, you can:

- Introduce new storage subsystem technologies with minimal disruption of service.
- Reclaim z/OS UCBs by simplifying the migration of datasets to larger volumes (combining volumes).
- Facilitate data migration while applications continue to run and fully access data being migrated, eliminating application downtime usually required when migrating data.
- Eliminate the need to coordinate application downtime across the business, and eliminate the costly impact of such downtime on the business.
- Improve application performance by facilitating the relocation of poor performing datasets to lesser used volumes/storage arrays.
- Ensure all metadata always accurately reflects the location and status of datasets being migrated.

Refer to the *Dell EMC z/OS Migrator Product Guide* for detailed product information.

Volume migration using z/OS Migrator

EMC z/OS Migrator is a host-based data migration facility that performs traditional volume migrations as well as host-based volume mirroring. Together, these capabilities are referred to as the volume mirror and migrator functions of z/OS Migrator.

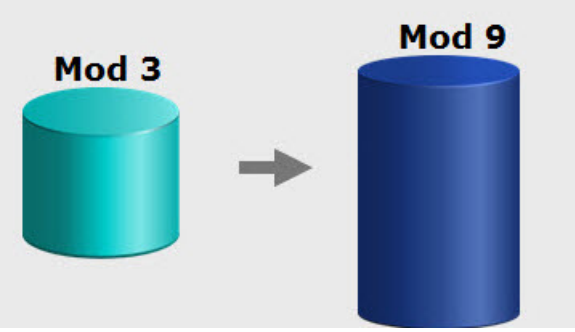


Figure 24. z/OS volume migration

Volume level data migration facilities move logical volumes in their entirety. z/OS Migrator volume migration is performed on a track for track basis without regard to the logical contents of the volumes involved. Volume migrations end in a volume swap which is entirely non-disruptive to any applications using the data on the volumes.

Volume migrator

Volume migration provides host-based services for data migration at the volume level on mainframe systems. It provides migration from third-party devices to devices on Dell EMC arrays as well as migration between devices on Dell EMC arrays.

Volume mirror

Volume mirroring provides mainframe installations with volume-level mirroring from one device on a EMC array to another. It uses host resources (UCBs, CPU, and channels) to monitor channel programs scheduled to write to a specified primary volume and clones them to also write to a specified target volume (called a mirror volume).

After achieving a state of synchronization between the primary and mirror volumes, Volume Mirror maintains the volumes in a fully synchronized state indefinitely, unless interrupted by an operator command or by an I/O failure to a Volume Mirror device. Mirroring is controlled by the volume group. Mirroring may be suspended consistently for all volumes in the group.

Dataset migration using z/OS Migrator

In addition to volume migration, z/OS Migrator provides for logical migration, that is, the migration of individual datasets. In contrast to volume migration functions, z/OS Migrator performs dataset migrations with full awareness of the contents of the volume, and the metadata in the z/OS system that describe the datasets on the logical volume.

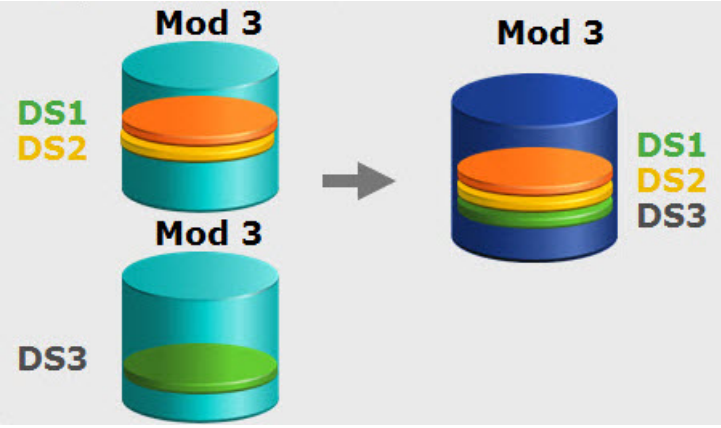


Figure 25. z/OS Migrator dataset migration

Thousands of datasets can either be selected individually or wild-carded. z/OS Migrator automatically manages all metadata during the migration process while applications continue to run.

A

Mainframe Error Reporting

This appendix lists the mainframe environmental errors.

Topics:

- Error reporting to the mainframe host
- SIM severity reporting

Error reporting to the mainframe host

HYPERMAX OS can detect and report the following error types to the mainframe host in the storage systems:

- Data Check — HYPERMAX OS detected an error in the bit pattern read from the disk. Data checks are due to hardware problems when writing or reading data, media defects, or random events.
- System or Program Check — HYPERMAX OS rejected the command. This type of error is indicated to the processor and is always returned to the requesting program.
- Overrun — HYPERMAX OS cannot receive data at the rate it is transmitted from the host. This error indicates a timing problem. Resubmitting the I/O operation usually corrects this error.
- Equipment Check —HYPERMAX OS detected an error in hardware operation.
- Environmental — HYPERMAX OS internal test detected an environmental error. Internal environmental tests monitor, check, and report failures of the critical hardware components. They run at the initial system power-up, upon every software reset event, and at least once every 24 hours during regular operations.

If an environmental test detects an error condition, it sets a flag to indicate a pending error and presents a unit check status to the host on the next I/O operation. The test that detected the error condition is then scheduled to run more frequently. If a device-level problem is detected, it is reported across all logical paths to the device experiencing the error. Subsequent failures of that device are not reported until the failure is fixed.

If a second failure is detected for a device while there is a pending error-reporting condition in effect, HYPERMAX OS reports the pending error on the next I/O and then the second error.

HYPERMAX OS reports error conditions to the host and to the Dell EMC Customer Support Center. When reporting to the host, HYPERMAX OS presents a unit check status in the status byte to the channel whenever it detects an error condition such as a data check, a command reject, an overrun, an equipment check, or an environmental error.

When presented with a unit check status, the host retrieves the sense data from the storage array and, if logging action has been requested, places it in the Error Recording Data Set (ERDS). The EREP (Environment Recording, Editing, and Printing) program prints the error information. The sense data identifies the condition that caused the interruption and indicates the type of error and its origin. The sense data format depends on the mainframe operating system. For 2105, 2107, or 3990 controller emulations, the sense data is returned in the SIM format.

SIM severity reporting

HYPERMAX OS supports SIM severity reporting that enables filtering of SIM severity alerts reported to the multiple virtual storage (MVS) console.

- All SIM severity alerts are reported by default to the EREP (Environmental Record Editing and Printing program).
- ACUTE, SERIOUS, and MODERATE alerts are reported by default to the MVS console.

The following table lists the default settings for SIM severity reporting.

Table 17. SIM severity alerts

Severity	Description
SERVICE	No system or application performance degradation is expected. No system or application outage has occurred.

Table 17. SIM severity alerts (continued)

Severity	Description
MODERATE	Performance degradation is possible in a heavily loaded environment. No system or application outage has occurred.
SERIOUS	A primary I/O subsystem resource is disabled. Significant performance degradation is possible. System or application outage may have occurred.
ACUTE	A major I/O subsystem resource is disabled, or damage to the product is possible. Performance may be severely degraded. System or application outage may have occurred.
REMOTE SERVICE	EMC Customer Support Center is performing service/ maintenance operations on the system.
REMOTE FAILED	The Service Processor cannot communicate with the EMC Customer Support Center.

Environmental errors

The following table lists the environmental errors in SIM format for HYPERMAX OS 5977 or later.

Table 18. Environmental errors reported as SIM messages

Hex code	Severity level	Description	SIM reference code
04DD	MODERATE	MMCS health check error	24DD
043E	MODERATE	An SRDF Consistency Group was suspended.	E43E
044D	MODERATE	An SRDF path was lost.	E44D
044E	SERVICE	An SRDF path is operational after a previous failure.	E44E
0461	NONE	The M2 is resynchronized with the M1 device. This event occurs once the M2 device is brought back to a Ready state. [a]	E461
0462	NONE	The M1 is resynchronized with the M2 device. This event occurs once the M1 device is brought back to a Ready state. [a]	E462
0463	SERIOUS	One of the back-end directors failed into the IMPL Monitor state.	2463
0465	NONE	Device resynchronization process has started. [a]	E465
0467	MODERATE	The remote storage system reported an SRDF error across the SRDF links.	E467
046D	MODERATE	An SRDF group is lost. This event happens, for example, when all SRDF links fail.	E46D
046E	SERVICE	An SRDF group is up and operational.	E46E

Table 18. Environmental errors reported as SIM messages (continued)

Hex code	Severity level	Description	SIM reference code
0470	ACUTE	OverTemp condition based on memory module temperature.	2470
0471	ACUTE	The Storage Resource Pool has exceeded its upper threshold value.	2471
0473	SERIOUS	A periodic environmental test (env_test9) detected the mirrored device in a Not Ready state.	E473
0474	SERIOUS	A periodic environmental est (env_test9) detected the mirrored device in a Write Disabled (WD) state.	E474
0475	SERIOUS	An SRDF R1 remote mirror is in a Not Ready state.	E475
0476	SERVICE	Service Processor has been reset.	2476
0477	REMOTE FAILED	The Service Processor could not call the EMC Customer Support Center (failed to call home) due to communication problems.	1477
047A	MODERATE	AC power lost to Power Zone A or B.	247A
047B	MODERATE	Drop devices after RDF Adapter dropped.	E47B
01BA 02BA 03BA 04BA	ACUTE	Power supply or enclosure SPS problem.	24BA
047C	ACUTE	The Storage Resource Pool has Not Ready or Inactive TDATs.	247C
047D	MODERATE	Either the SRDF group lost an SRDF link or the SRDF group is lost locally.	E47D
047E	SERVICE	An SRDF link recovered from failure. The SRDF link is operational.	E47E
047F	REMOTE SERVICE	The Service Processor successfully called the EMC Customer Support Center (called home) to report an error.	147F
0488	SERIOUS	Replication Data Pointer Meta Data Usage reached 90-99%.	E488
0489	ACUTE	Replication Data Pointer Meta Data Usage reached 100%.	E489

Table 18. Environmental errors reported as SIM messages (continued)

Hex code	Severity level	Description	SIM reference code
0492	MODERATE	Flash monitor or MMCS drive error.	2492
04BE	MODERATE	Meta Data Paging file system mirror not ready.	24BE
04CA	MODERATE	An SRDF/A session dropped due to a non-user request. Possible reasons include fatal errors, SRDF link loss, or reaching the maximum SRDF/A host-response delay time.	E4CA
04D1	REMOTE SERVICE	Remote connection established. Remote control connected.	14D1
04D2	REMOTE SERVICE	Remote connection closed. Remote control rejected.	14D2
04D3	MODERATE	Flex filter problems.	24D3
04D4	REMOTE SERVICE	Remote connection closed. Remote control disconnected.	14D4
04DA	MODERATE	Problems with task/threads.	24DA
04DB	SERIOUS	SYMPL script generated error.	24DB
04DC	MODERATE	PC related problems.	24DC
04E0	REMOTE FAILED	Communications problems.	14E0
04E1	SERIOUS	Problems in error polling.	24E1
052F	None	A sync SRDF write failure occurred.	E42F
3D10	SERIOUS	A SnapVX snapshot failed.	E410

a. Dell EMC recommendation: NONE.

Operator messages

Error messages

On z/OS, SIM messages are displayed as IEA480E Service Alert Error messages. They are formatted as shown below:

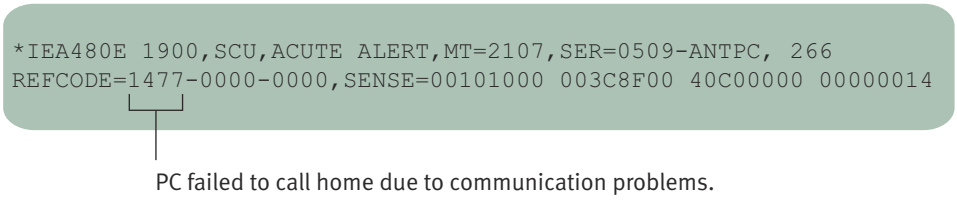


Figure 26. z/OS IEA480E acute alert error message format (call home failure)

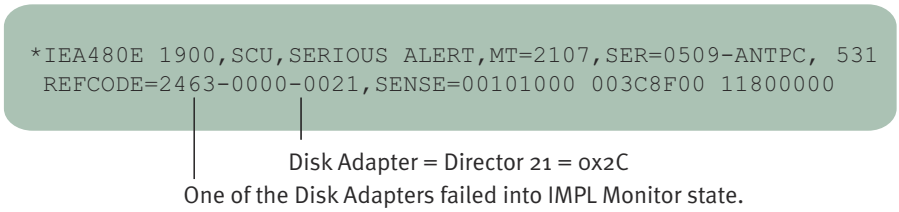


Figure 27. z/OS IEA480E service alert error message format (Disk Adapter failure)

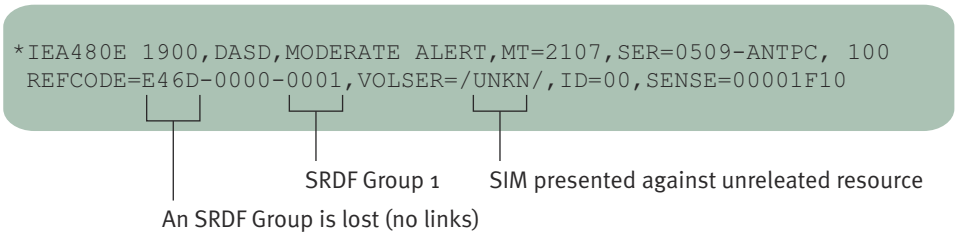


Figure 28. z/OS IEA480E service alert error message format (SRDF Group lost/SIM presented against unrelated resource)

Event messages

The storage array also reports events to the host and to the service processor. These events are:

- The mirror-2 volume has synchronized with the source volume.
- The mirror-1 volume has synchronized with the target volume.
- Device resynchronization process has begun.

On z/OS, these events are displayed as IEA480E Service Alert Error messages. They are formatted as shown below:

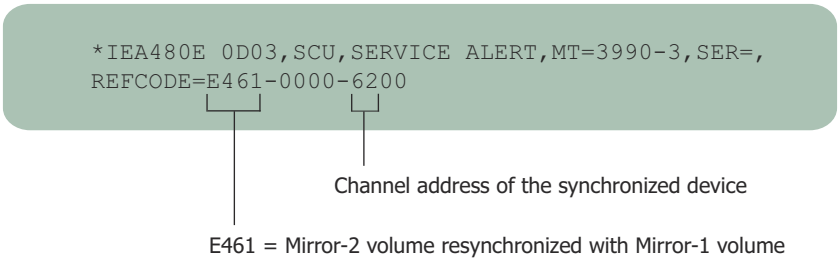


Figure 29. z/OS IEA480E service alert error message format (mirror-2 resynchronization)

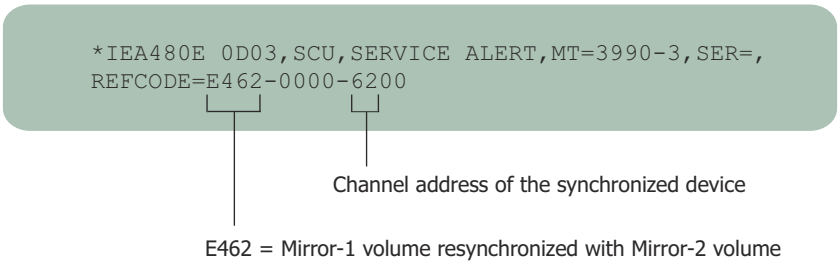


Figure 30. z/OS IEA480E service alert error message format (mirror-1 resynchronization)

B

Licensing

This appendix is an overview of licensing on arrays running HYPERMAX OS.

Topics:

- eLicensing
- Open systems licenses

eLicensing

Arrays running HYPERMAX OS use *Electronic Licenses* (eLicenses).

NOTE: For more information on eLicensing, refer to Dell EMC Knowledgebase article 335235 on the Dell EMC Online Support website.

You obtain license files from Dell EMC Online Support, copy them to a Solutions Enabler or a Unisphere host, and push them out to your arrays. The following figure illustrates the process of requesting and obtaining your eLicense.

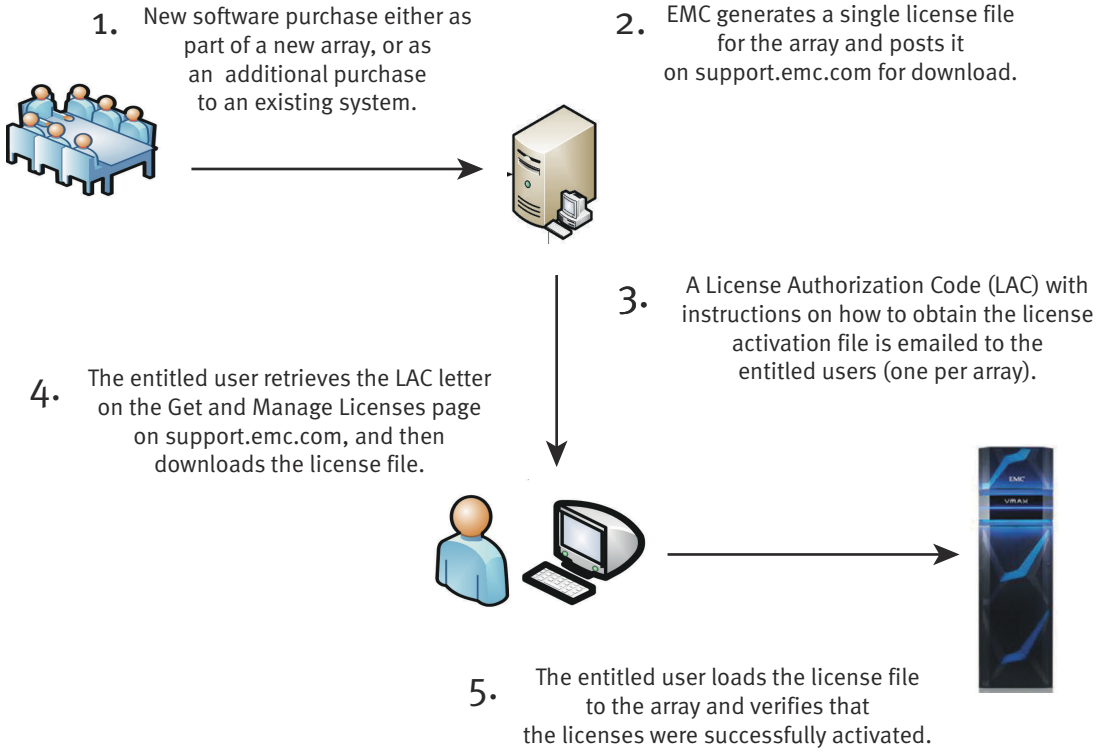


Figure 31. eLicensing process

NOTE: To install array licenses, follow the procedure described in the *Solutions Enabler Installation Guide* and *Unisphere Online Help*.

Each license file fully defines all of the entitlements for a specific system, including the license type and the licensed capacity. To add a feature or increase the licensed capacity, obtain and install a new license file.

Most array licenses are array-based, meaning that they are stored internally in the system feature registration database on the array. However, there are a number of licenses that are host-based.

Array-based eLicenses are available in the following forms:

- An *individual license* enables a single feature.
- A *license suite* is a single license that enables multiple features. License suites are available only if all features are enabled.
- A *license pack* is a collection of license suites that fit a particular purpose.

To view effective licenses and detailed usage reports, use Solutions Enabler, Unisphere, Mainframe Enablers, Transaction Processing Facility (TPF), or IBM i platform console.

Capacity measurements

Array-based licenses include a *capacity licensed* value that defines the scope of the license. The method for measuring this value depends on the license's *capacity type* (Usable or Registered).

Not all product titles are available in all capacity types, as shown below.

Table 19. VMAX All Flash product title capacity types

Usable	Registered	Other
All F software package titles	Storage Direct	PowerPath (if purchased separately)
All FX software package titles		Events and Retention Suite
All zF software package titles		
All zFX software package titles		
RecoverPoint		

Usable capacity

Usable Capacity is defined as the amount of storage available for use on an array. The usable capacity is calculated as the sum of all Storage Resource Pool (SRP) capacities available for use. This capacity does not include any external storage capacity.

Registered capacity

Registered capacity is the amount of user data managed or protected by each particular product title. It is independent of the type or size of the disks in the array.

The methods for measuring registered capacity depends on whether the licenses are part of a bundle or individual.

Registered capacity licenses

Registered capacity is measured according to the following:

- Storage Direct
  - The registered capacity of this license is the sum of all Data Domain encapsulated devices that are link targets. When there are TimeFinder sessions present on an array with only a Storage Direct license and no TimeFinder license, the capacity is calculated as the sum of all Data Domain encapsulated devices with link targets and the sum of all TimeFinder allocated source devices and delta RDPs.

Open systems licenses

This section details the licenses available in an open system environment.

License suites

This table lists the license suites available in an open systems environment.

Table 20. VMAX All Flash license suites

License suite	Includes	Allows you to	With the command
All Flash F	• HYPERMAX OS • Priority Controls • OR-DM • Unisphere for VMAX • FAST • SL Provisioning • Workload Planner • Database Storage Analyzer	Create time windows	symoptmz symtw
		• Add disk group tiers to FAST policies • Enable FAST • Set the following FAST parameters: ○ Swap Non-Visible Devices ○ Allow Only Swap ○ User Approval Mode ○ Maximum Devices to Move ○ Maximum Simultaneous Devices ○ Workload Period ○ Minimum Performance Period • Add virtual pool (VP) tiers to FAST policies • Set the following FAST VP-specific parameters: ○ Thin Data Move Mode ○ Thin Relocation Rate ○ Pool Reservation Capacity • Set the following FAST parameters: ○ Workload Period ○ Minimum Performance Period	symfast
		Perform SL-based provisioning	symconfigure symsg symcfg
	AppSync	Manage protection and replication for critical applications and databases for Microsoft, Oracle and VMware environments.	
	• TimeFinder/Snap	Create new native clone sessions	symclone

Table 20. VMAX All Flash license suites (continued)

License suite	Includes	Allows you to	With the command
	• TimeFinder/SnapVX • SnapSure	Create new TimeFinder/ Clone emulations	symmir
		• Create new sessions • Duplicate existing sessions	symsnap
		• Create snap pools • Create SAVE devices	symconfigure
		• Perform SnapVX Establish operations • Perform SnapVX snapshot Link operations	symsnapvx
All Flash FX	All Flash F Suite	Perform tasks available in the All Flash F suite.	
	• SRDF • SRDF/Asynchronous • SRDF/Synchronous • SRDF/Star • Replication for File	• Create new SRDF groups • Create dynamic SRDF pairs in Adaptive Copy mode	symrdf
		• Create SRDF devices • Convert non-SRDF devices to SRDF • Add SRDF mirrors to devices in Adaptive Copy mode Set the dynamic-SRDF capable attribute on devices Create SAVE devices	symconfigure
		• Create dynamic SRDF pairs in Asynchronous mode • Set SRDF pairs into Asynchronous mode	symrdf
		• Add SRDF mirrors to devices in Asynchronous mode Create RDFA_DSE pools Set any of the following SRDF/A attributes on an SRDF group: ○ Minimum Cycle Time ○ Transmit Idle ○ DSE attributes, including: ■ Associating an RDFA-DSE pool with an SRDF group DSE Threshold	symconfigure

Table 20. VMAX All Flash license suites (continued)

License suite	Includes	Allows you to	With the command
		DSE Autostart ○ Write Pacing attributes, including: ■ Write Pacing Threshold ■ Write Pacing Autostart ■ Device Write Pacing exemption ■ TimeFinder Write Pacing Autostart	
		● Create dynamic SRDF pairs in Synchronous mode ● Set SRDF pairs into Synchronous mode	symrdf
		Add an SRDF mirror to a device in Synchronous mode	symconfigure
	D@RE	Encrypt data and protect it against unauthorized access unless valid keys are provided. This prevents data from being accessed and provides a mechanism to quickly shred data.	
	SRDF/Metro	● Place new SRDF device pairs into an SRDF/Metro configuration. ● Synchronize device pairs.	
	VIPR Suite (Controller and SRM)	Automate storage provisioning and reclamation tasks to improve operational efficiency.	

Individual licenses

These items are available for arrays running HYPERMAX OS and are not in any of the license suites:

Table 21. Individual licenses for open systems environment

License	Allows you to	With the command
Storage Direct	Store and retrieve backup data within an integrated environment containing arrays running HYPERMAX OS and Data Domain arrays.	
RecoverPoint	Protect data integrity at local and remote sites, and recover data from a point in time using journaling technology.	

Ecosystem licenses

These licenses do not apply to arrays:

Table 22. Individual licenses for open systems environment

License	Allows you to
PowerPath	Automate data path failover and recovery to ensure applications are always available and remain operational.
Events and Retention Suite	• Protect data from unwanted changes, deletions and malicious activity. • Encrypt data where it is created for protection anywhere outside the server. • Maintain data confidentiality for selected data at rest and enforce retention at the file-level to meet compliance requirements. • Integrate with third-party anti-virus checking, quota management, and auditing applications.